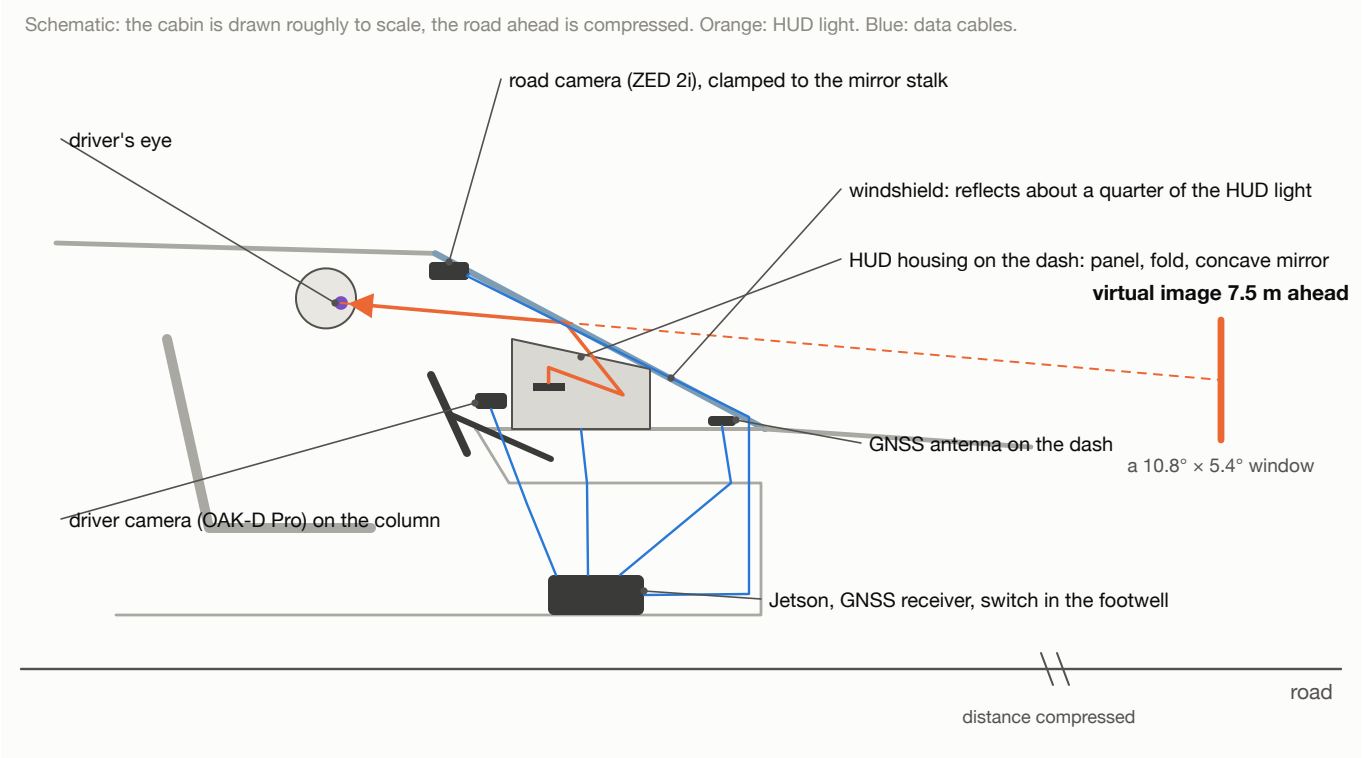
WINDSHIELD AR HUD · HUD-ONLY TFT LCD EDITION

Build Guide

What the head-up display does, the problems it has to solve, the parts that solve them, and how to build, wire, calibrate and prove every stage, from an empty bench to a daylight drive.



Edition of 2026-10-25. Companion to `docs/ARHUD-PLAN-TFT.pdf` (the plan: requirements, derivations, sources) and its condensed specification `docs/ARHUD-PLAN-TFT-SPEC.md`. Numbers are imported from the plan's `calc.py`; housing data from `docs/arhud-cad`; optical layout from `docs/arhud-sim`. Source of this document: `docs/arhud-build-src`.

BEFORE ANYTHING ELSE

How to use this guide

The guide has three parts. **Part I** explains the idea: what the driver sees, the problems that stand between an ordinary display and that picture, and the component that solves each one. Read it once, end to end, before buying anything; every later instruction refers back to it. **Part II** sets up the workshop and lists every material, tool and instrument, with the build stage that first needs it. **Part III** is the build, in fourteen build stages (B1 to B14), each with the same layout: what the stage solves, what you will have at the end, materials, tools, a numbered procedure with checks, the acceptance test, and what to do when something goes wrong. The appendices hold wiring schedules, pin maps, file formats, print settings and blank test sheets.

Two words are used carefully. A **build stage** (B1 to B14) is a chapter of this guide. An **optical stage** (Stage 0, Stage 1, Stage 2) is one of the plan's three ways of making the picture: a bench beamsplitter, an in-car Fresnel magnifier and combiner, and the final relay mirror into the windshield. Gates **G0 to G7** are the plan's pass/fail tests; a build stage is finished when its gate passes, and a failed gate is recorded, not waived.

Conventions

MARK	MEANING
ASSUMED	A design value that a named measurement must replace. The guide says which measurement, and where in the build it happens.
VERIFY	Not confirmed from a primary source or not yet tested on this hardware. Check before relying on it.
DECISION	A choice the builder must make at that point; the options and their consequences are given.
☐	A check. Tick it in the printed copy; every check of a stage must be ticked before its gate is run.
Box with a heavy left rule	A safety warning. These are not optional.
Prices	US dollars, last seen on vendor pages on 2026-10-21 for the plan. They are not quotes.
Units and frames	Millimetres for parts, metres for the car. Car frame as in the plan: x forward, y right, z down, origin at the road camera's left lens. The simulator and CAD use x forward, y up, z right with the origin on the road under the nominal eye; the guide says which frame a number is in when it matters.

DESIGN BASIS: WHAT THIS GUIDE BUILDS, AND WHERE IT DIFFERS FROM THE PLAN PDF

The guide builds the plan's design (HUD only, no driving functions, one Jetson, a TFT panel with a custom backlight, a relay mirror into the stock windshield) to the layout in the simulator (`docs/arhud-sim/js/geom.js`) and the printed housing in `docs/arhud-cad`, because those are the only geometry with print files. Where the simulations found a defect in the plan's numbers, the guide follows the simulation and says so. None of these findings has been measured on hardware yet; the real-car ray trace in B1 supersedes all of them.

ITEM	PLAN PDF	THIS GUIDE, AND WHY
Concave mirror figure	Sphere, R = 660 mm	Not a sphere. At the 15.5° incidence the fold needs, a sphere splits the image into two focal distances (about 52.7 m and 4.1 m, simulator). A toroid with R_t 685 / R_s 636 mm fixes that for a flat windshield; the Cycles cross-check then found the toroid's image surface strongly curved (about 6 to 19 m across the window), which a freeform mirror with a tilted panel flattens to 7.4 to 7.6 m. Order the figure the B1 trace produces for the chosen car.
Eye-to-mirror path	1 m (assumed)	1.06 m (glass to mirror 310 mm). At 1 m the top of a 200 mm mirror enters a 28° windshield; 1.06 m clears it by about 24 mm.
Fold to panel	not specified	50 mm in the CAD. The CAD's own check finds 42 of 690 traced rays grazing the front edge of the panel's cell glass at this spacing; 65 to 70 mm clears it. A **DECISION** in B11, before printing.
Window and distance	10.8° × 5.4° at 7.5 m	Traced through the hardware: about 8.6 m and 8.7° × 4.6° for the toroid; the freeform fit gives about 9.1° × 3.6°. Calibrate to the window you measure, not to the plan's numbers.
Projection in the renderer	one virtual plane plus a 2-D warp	A lookup table from the ray trace (eye position to panel point), refined by the photographed grid. With head motion the single plane left 15 to 17′ of error in simulation; the lookup table left about 1.5′.
Night white level	≤ 50 cd/m²	Start at 3 cd/m² and tune by eye. 50 is about thirty times a headlit road; TI's guidance is under 3. Analogue dimming, one string off, and grey levels on the panel cover the range (B5.2).
LED dimming	PWM above 2 kHz, RCD-24 class driver	Analogue current dimming with the RCD-24, its PWM/ON/OFF pin used only as a fail-safe enable. The RCD-24's datasheet limits PWM to 1 kHz (linear only to 200 Hz), so it cannot meet the plan's 2 kHz; current dimming has no flicker at all.
Power at key-off	ignition-sensed converter on the accessory feed	The converter's input must come from an always-live socket, with only its ignition sense on a switched one; on a switched input the shutdown delay cannot work (B7.2).
Lane points	drawn per frame	Smoothed over time before drawing. Raw per-frame noise made the lane lines wobble visibly in simulation.
Ribbon fill	translucent	At least about 39% of white at noon; at 32% the road simulation measured a contrast of 1.01 (invisible).
Sun-watch patch	21° × 11°	39° × 13.5°: the patch of sky that can focus onto any part of the cell, from the Cycles sun trace, is larger than the patch for the lit crop alone.

ITEM	PLAN PDF	THIS GUIDE, AND WHY
Panel luminance target	10,000 cd/m² (1.3× the 7,716 needed)	About 13,000 cd/m². The plan's path omits the dust-cover glass, its polariser, the IR-cut plate and the retarder; with them the 10,000 design gives roughly 1,250–1,450 cd/m² at the eye. Measure each part's transmission (B10) and recompute.
IR-cut glass	text: in front of the panel; backlight figure: under the cell	A hot-mirror IR-cut plate in front of the cell, held on the panel mask. Sunlight reaches the cell from the fold side, so glass under the cell cannot protect it; a plain cover glass seals the backlight instead. White LEDs emit almost no infrared.
Housing	printed, matte black	32 printed parts in ASA (39 pieces), each within a 250 mm cube; ASA because dash tops pass 80 °C and PETG softens near 75 °C.

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PART I

The idea

What the driver sees, why that is hard, and which part of the machine exists to solve each difficulty. No soldering yet: this part is the map that every later instruction refers to.

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1 What you are building
2 The problems, and the parts that solve them
3 One frame, from photons in to photons out
4 The build at a glance

1 What you are building

A head-up display that draws on the road. The driver looks through the windshield as usual and sees, floating over the lane ahead, a translucent ribbon lying exactly on their lane, thin lines over the painted lane markings, a chevron under the car in front with the gap to it, outlines around pedestrians and cyclists, and a small fixed block with the speed and the last speed-limit sign.

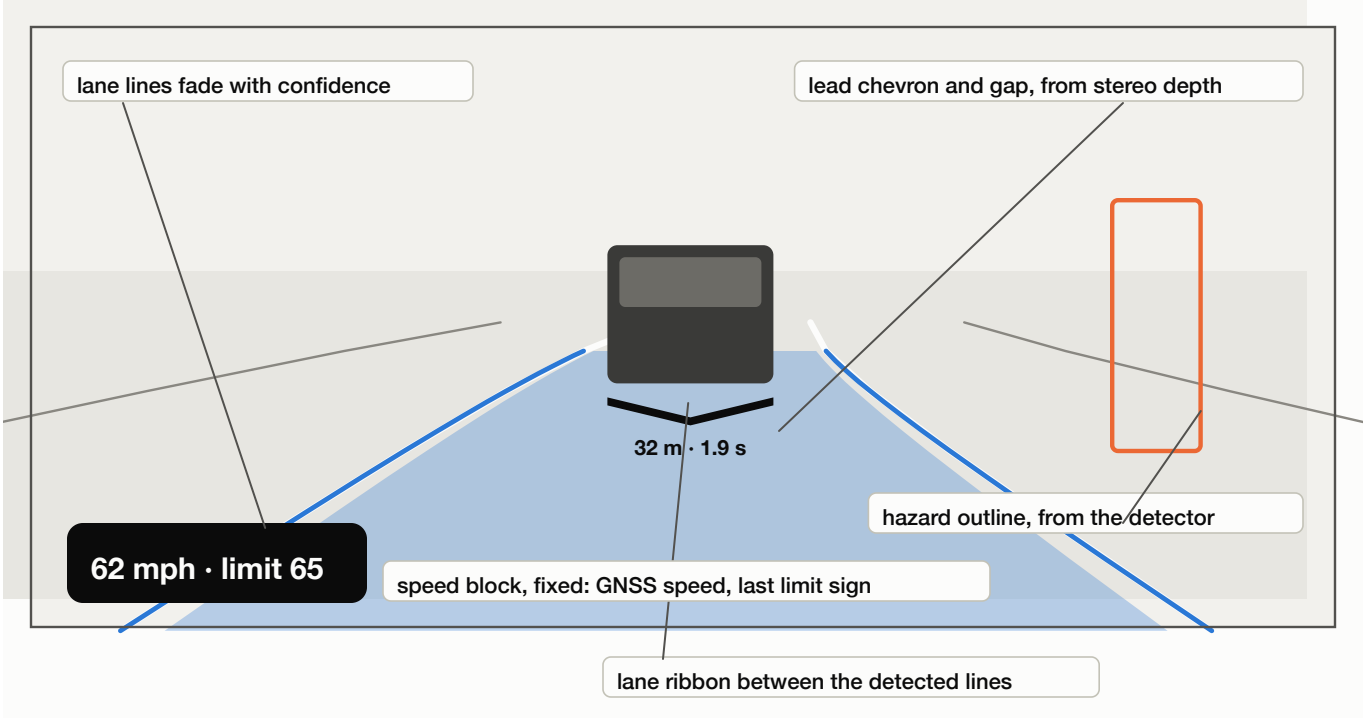


Figure 1.1. What the driver sees through the HUD window (illustrative). Everything except the speed block is *conformal*: drawn so that it lies on the real object it describes, from the driver's own eye.

1.1 What makes it different from an ordinary HUD

A factory head-up display shows numbers: a speed and an arrow, floating about two metres ahead, in a small box that does not need to line up with anything. This one has to line up with the world. A lane line drawn half a degree to the side of the paint looks wrong; a chevron that trails a braking car by a metre looks broken. Three consequences follow, and they shape the whole machine.

- **The picture must appear far away.** If the image floated two metres ahead, the eye would have to refocus between the image and the road, and the two eyes would see the mark on two different spots of the road. The design puts the image 7.5 m ahead, where focus and the two eyes' views almost agree with the road from about 10 m out.
- **The machine must know where things are, where the driver's eye is, and how old its information is,** all to a fraction of a degree, 30 to 60 times a second.
- **The picture must be very bright.** The windshield reflects only about a quarter of the light sent at it, and the background at noon is a sunlit sky.

1.2 The design in one paragraph

One computer, an NVIDIA Jetson Orin NX, does everything. A stereo road camera high on the windshield gives it images and depth; neural networks find the lanes and the vehicles; a satellite receiver with an inertial unit gives speed and turning rate; a second camera watches the driver's face and reports where the eyes are. The Jetson draws the overlay for that eye position, 60 times a second, onto a small 5.5-inch TFT LCD panel. Only a 60 × 30 mm patch of that panel is used, and a custom LED backlight behind it makes it about twenty times brighter than the panel's stock backlight. Light from the patch goes up to a fold mirror, forward to a large concave mirror, up to the windshield, and back to the driver's eye. The concave mirror works like a magnifying glass, so the eye sees the patch as a window about 1.4 m wide, 7.5 m ahead, over the road. All of the optics sit in a printed housing on the dash, which lifts off without tools. The only electrical connection to the car is a fused 12 V accessory feed.

1.3 The targets that define "working"

These are the plan's acceptance criteria, abbreviated. B13 turns the full list into a test sheet.

QUANTITY	TARGET	WHY THAT NUMBER
Virtual image distance	≥ 6 m, design 7.5 m	Far enough that eye focus and the two eyes' views agree with the road beyond about 10 m.
Field of view	10.8° × 5.4° (10° × 4° to 13° × 5° accepted)	Covers one lane from about 20 m out and the lead car to 80 m.
Eye box	≥ 120 × 60 mm untracked; design 130 × 70 mm	The driver's head moves; the whole window must stay visible.
Luminance at the eye, noon, sky behind	≥ 1,500 cd/m²	A clear sky is about 10,000 cd/m²; the overlay must add at least 15% to be seen.
Static registration at 30 m	≤ 0.5°	About a quarter of a lane width at 30 m.
Registration with ±75 mm head motion	≤ 0.3°	Ordinary head sway must not break alignment.
Motion to photon, whole chain	≤ 90 ms, compensated error ≤ 0.3°	At 30 m/s the car moves 2.7 m in 90 ms.
Connection to the car	12 V accessory feed only, fused	Nothing in the HUD can talk to the car's computers or move an actuator.

WHAT IT IS NOT

It does not drive, steer, brake or warn on its own authority. It reads nothing from the car's buses and writes nothing to them. The driver drives; a separate safety driver is required for demonstrations; demonstrations happen on a closed course or private lot until the university's risk office signs off anything else (plan, section 11).

2 The problems, and the parts that solve them

It helps to think of the HUD as a chain of answers to six questions asked every frame, with a layer of protection underneath that runs all the time. Figure 2.1 is the map; the sections after it take one problem at a time. Each section says what the problem is in plain terms, why it is hard (with the number that makes it hard), the idea that solves it, the parts that idea needs, and the build stage where you make those parts.

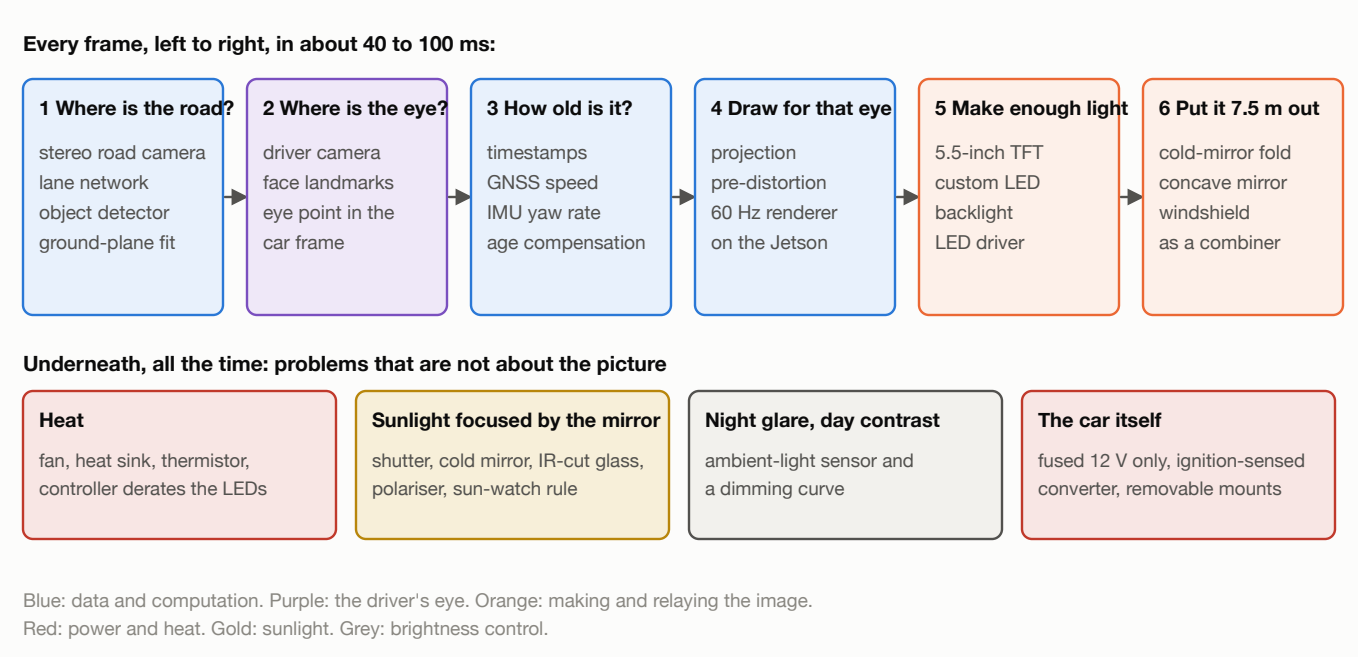


Figure 2.1. The six problems the HUD solves every frame, and the four it guards against all the time. The colours are used the same way in every diagram in this guide.

2.1 Putting a picture out on the road

THE PROBLEM

A screen on the dash is half a metre from the eye. Light reflected off the windshield from that screen would appear about half a metre beyond the glass: the driver would see a blurred overlay while looking at the road, and each eye would see it over a different part of the road.

THE IDEA

A magnifying glass. When an object sits just inside the focal point of a converging lens or concave mirror, the light leaving it spreads as if it came from a much larger object much farther away: a *virtual image*. Put the panel 316 mm from a concave mirror of focal length 330 mm and the image appears 7.5 m away and 23.7 times larger. The windshield then acts as a partly reflecting mirror, the *combiner*, that folds that light back to the driver while letting the road show through.

WHY IT IS HARD	The nearer the panel is to the focal point, the farther the image and the more sensitive it is: the panel sits only 14 mm inside focus, so a millimetre of panel position moves the image by metres. A mirror used at an angle has different focal lengths in its two directions (astigmatism), and the windshield's own curvature adds more. That is why the mirror's exact shape comes from a ray trace with the real car's windshield, and why the mirror is the longest lead-time item in the project.
PARTS	TFT panel (the object), cold-mirror fold (turns the beam so it fits in the dash), custom concave mirror about 300 × 200 mm (the magnifier), the stock windshield (the combiner), a printed housing holding them in place.
BUILT IN	B1 (measure, trace), B10 (mirror and fold), B11 (housing), B12 (assembly), B13 (calibration). Optical Stages 0 and 1 (B6, B8) are simpler versions that need no custom mirror.

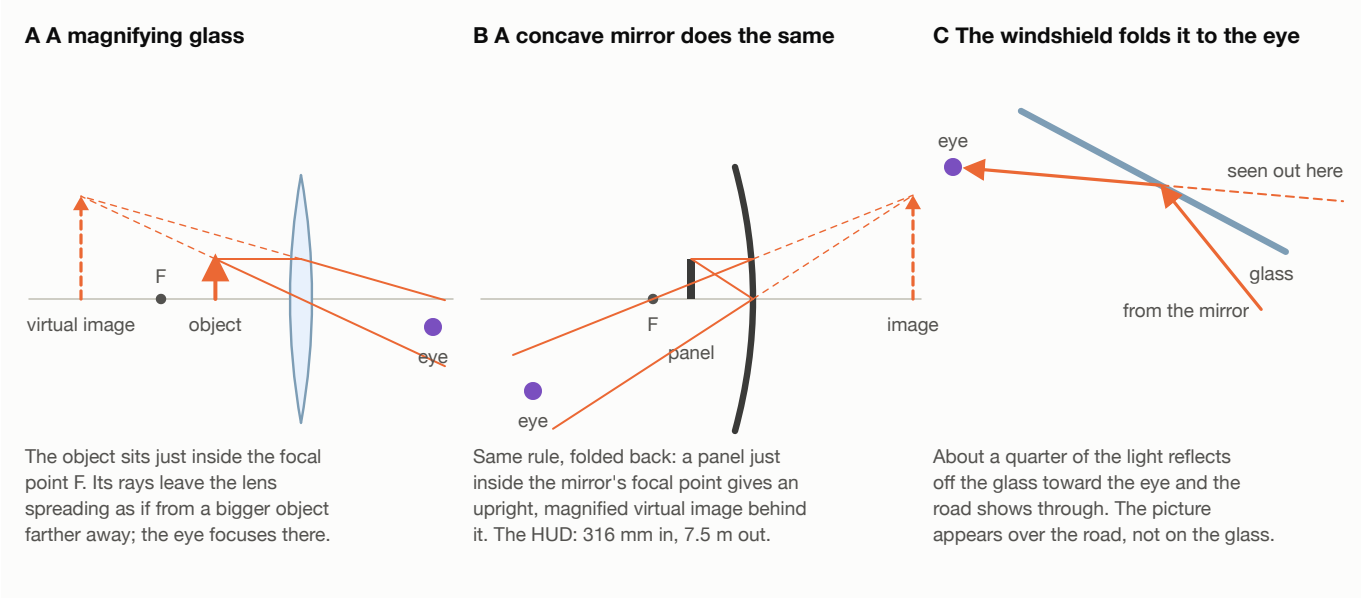


Figure 2.2. The virtual image, three ways. A lens and a concave mirror do the same job; the windshield only folds the result toward the eye.

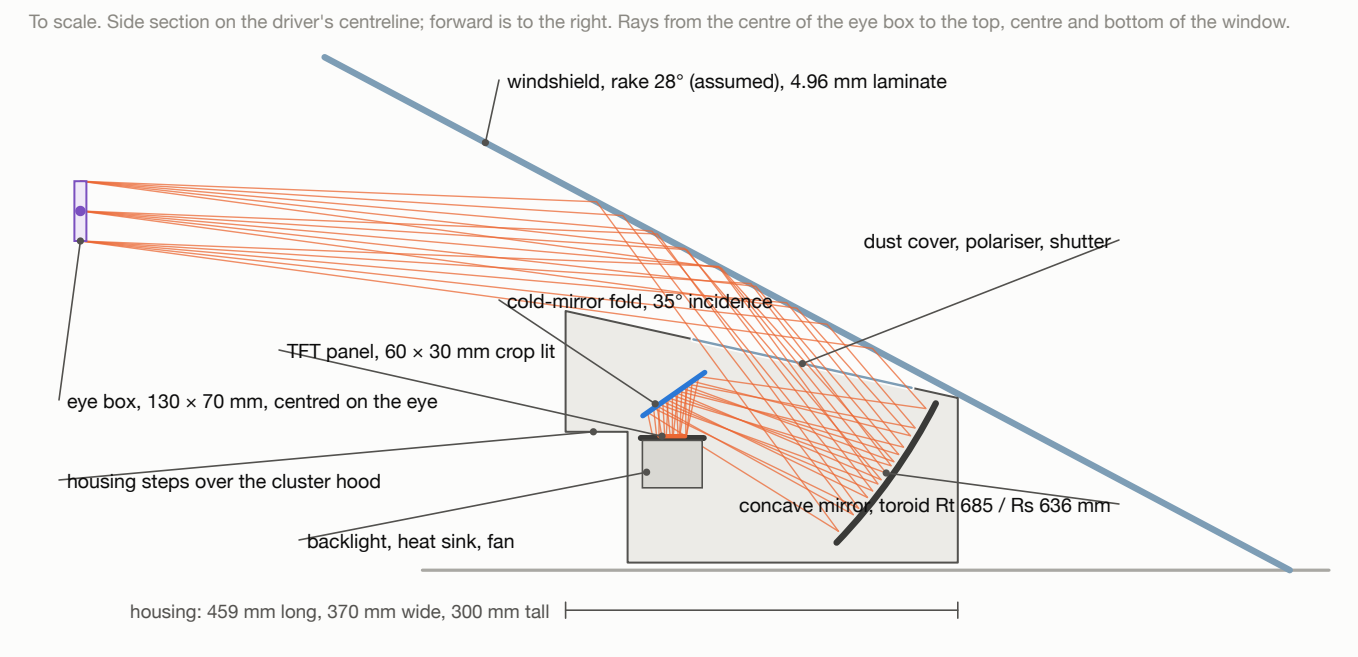


Figure 2.3. The real thing, to scale: the layout the printed housing was designed around. Light leaves the 60 × 30 mm crop going up, reflects forward off the fold, back and up off the concave mirror, and off the windshield into the eye box.

2.2 Knowing where the road is

THE PROBLEM	To draw a ribbon on the lane, the computer needs the lane's position in three dimensions in front of the car, not just where it appears in a picture. To put a chevron under the car ahead, it needs that car's direction and distance.
THE IDEA	A *stereo* camera: two cameras 12 cm apart see nearby things shifted more than distant things, which gives a depth for every pixel, the way two eyes do. A neural network finds the lane lines in the left image; another finds and labels vehicles, people and signs; a tracker gives each one a stable identity across frames. Each frame, the software fits a flat plane to the depth points on the road just ahead (RANSAC, a fit that ignores outliers such as cars), and places each lane pixel where its line of sight meets that plane. Direction comes from the image and is very accurate; distance comes from stereo depth.
WHY IT IS HARD	Lane networks are trained on other people's roads and may not generalise to the demo course; that is why the build records the course early and scores the networks on 300 hand-labelled frames before anything else depends on them (gate G1). The networks, the depth and the renderer must share one embedded GPU in about 33 ms per frame; the plan's estimate is 24–50 ms, so the first job on the Jetson is to measure it.
PARTS	Stereolabs ZED 2i stereo camera with 4 mm lenses (clamped high on the windshield's centreline at the mirror stalk); the Jetson's GPU running a lane network (UFLDv2 or CLRNet), YOLO11s object detection and ByteTrack tracking as TensorRT FP16 engines.

BUILT IN	B2 (Jetson), B3 (camera mount and recordings), B4 (perception and gate G1).

2.3 Knowing where the driver's eye is

THE PROBLEM	A mark lines up with a road point only along the line from the eye to that point. If the driver's head moves, the right place for the mark moves too (Figure 2.4). And the road camera is not where the eye is: it sits about 45 cm away, so every position it measures must be re-expressed as seen from the eye.
THE IDEA	Measure the eye. A second camera faces the driver, finds the face and the eye landmarks on its own processor, measures their distance with its own stereo pair, and reports the midpoint between the eyes in the car's coordinate frame 30 times a second. The renderer projects every mark for that eye position.
WHY IT IS HARD	With the image at 7.5 m, an untracked 150 mm head shift misplaces a mark by 0.15 × |1/7.5 – 1/d| radians for a point at distance d: nothing at 7.5 m, but 0.29° at 10 m, 0.57° at 15 m, 0.86° at 30 m and 1.15° far away, against a 0.3° target. (The plan's table lists the same numbers in reverse order; the simulator found the correction.) At night the face is dark, so the camera brings its own infrared illuminator. Both camera positions must be known in one shared frame, which is what calibration (B7, B13) establishes.
PARTS	Luxonis OAK-D Pro on the steering-column shroud (global-shutter stereo pair, infrared flood and dot projector, on-camera neural processor); a 30 m target board and a camera at the eye point for calibration.
BUILT IN	B3 (mount), B4 (head-tracking process), B7 and B13 (calibration).

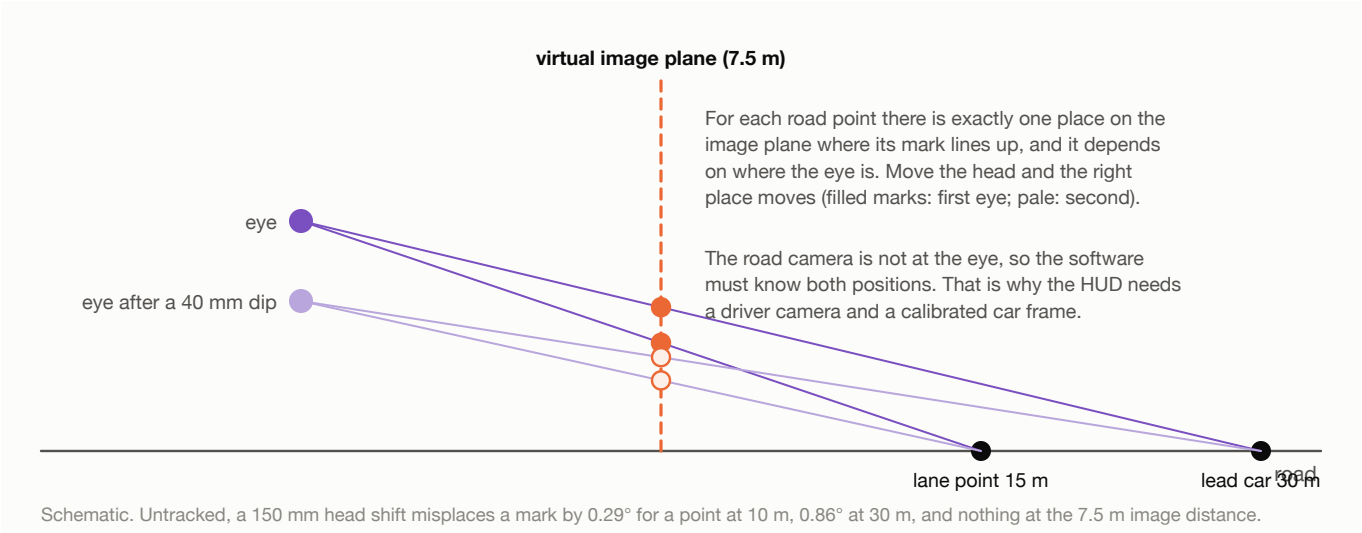


Figure 2.4. Registration: the mark belongs where the line from the eye to the road point crosses the image plane. A dip of the head moves every mark, by more for near points than for far ones.

2.4 Everything is late

THE PROBLEM	From the moment the road camera exposes a frame to the moment the panel lights up, 37 to 99 ms pass. At 30 m/s the car has moved 1.1 to 3.0 m, and if it is turning, it has rotated. Drawn as captured, every mark would lag behind the paint and swing wide in curves.
THE IDEA	Stamp every measurement with the time its camera frame was captured, on one shared clock. Measure the car's own motion independently, at a high rate, from satellite speed and a gyroscope. Just before drawing, move each point back by the distance travelled and the angle turned since its capture: *age compensation*. For the plan's worked example (a point 10 m ahead, 30 m/s, 100 ms old) the uncompensated error is 2.9°; after compensation it is 0.003° for an unmodelled 1 m/s² acceleration.
WHY IT IS HARD	Latency cannot be measured from recordings; only a physical test (a light flashed at the camera, a photodiode on the panel) measures the real chain. And the sign of the correction is easy to get backwards, which doubles the error instead of removing it, so it is checked on a recording before any drive.
PARTS	u-blox ZED-F9R GNSS receiver with its built-in inertial unit and dead reckoning (speed to 0.05 m/s); an active multi-band antenna; the ZED 2i's own 400 Hz inertial unit for yaw rate; a latency probe (LED, photodiode, microcontroller).
BUILT IN	B3 (receiver and antenna), B4 (ego-motion process), B6 (latency probe and gate G2).

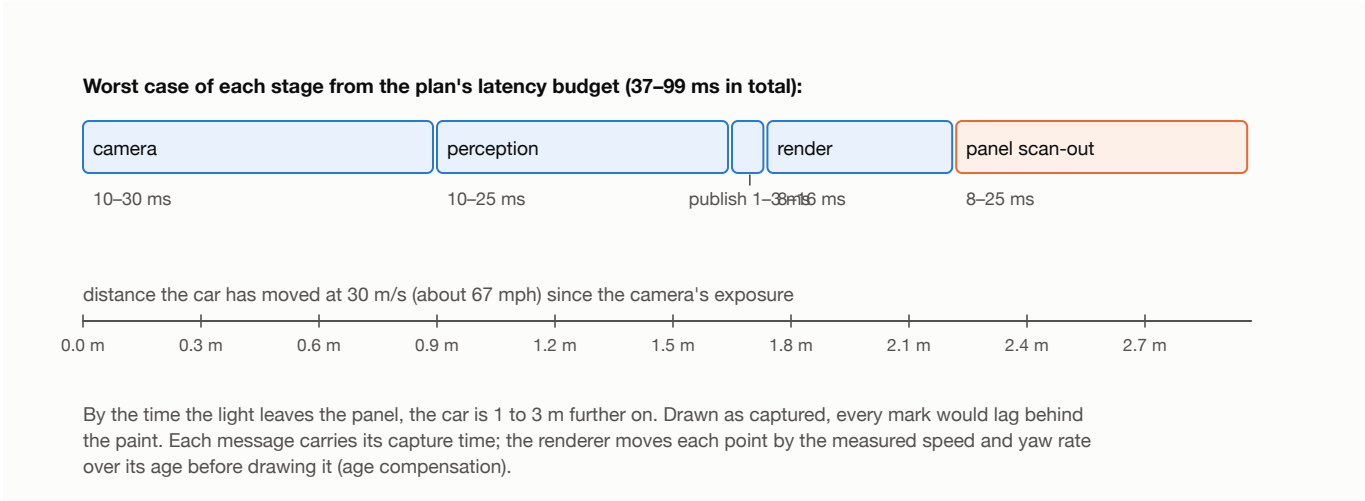
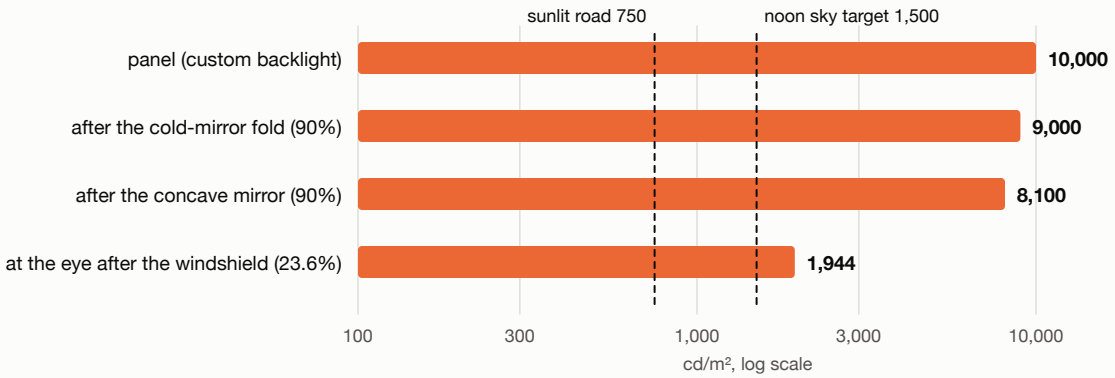


Figure 2.5. Where the time goes in one frame, at the high end of each stage's estimate, and how far the car moves meanwhile.

2.5 Making enough light

THE PROBLEM	The overlay must be visible against a noon sky of about 10,000 cd/m². Light from the panel loses 10% at each of the two mirrors and then only 23.6% of it reflects off the windshield toward the eye (for light polarised the right way; almost none for the other). Overall 19.4% of the panel's luminance reaches the eye. To deliver 1,500 cd/m² at the eye, the panel needs 7,716 cd/m². The stock panel gives 450. No buyable small panel gives more than about 1,600.
THE IDEA	A liquid-crystal panel makes no light; it is an array of shutters in front of a backlight, passing only about 6% of it. So build a much brighter backlight, but only behind the 60 × 30 mm crop that is actually used: eight high-power white LEDs in a white mixing box, a diffuser to even them, two crossed prism films to aim the light into the narrow cone the eye box needs, and a clear cover glass. About 12.6 W of LED power (half that with the films) gives a 10,000 cd/m² panel; lighting the whole panel would take 58 W. The plan's budget leaves out the dust cover, its polariser and the IR-cut plate, which B10 adds back; this guide therefore aims the backlight at about 13,000 cd/m².
WHY IT IS HARD	Uniformity (eight point sources must look like one even surface), heat (the cell absorbs about 3.0 W in 18 cm², and liquid crystal stops working near 75 °C), and flicker (cameras filming the HUD see beats from pulse-width dimming, which is why this guide dims the LEDs by their current instead).
PARTS	Wisecoco 5.5-inch 1920 × 1080 IPS HDMI kit (two: one to take apart, one stock), white LEDs on an aluminium board, finned heat sink and 40 mm fan, mixing cavity, diffuser plate, two prism films, cover glass, two constant-current LED drivers with analogue dimming, a thermistor on the cell.
BUILT IN	B5 (driver and controller), B9 (backlight, measurements, gate G5).



The windshield is the big loss: it reflects about a quarter of s-polarised light and almost none of p. The stock panel (450 cd/m²) would give 87 cd/m² at the eye, invisible at noon; the custom backlight brings it to 10,000 (13,000 in this guide, to cover the losses B10 adds).

Figure 2.6. The light budget from panel to eye, on a log scale. Each step is a multiplication, so the bars shrink by the same ratio on this axis; the windshield is the big loss.

2.6 Keeping the panel cool

THE PROBLEM	The LEDs turn most of their power into heat at the board, and the cell absorbs the light it blocks, about 3.0 W, or 1,641 W/m² over its own area, more than direct sunlight. Consumer liquid crystal goes dark (clears) near 75 °C and the kit panel's temperature rating is not published.
THE IDEA	Remove the heat, measure the temperature, and back off before damage. The heat sink and fan pull cabin air through the housing floor; a thermistor taped to the back of the cell lets a small independent controller halve the backlight above one temperature and switch it off (and close the shutter) above another, even if the Jetson has crashed.
PARTS	Heat sink, 40 mm fan, 10 kΩ NTC thermistor, the HUD controller board (Raspberry Pi Pico), a type-K thermocouple and MAX31855 amplifier for the tests.
BUILT IN	B5 (controller), B9 (30-minute thermal test at 40 °C).

2.7 The mirror is also a burning glass

THE PROBLEM	Light runs both ways. Sunlight that enters through the windshield from the right part of the sky retraces the HUD's path backwards: it reaches the concave mirror, which focuses it, like a magnifying glass on paper, onto the panel. The plan's trace puts about 42 W into a spot about 6 mm across. The direction is specific: about 51° above the horizon *behind* the car for a 28° windshield, which is a summer midday drive away from the sun, or a car parked facing away from it.

THE IDEA	Five layers. A shutter over the aperture that is closed whenever the HUD is off, and that closes itself when power fails. A sun-watch rule: from GNSS time, position and heading the Jetson computes where the sun is relative to the car and closes the shutter when the sun enters the dangerous patch of sky. A cold-mirror fold (reflects visible light, passes infrared to an absorber), an infrared-cut glass in front of the cell and a polariser on the dust cover, which together pass about an eighth of the solar power. The thermistor cut-off. And a thermocouple test of the empty housing in the sun before the panel is ever exposed.
WHY IT IS HARD	Even after the filters the plan estimates about 170 kW/m² at the spot, 4 times the peak TI computes for a production HUD; one HUD maker's patent puts time to damage at minutes. The defences must be built before the optics see daylight, not after.
PARTS	Two-leaf aluminium shutter, SG90 servo and a return spring, cold-mirror glass with an absorber plate, IR-cut glass, an s-pass polariser laminated to the dust cover, the controller and its thermistor.
BUILT IN	B5 (controller rules), B10 (filters), B11 and B12 (shutter, thermocouple test), B4 (sun-watch computation).

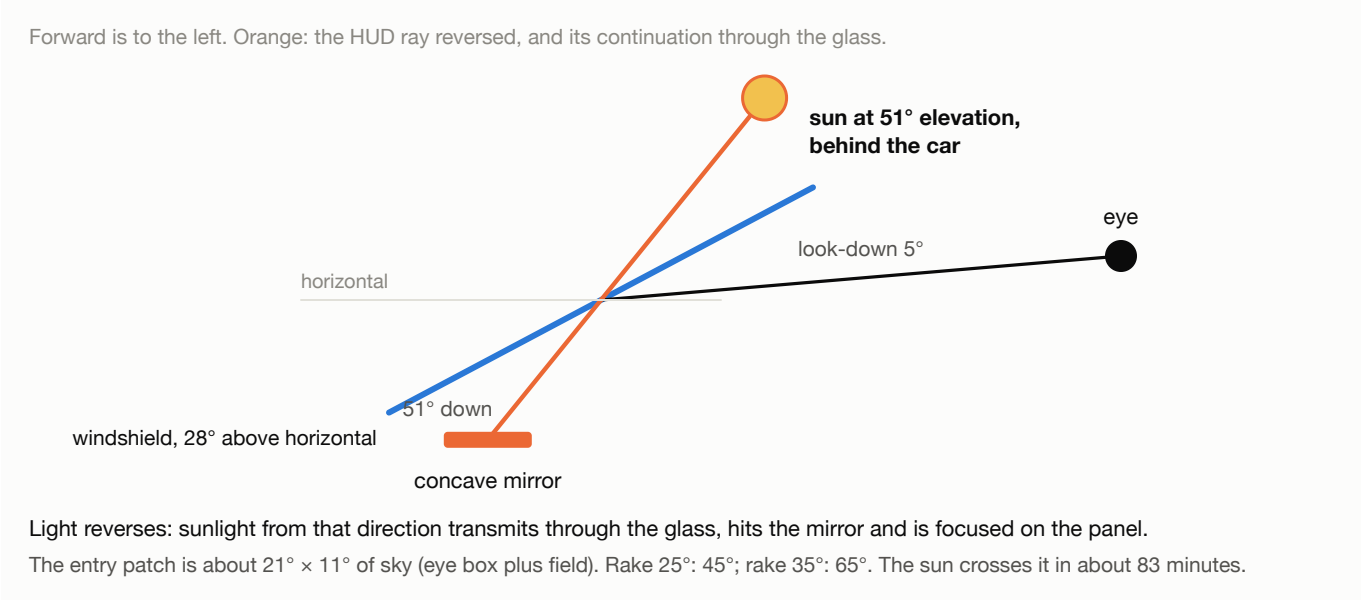


Figure 2.7. From the plan: the HUD's own ray, reversed, continues through the glass to the sky behind the car. Sunlight from there is focused onto the panel. Forward is to the left in this figure.

2.8 Polarisation: the windshield prefers one direction of light

THE PROBLEM	At the windshield's steep angle, glass reflects light polarised parallel to its surface (*s*, here across the car) about nineteen times better than the other direction (*p*): 23.6% against 1.2%. An LCD's output is already polarised, often at 45° to its edges. Sent the wrong way, most of the light passes straight through the windshield.
THE IDEA	Turn the panel's polarisation to s before it leaves the housing. Measure the panel's axis with a reference polariser; if it is not across the car, lay a half-wave retarder film over the crop with its fast axis halfway between the two directions. Keep the fold and the mirror's planes of incidence in the same vertical plane as the windshield's, so s stays s. Laminate an s-pass polariser to the dust cover: it passes the HUD image unchanged and halves unpolarised sunlight.
CONSEQUENCE FOR THE DRIVER	Polarised sunglasses block s light, so they black out the HUD. Demo rule: no polarised eyewear.
BUILT IN	B9 (measure the axis, fit the retarder), B10 (dust-cover polariser).

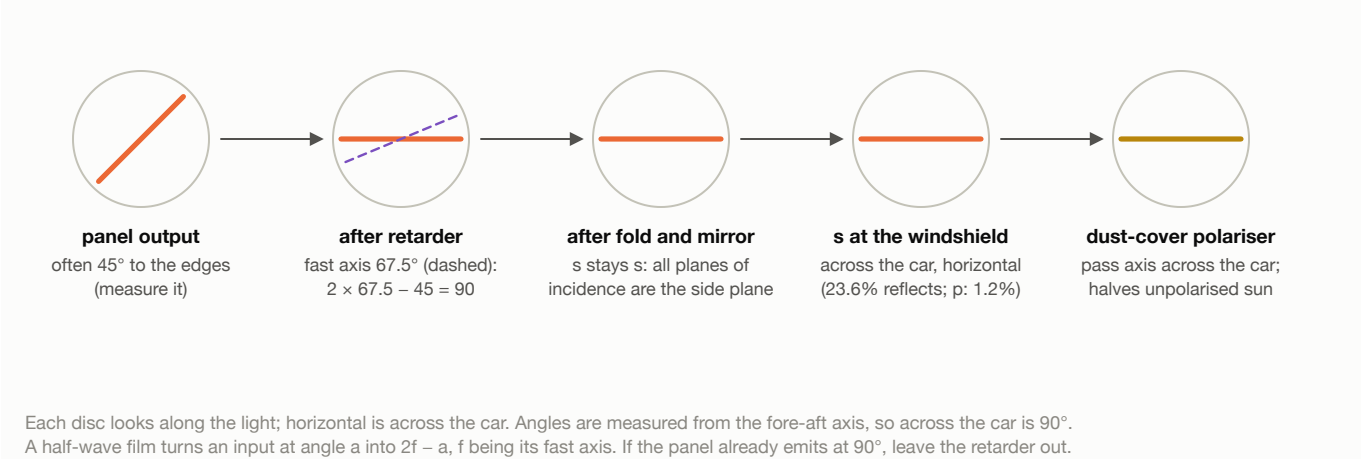


Figure 2.8. Polarisation along the path, for a panel that emits at 45° to the car's axes.

2.9 Two reflections from one windshield

THE PROBLEM	A laminated windshield has two glass surfaces, and each reflects. The driver sees two copies of the picture offset slightly: a *ghost*. Production HUD cars use a wedge-shaped interlayer to merge them.

THE IDEA	Distance helps. The two reflections leave parallel, 3.1 mm apart, so the angle between the copies falls with image distance: 5.4′ at 2 m but 1.4′ at 7.5 m, about 2.1 pixels. The design accepts it on the stock windshield and draws lines at least 6 px wide, ribbons as filled areas and heavy text, so the ghost hides inside the strokes. A wedge windshield ($600–1,500) is bought only if the B13 photograph says so.
BUILT IN	B13 (ghost photographs at three eye heights).

2.10 Bending the picture so it looks straight

THE PROBLEM	A tilted curved mirror and a curved windshield stretch the picture: the top and bottom have different widths (keystone), lines bow, and the image is not at one distance everywhere. A rectangle drawn on the panel arrives as a distorted shape.
THE IDEA	Distort the drawing in advance by exactly the opposite amount. Photograph a known grid through the HUD from the eye point, compare it with a real grid on the ground, and store the correction; the renderer applies it as a last full-screen pass. In this guide the correction is a lookup table computed from the ray trace (eye position to panel point) and refined by the photographs, because the image is not a flat plane.
BUILT IN	B8 (Stage 1 version), B13 (final calibration).

2.11 Too bright at night, too dim at noon

THE PROBLEM	The same overlay that must beat a noon sky would dazzle at night, and the liquid-crystal cell leaks about 1/1000 of its backlight even when black, which shows as a faint grey rectangle on a dark road.
THE IDEA	An ambient-light sensor at the dust cover and a dimming curve: the controller sets the backlight from the measured light level, from full power at 10,000 lux or more down to a night level several hundred times lower. The LED drivers' analogue current control covers about 20:1 without flicker; switching off one of the two LED strings and drawing white as grey on the panel cover the rest.
PARTS	TSL2591 (or VEML7700) light sensor in a pod behind the dust cover; the controller; the LED driver's dim input.
BUILT IN	B5, tuned in B13.

2.12 Living in someone's car

THE PROBLEM	The HUD must not damage the car, drain its battery, corrupt its own disk at key-off, distract the driver more than it helps, or break the law on windshield obstructions.
THE IDEA	One fused 12 V accessory feed and nothing else. An ignition-sensed converter that keeps the Jetson running for a clean shutdown after the key turns off and cuts off before the battery runs flat. Every mount clamps or sits; nothing is bonded to the glass; the housing lifts off. A written installation checklist is signed before every drive.
PARTS	Mini-Box DCDC-USB-200 converter, inline fuse and accessory plug, clamps, levelling feet, the checklist (Appendix F).
BUILT IN	B7.

2.13 Showing other people

Only the driver can see a HUD. Passengers and an audience watch a second monitor showing the rendered frame, the driver camera with the tracked eye point, and live numbers (registration error, data age, contrast) next to their targets. The Jetson's carrier has a single HDMI output, which drives the panel, so the audience view is a video stream over Ethernet to a mini PC that drives the second monitor (B7).

2.14 Summary: every part and why it is there

PART	PROBLEM IT SOLVES	BUILT IN
Jetson Orin NX (reComputer J4012)	Runs perception, ego-motion, head tracking, rendering	B2
ZED 2i stereo camera, 4 mm	Where the road and objects are (2.2); its IMU gives yaw rate (2.4)	B3, B4
OAK-D Pro	Where the eye is (2.3)	B3, B4
u-blox ZED-F9R and antenna	Speed for age compensation (2.4); time, position, heading for sun-watch (2.7)	B3, B4
5.5-inch 1080p TFT kit	The picture (2.1)	B9
Custom LED backlight and driver	Daylight brightness (2.5)	B5, B9
Heat sink, fan, thermistor	Heat (2.6)	B9
Half-wave retarder	Polarisation to s (2.8)	B9
Cold-mirror fold and absorber	Fits the path in the dash (2.1); dumps solar infrared (2.7)	B10, B12
Hot-mirror IR-cut glass, in front of the cell	Keeps solar infrared off the cell (2.7)	B10, B12
Concave mirror (custom)	The magnifier: image 7.5 m out (2.1)	B10, B12
Dust cover with s-pass polariser	Keeps dust out; halves sunlight (2.7, 2.8)	B10, B11
Shutter, servo, spring	Sun protection when off or in the sun patch (2.7)	B11, B12
HUD controller (Pico) and sensors	Dimming (2.11), thermal and shutter fail-safes (2.6, 2.7)	B5

PART	PROBLEM IT SOLVES	BUILT IN
Printed ASA housing	Holds the optics in alignment; blocks stray light; removable (2.1, 2.12)	B11
DC-DC converter, fuse, plug	Safe power, clean shutdown (2.12)	B7
Mini PC 2, monitor 2, switch	Audience view (2.13)	B7
Beamsplitter glass, Fresnel lens	Optical Stages 0 and 1: software and in-car tests before the mirror arrives	B6, B8
Luminance meter, latency probe, target board, rangefinder	Proving the targets in 1.3	B6, B9, B13

3 One frame, from photons in to photons out

Follow one frame through the finished system. The software is a set of small programs (processes), one per job, that pass messages to each other; Figure 3.1 is the plan's view of them, and the table after it lists each one.

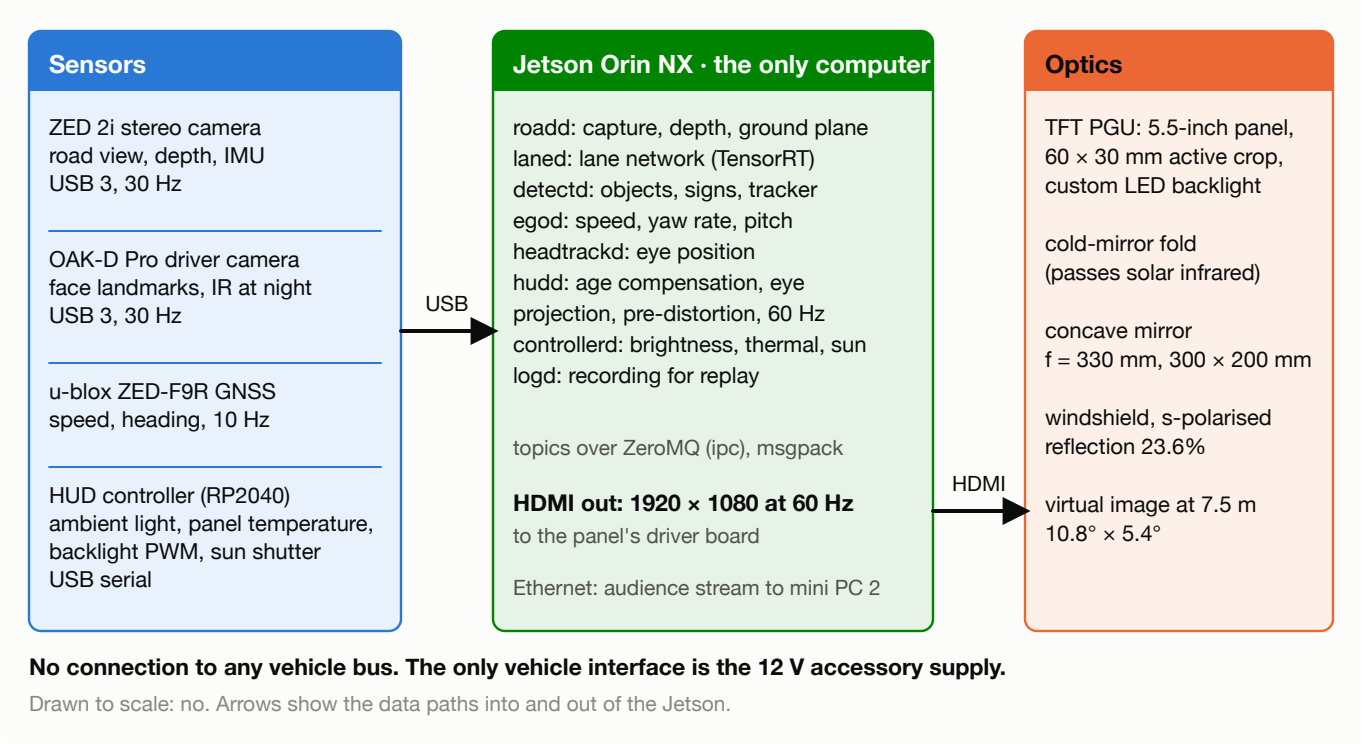


Figure 3.1. From the plan: the single-computer architecture. Sensors feed the Jetson over USB; the Jetson drives the panel over HDMI and the audience PC over Ethernet.

1. **t = 0 ms.** The ZED 2i exposes a stereo pair. `roadd` receives it 10–30 ms later, stamps it with the capture time on the Jetson's monotonic clock, computes depth, and fits the road plane.
2. **Perception.** `laned` runs the lane network on the left image and publishes up to four lines as image points with confidences; `detectd` runs YOLO11s and the tracker and publishes boxes with track identities. `roadd` lifts both into the car frame: lane points onto the fitted plane, objects at their stereo range. Each message carries the original capture time.
3. **Meanwhile, at 100 Hz,** `egod` fuses GNSS speed with the gyroscope into speed, yaw rate and pitch, and once a second computes the sun's direction in the car frame. `headtrackd` publishes the eye midpoint at 30 Hz from the OAK-D Pro. `controllerd` reports the ambient light, cell temperature and shutter state from the Pico.
4. **Draw.** `hudd`, the renderer, wakes on the panel's 60 Hz refresh. For each lane point and object it computes the age (now minus capture time), moves it by the distance travelled and angle turned over that age, finds where the line from the current eye position to it lands on the panel (lookup table, then pre-distortion), and draws the ribbon, lines, chevron, hazards and speed block into the 946 × 473 pixel crop. Anything too old fades out by rule rather than being drawn stale.

5 **Light.** The HDMI board scans the frame onto the panel (8–25 ms). The custom backlight shines through the crop. The light goes up to the fold, forward to the concave mirror, up through the dust cover and shutter opening, off the windshield, into the eye. The eye sees the window about 7.5 m ahead with the marks lying on the road.

6 **Record and show.** `logd` records every message with its timestamps; `audienced` streams the frame and the driver camera to the audience PC.

PROCESS	JOB	PUBLISHES
`roadd`	Owns the ZED 2i: capture at 30 Hz, stereo depth, RANSAC road plane, IMU stream; lifts lanes and boxes to the car frame; records SVO files on request	`lanes` , `objects` , `imu`
`laned`	Lane network (TensorRT FP16) on the left image	`lanes2d`
`detectd`	Object detector (TensorRT FP16), tracker, sign classifier	`objects2d`
`egod`	Fuses GNSS and IMU at 100 Hz; sun direction at 1 Hz	`ego` , `gnss` , `sunWatch`
`headtrackd`	Owns the OAK-D Pro: face, landmarks, depth at the eyes; eye midpoint in the car frame	`eyePose`
`hudd`	Renderer: age compensation, projection for the eye, 60 Hz draw with vsync, pre-distortion, full-screen HDMI	`hudFrame` , `hudStats`
`controllerd`	Serial link to the Pico: light, temperature, backlight duty, shutter	`health`
`logd`	Records all topics; triggers camera recordings	files
`audienced`	Composites the HUD frame, driver camera and live numbers; MJPEG stream to the audience PC	HTTP stream

Messages travel over ZeroMQ publish/subscribe sockets on the local machine, encoded with msgpack. Every message carries its capture time, its publish time and a sequence number. Empty messages are still published, so that silence (a crashed process) is distinguishable from absence (no car ahead). A supervisor restarts a crashed process after one second and marks its topics stale. Appendix D lists every field.

4 The build at a glance

The build splits into fourteen stages. Two chains run in parallel from the first week. The **software chain** (Jetson, sensors, perception, renderer, live drive) needs only a camera and a passenger seat, so it starts as soon as the camera arrives. The **optics chain** (car, measurement, ray trace, mirror order, mirror lead time, housing, assembly, daylight tests) is the critical path, because the mirror's shape depends on the car's windshield and the mirror's lead time has not been quoted. The picture generator is a third, shorter chain that must be finished before the mirror arrives.

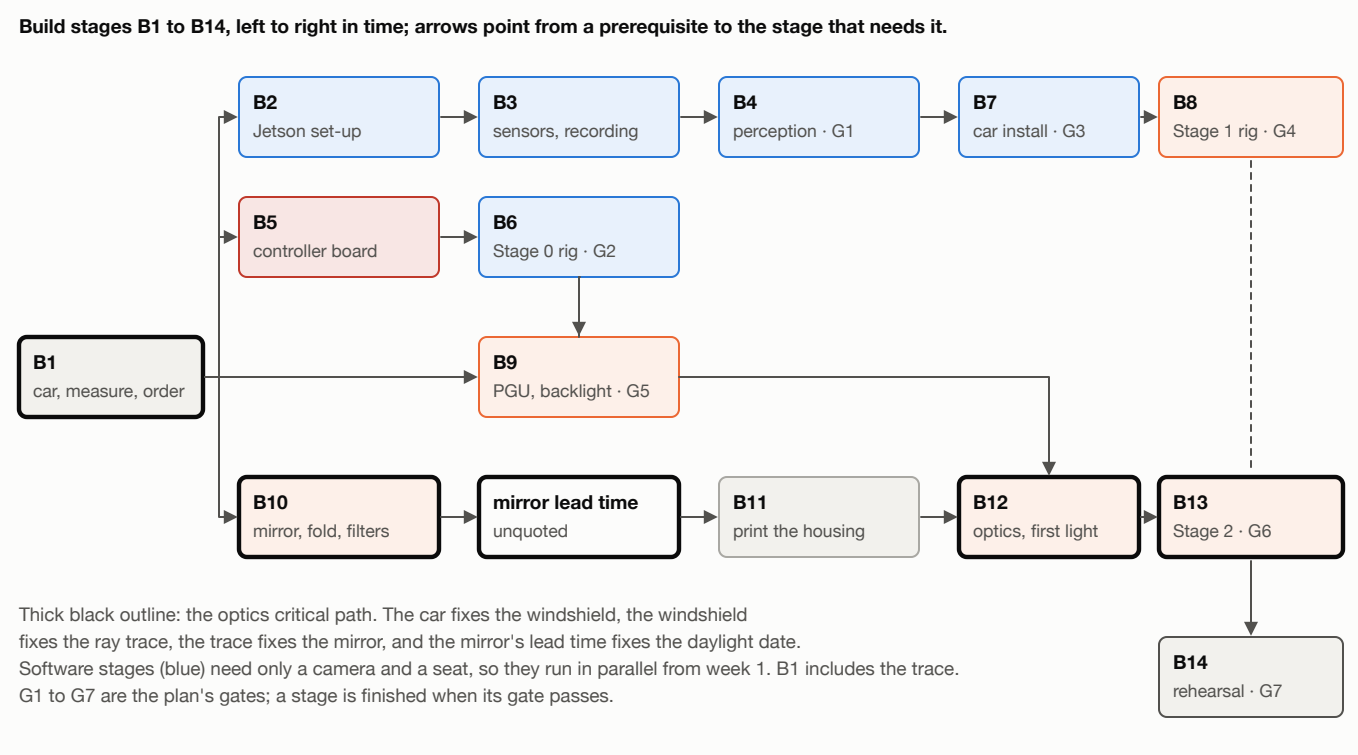


Figure 4.1. Build stages and dependencies. Gates are shown with the stage that closes them.

STAGE	TITLE	WEEKS	AT THE END YOU HAVE	GATE
B1	Choose and measure the car; ray trace; order	0–1	A named car with measured windshield, a ray trace, the first orders placed, mirror quotes requested, venue and sign-off agreed	G0
B2	Prepare the Jetson	1	JetPack 6.2, SDKs, engines, a measured GPU budget	
B3	Sensors and recordings	1–2	Cameras and GNSS mounted for recording; the course recorded by day, dusk and night; 300 frames labelled	
B4	Perception software	1–4	Lanes, objects, ego-motion, head pose, sun-watch publishing on the Jetson; perception scored	G1
B5	HUD controller and LED driver	1–3	A tested Pico board with sensors, analogue LED dimming, servo, fail-safe firmware	
B6	Stage 0 bench rig, renderer, latency	1–3	The overlay rendered at 60 fps through a beamsplitter; latency measured by photodiode	G2

STAGE	TITLE	WEEKS	AT THE END YOU HAVE	GATE
B7	Install in the car; overlay on a monitor	3–5	Everything but the optics in the car; overlay on the camera image within 0.5° for 20 min	G3
B8	Stage 1 combiner in the car	4–8	An in-car HUD at 4.2 m for night and overcast drives; the demo fallback	G4
B9	Picture generator and backlight	4–8	A measured 10,000 cd/m² PGU, thermally tested, polarisation oriented	G5
B10	Mirror, fold and filters	1–10	The custom mirror (ordered after B1's trace, then lead time), fold, IR-cut glass, dust cover, shutter leaves, all inspected	
B11	Print and assemble the housing	5–9	The 39 printed pieces, inserts, body, bay, mounts, lid, shutter	
B12	Optics into the housing; first light; sun test	8–10	An aligned HUD unit on the bench, empty-housing thermocouple test passed	
B13	Stage 2 in the car: calibrate and prove it	8–12	Every acceptance row passing at noon	G6
B14	Hazards, polish, rehearsal	8–12	The demo script run twice without touching the Jetson	G7

Weeks count from the day the car and venue are agreed, as in the plan. They are planned windows, not measured durations: the mirror's lead time, the camera and GNSS delivery times, the time to complete the ray trace, and the number of backlight build-and-measure turns are unknown until B1. Re-draw this table when they are known.

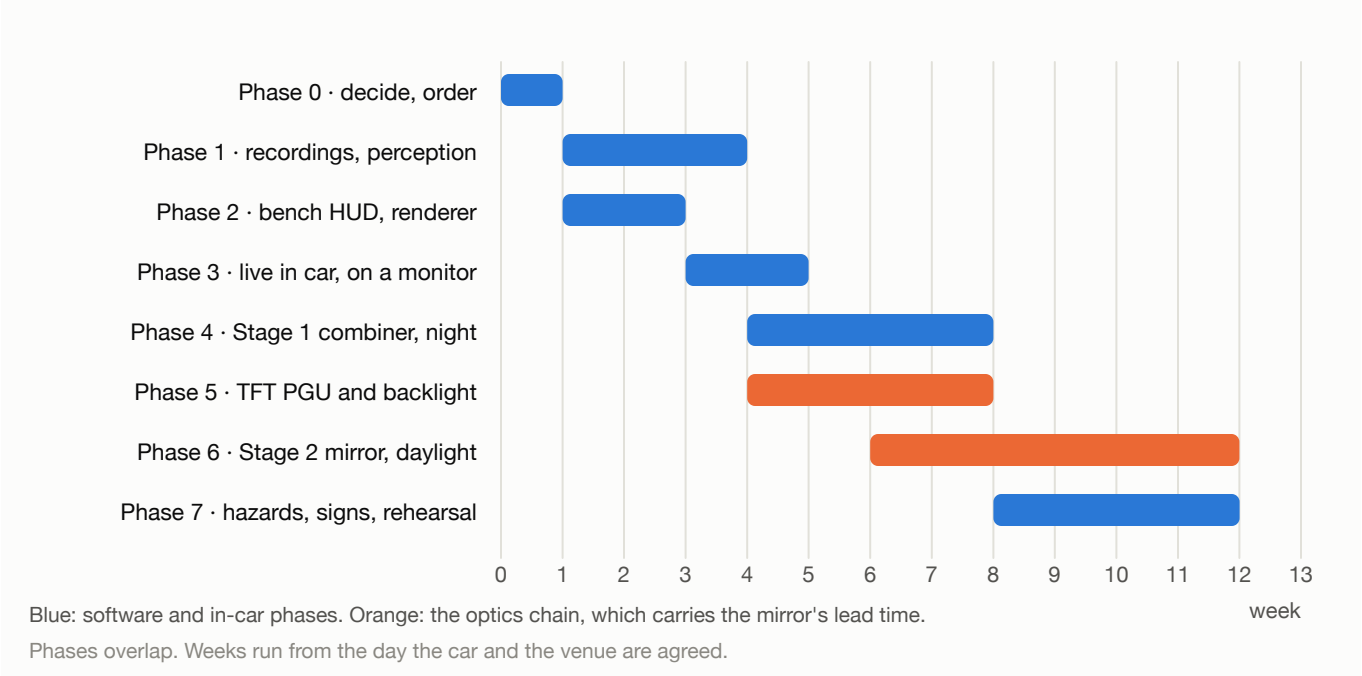


Figure 4.2. From the plan: phase windows over weeks 0–13. Blue bars are software and in-car work, orange the optics chain that carries the mirror's lead time.

PART II

Before you start

The workshop, the tools, the safety rules that apply to every stage, and every material the build consumes, with the stage that first needs it.

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5 Workspace, tools and safety
6 Materials: the complete bill

5 Workspace, tools and safety

5.1 The spaces you need

SPACE	WHAT HAPPENS THERE	REQUIREMENTS
Electronics bench	Jetson set-up, controller board, LED driver, Stage 0 rig, renderer development	ESD mat and wrist strap, bench supply, ventilation for soldering fumes, a darkened corner for the Stage 0 rig
Optics bench	Panel teardown, backlight build, luminance and polarisation measurements, mirror inspection, housing alignment	Clean, low-dust table; blackout curtain or a room that can be made dark; a 2 m clear line for the meter
Print room	Printing 39 housing pieces in ASA (about 248 printer hours in all)	Enclosed printer with a 250 mm cube build volume, extraction or a well-ventilated room (ASA emits styrene)
The car, parked	Measurements, installation, calibration	Level ground with 35 m clear ahead of the car for the target board; shade for daytime work; a socket for the bench supply
Closed course or private lot	Recordings, live drives, rehearsals	Agreed in B1 with the venue; a safety driver who is not the presenter

5.2 Tools

Every tool the procedures use, grouped by trade. The stage column says where it is first needed.

TOOL	SPECIFICATION OR USE	FIRST IN
Electronics		
Temperature-controlled soldering station	Fine conical and 2 mm chisel tips; 60/40 or lead-free solder, flux pen	B5
Heat-set insert tips	For M3, M4 and M5 inserts (the CAD uses 196 fasteners; inserts are pressed at 245 °C)	B11
Digital multimeter	Voltage, current to 10 A, resistance, continuity, diode test	B5
Bench power supply	0–30 V, 0–5 A, adjustable current limit (for first LED light-up at low current)	B5
Oscilloscope (or a logic analyser)	To confirm the LED current is free of ripple, the dimming filter's output, and the servo pulses; a cheap two-channel scope is enough	B5
Wire strippers, flush cutters, ferrule crimper, JST/Dupont crimper	0.25–1.5 mm^2 ferrules for screw terminals	B5
Heat-shrink assortment and heat gun	Every joint insulated; adhesive-lined for the 12 V lead	B5
ESD mat and wrist strap	Panel teardown and Jetson handling	B2
Mechanical		

TOOL	SPECIFICATION OR USE	FIRST IN
Hex keys 1.5–4 mm, with a 3 mm ball-end key at least 150 mm long	M3 button heads take 2 mm; the M4 mirror adjusters take 3 mm through the front-wall access holes	B11
Digital calipers, 150 mm; steel rule, 300 mm; tape, 5 m	Measurement of parts and the dash	B1
Machinist's straightedge, 300 mm (ground edge) and feeler gauges	Windshield curvature by sagitta	B1
Digital angle gauge (inclinometer), 0.1° resolution	Windshield rake, camera pitch, housing level	B1
Contour gauge, 250 mm	Dash-top profile for the housing's feet	B1
Laser rangefinder, 50 m class	Surveying the 30 m target board and the eye point	B1
30 m tape, chalk line, string, two stakes	Car centreline and target set-up	B7
Deburring tool, needle files, hobby knife, flush cutters	Print clean-up	B11
Spring clamps and a small vice	Holding parts while inserts cool and adhesives cure	B11
Optics and measurement		
Powder-free nitrile gloves, lint-free wipes, lens tissue, isopropyl alcohol, a hand air blower	Handling the cell, films and mirrors; never touch a first-surface coating	B9
Reference linear polariser (a 50 mm sheet with a marked axis)	Finding the panel's polarisation axis	B9
Spot luminance meter: Konica Minolta LS-150 (1° spot, focuses 1 m to infinity) or Sekonic L-858D if it focuses on a virtual image VERIFY	Every luminance and contrast row	B6
Type-K thermocouple, MAX31855 amplifier, IR thermometer	Cell temperature and the sun test	B5
Camera with manual focus and manual exposure (mirrorless or a phone with a manual-camera app), tripod, a 300 mm macro focusing rail or linear rail	Photographing the HUD from the eye point; eye-box sweeps; ghost photographs	B6
Latency probe (built in B6)	Motion-to-photon timing	B6
Insulated box with a 100–200 W heater and a thermostat controller	40 °C thermal test of the PGU	B9
Welding glass (shade 5 or darker) or an ND filter for viewing the bare backlight	The bare backlight is about 166,667 cd/m²	B9
Computers		
x86 Ubuntu 22.04 host (mini PC 1 works) with a USB-C data cable	Flashing the Jetson; renderer bench; nightly replay	B2
A laptop on the course	SSH to the Jetson for start-up and logs	B3

5.3 Safety rules for the whole build

THE CONCAVE MIRROR CONCENTRATES SUNLIGHT

A 300 mm concave mirror at about f/1.1 is a burning glass. Never carry it, leave it, or point it with its face toward the sun, indoors near a window or outdoors. Store it face down on a soft cloth, or with its cover on. Once installed, the shutter is closed whenever the HUD is off; never defeat the shutter's spring or wedge it open for a test in daylight. Keep hands, cables and labels out of the focal region (about 330 mm in front of the mirror) when the shutter is open outdoors.

THE BARE BACKLIGHT IS VERY BRIGHT

Without the cell over it, the backlight runs at up to about 166,667 cd/m², around sixteen times a clear sky. Do not look at it directly at full power; view it through welding glass, or bring it up at low current first. Point the luminance meter, not your eye.

CAR ELECTRICS AND DRIVING

The fuse goes within 150 mm of the power source, before anything else. Never connect anything to the OBD-II port, the CAN bus or any vehicle wiring other than the accessory socket. Recordings are made from the passenger seat or by a second person; nobody operates the laptop while driving. Every drive starts with the signed installation checklist (Appendix F). Drives happen on the agreed closed course or lot, with a safety driver who is not the presenter, until the sign-off says otherwise. No polarised sunglasses while the HUD is on.

WORKSHOP HAZARDS

Heat-set insert tips and soldering irons are at 245–350 °C; use a stand and keep the cable clear. ASA prints emit styrene; run the printer enclosed with extraction or in a ventilated room, and do not sleep in the same room as a running ASA print. Wear eye protection when cutting the dust-cover glass, cleaving films, or crimping. The laser rangefinder is a visible laser; never aim it at people or cars. Lithium and lead-acid sources can deliver hundreds of amps into a short: insulate every joint before connecting to the car.

STATIC AND THE LCD CELL

The panel's driver chips sit on the glass ledge and its flexible circuit is fragile. Work on the ESD mat with the wrist strap, never bend the flex near the glass, and never press on the polariser surface; a fingerprint on a polariser shows in the image at 23× magnification.

5.4 Records you keep

Keep a bound build log (or a text file in the repository) with the date, the stage, every measurement with its instrument, every decision with its reason, and every gate result. The appendices include blank sheets for the gates (Appendix E). Commit the calibration file, the dimming curve, the ray-trace inputs and outputs, and the CAD's regenerated files to the repository after each stage, so the build can be reproduced or repaired.

6 Materials: the complete bill

This chapter lists everything the build consumes, grouped by where it is used, with the build stage that first needs it. Prices are the plan's (last seen on 2026-10-21, not quotes); detailed items carry the plan cost line they fall under rather than a new price. The printed parts and fasteners are listed in full in B11, generated from the CAD.

6.1 The plan's cost lines

ITEM (PLAN LIST)	LAST SEEN, USD	BASIS
Stereolabs ZED 2i, 4 mm lens	$499	firm
Luxonis OAK-D Pro (driver camera)	$349–429	range
u-blox ZED-F9R board (SparkFun) + antenna	$300–380	range
Panels: 5.5-inch 1080p kits ×2, 5-inch 1,100-nit module	$190–265	range
Backlight: LEDs, driver, films, heat sink, fan	$150–300	estimate
Half-wave retarder and polariser sheets	$150–450	range
Cold mirror (fold) and IR-cut glass	$150–400	estimate
Concave mirror, 300 × 200 mm (quote)	$1,000–4,000	estimate
Teleprompter beamsplitters ×2	$110–1,030	range
Fresnel lenses 300 mm ×2	$160–320	range
HUD controller: RP2040, sensors, shutter servo	$40–80	range
DC-DC converter with ignition sense	$89	firm
USB-C PD charger, Ethernet switch, cables	$90–120	range
Housing: printing, dust cover, mounts	$150–300	estimate
Luminance meter (spot meter to lab meter)	$690–4,000	range
Latency probe, rangefinder, target board	$300	estimate
Total of the listed ranges	**$4,417–12,962**	

Optional, not in the total: a wedge-PVB windshield, installed ($600–1,500), only if the B13 ghost photographs call for it; an automotive HUD-grade TFT module ($200–800) if one can be bought. Already owned and not priced: the Jetson (reComputer J4012), two YODOIT 16-inch 1,200-nit monitors, two ACEMAGIC mini PCs.

6.2 Itemised list

ITEM	QTY	SPECIFICATION	PLAN COST LINE	FIRST
Compute and sensing				
Seeed reComputer J4012 (Orin NX 16 GB)	1	Owned. JetPack 6.2	owned	B2
NVMe SSD, M.2 2280	1	1 TB or more, if the fitted drive is smaller	cables and small parts	B2

ITEM	QTY	SPECIFICATION	PLAN COST LINE	FIRST
Stereolabs ZED 2i, 4 mm lenses, and its locking USB-C cable (1.5 or 3 m)	1	Stereo road camera, USB 3, 400 Hz IMU; the cable is sold separately ($20–30)	$499 + cable	B3
Luxonis OAK-D Pro and its USB-C Y-adapter	1	Driver camera, IR flood and dot projector, on-device inference; external power through the Y-adapter	$349–429	B3
u-blox ZED-F9R breakout (SparkFun) and active multi-band antenna	1	Dead-reckoning GNSS; a NEO-M9N ($71) is the budget fallback without dead reckoning	$300–380	B3
Mirror-stalk camera clamp, ball head, footwell base plate	1 each	Section B3.1	housing and mounts	B3
Picture generator				
Wisecoco 5.5-inch 1920 × 1080 IPS HDMI kit	2	One taken apart, one stock	panels	B6
Newhaven 5-inch 1,100-nit HDMI module (800 × 480)	1	Stage 1 at night until the backlight exists	panels	B8
High-power white LEDs, 5,000–5,700 K	12	Cree XP-G4 or Lumileds HL2X class on 20 mm stars, or one custom aluminium board of 8 (B5.2)	backlight	B9
RECOM RCD-24-0.50 constant-current LED driver	2	One per string; analogue dimming (B5.2)	backlight	B5
Dual rail-to-rail op-amp, NPN transistors, passives	1 set	Dimming and enable circuits (B5.1)	controller	B5
Heat sink, 40 mm fan, thermal pad, nylon washers	1 set	B9.1	backlight	B9
Opal acrylic 2 mm, white reflector film, clear 1–2 mm glass	1 each	Cavity, diffuser, cover glass (B9.1)	backlight	B9
Prism films	2	Salvaged from the stock backlight, or 3M BEF-class	backlight	B9
Half-wave retarder film	1 sheet	Only if the axis needs it; plan: Edmund WP280, American Polarizers, Knight Optical	retarder and polariser	B9
Optics				
Custom concave mirror	1	From the B1 trace (B10.1)	$1,000–4,000 (quote)	B10
Cold-mirror fold glass and absorber plate	1 each	B10.1	cold mirror and IR-cut	B10
Hot-mirror IR-cut plate	1	B10.1	cold mirror and IR-cut	B10
Dust-cover glass, polariser sheet, optically clear adhesive	1 each	B10.1	retarder and polariser; housing	B10
Teleprompter beamsplitter glass, 70/30	2	Stage 0 and the Stage 1 combiner	$110–1,030	B6
Fresnel lens 300 mm, f ≈ 300 mm	2	Stage 1	$160–320	B8
Controller, power and wiring (details in B5 and B7)				
Raspberry Pi Pico (RP2040)	2	Controller and latency probe	controller	B5

ITEM	QTY	SPECIFICATION	PLAN COST LINE	FIRST
TSL2591 light-sensor breakout (VEML7700 alternative)	1	I²C	controller	B5
MAX31855K thermocouple amplifier and type-K bead thermocouples	1 + 3	Sun test, thermal test	controller	B5
10 kΩ NTC thermistors (B ≈ 3950), 10 kΩ 0.1% resistors, 100 nF capacitors	3 each	Cell temperature	controller	B5
SG90 micro servo	2	Shutter, plus a spare	controller	B5
5 V buck converter, 3 A, wide input; 470 µF capacitor	1	Servo and panel-board supply	controller	B5
Perfboard, headers, JST-XH connectors, screw terminals	1 set	Controller carrier	controller	B5
Mini-Box DCDC-USB-200	1	Ignition-sensed converter	$89	B7
12 V accessory plug, inline blade-fuse holder, fuses	3 sets	Feed 1 (always-live), ignition sense, feed 2 (B7.2)	cables and small parts	B7
Wire: 1.0–1.5 mm² (16–18 AWG) red and black; 0.25 mm² signal; shielded 4-core	5 m each	Power and sensor leads	cables and small parts	B5
USB-C PD car charger (100 W), gigabit switch, Ethernet cables, HDMI cable	1 each	Audience link and panel	$90–120	B7
Housing (full list in B11)				
ASA filament, matte black	5 kg	32 printed parts	housing	B11
TPU 95A filament	small spool	Grommets	housing	B11
Heat-set inserts, screws, springs, feet	per B11	196 fasteners in all	housing	B11
Shutter leaves (laser cut, anodised)	2	From the CAD's DXF	housing	B11
Neutral-cure RTV, VHB tape, foam gasket, flock paper	1 each	Bonding and sealing	housing	B11
Instruments and test fixtures				
Spot luminance meter	1	Konica Minolta LS-150 assumed; Sekonic L-858D if it focuses on a virtual image	$690–4,000	B6
Latency probe parts, laser rangefinder, 30 m target board	1 each	B6.1, B13.1	$300	B6
Angle gauge, straightedge, feeler gauges, contour gauge, calipers, rail, reference polariser, welding glass, test box	1 each	Chapter 5	workshop	B1

6.3 When to order

WHEN	ORDER	WAITS ON
Week 0	ZED 2i, OAK-D Pro, GNSS board and antenna, both panel kits, controller parts, DC-DC converter, cheap beamsplitter, Fresnel lenses, retarder and polariser films, LEDs and drivers, ASA filament and inserts	Nothing
Week 0–1	Mirror quotes (B1.7)	The B1 trace for the request; the order waits on acceptance
Week 1–2	Luminance meter: try the spot meter on the Stage 0 image first	The Stage 0 rig
After the trace	Concave mirror; cold mirror, hot mirror and dust-cover glass in the trace's sizes; shutter leaves after the CAD is regenerated	B1, B11.1
After gate G5	Better combiner glass if Stage 1 will be shown; collimation parts if brightness falls short	B9
After the ghost photographs	Wedge-PVB windshield, if needed	B13.6

PART III

The build, stage by stage

Fourteen build stages, each with what it solves, what it needs, a numbered procedure with checks, and the test that closes it. Software stages B2 to B7 and the hardware stages B5, B9 to B11 overlap in time; the order of chapters is the order in which each stage starts.

B1 Car, measurements, ray trace, orders · B2 Jetson · B3 Sensors and recordings · B4 Perception · B5 Controller and LED driver · B6 Stage 0 rig, renderer, latency · B7 Car installation · B8 Stage 1 combiner · B9 Picture generator · B10 Mirror, fold, filters · B11 Housing · B12 Optics and first light · B13 Stage 2 calibration and proof · B14 Rehearsal

B1 Choose and measure the car, run the ray trace, order

SOLVES	WEEKS	CLOSES
Fixes the geometry every optical part depends on (2.1, 2.7, 2.9)	0–1; the trace may take longer	Gate G0: car named, orders placed, venue agreed, checklist exists
NEEDS	**PRODUCES**	**PEOPLE**
Access to the candidate car; the project repository; the tool venvs	A measurement record, ray-trace outputs, the mirror request for quote, the first purchase orders	Builder, the car's owner, the regular driver, whoever signs off installation

Why this comes first. The windshield is an optical element of the HUD: its angle sets where the mirror must sit and where the sun can enter, its curvature changes the image distance and shape, and its thickness sets the ghost. The mirror's shape therefore cannot be specified until the car is chosen and measured, and the mirror has the longest lead time in the project. Every week spent here before ordering the mirror is a week on the critical path, so do the measurements in one session and run the trace immediately.

B1.1 Materials and tools

ITEM	USE
Digital angle gauge, 0.1°	Rake, floor level
300 mm ground straightedge, feeler gauges 0.05–1 mm	Windshield curvature by sagitta
Laser pointer (class 2), white card, 150 mm rule	Laminate thickness by the double-spot method; finding the HUD patch
Tripod with a ball head; a small bubble level	Holding the laser at the eye point
Contour gauge, tape measure, plumb bob, painter's tape, marker	Dash profile, eye point, datum marks
Laser rangefinder	Eye-to-glass distance; car dimensions
Camera	Photographs of the dash, glass labels, sensor positions
Optional: a handheld 3-D scanner or a phone LiDAR / photogrammetry app	A scan of the dash and the inner glass for the CAD (`docs/arhud-blender/import_scan.py` cleans and fits it)

B1.2 Choose the car

Use this table to compare candidates. The HUD can be built for any car that passes every row; the numbers in this guide assume the plan's 28° rake, 750 mm eye-to-glass distance and a left-hand-drive car.

CRITERION	REQUIREMENT AND REASON
Windshield rake	25–35° above horizontal. Flatter glass moves the mirror forward and lowers the sun-entry elevation (45° at 25°); more upright glass raises it (65° at 35°) and shortens the path.

CRITERION	REQUIREMENT AND REASON
Glass at the HUD patch and the camera	Clear: no dark frit band, no shade band, no heated wire grid, no rain-sensor housing at the patch about 750 mm ahead of the eye on a 5° downward line. A factory-HUD (wedge) windshield is acceptable and reduces the ghost; the trace must then use its wedge angle.
Dash top in front of the driver	A region about 460 mm fore-aft and 370 mm wide under the glass, which the housing can sit on through four levelling feet, with at least 8 mm between the housing's front top edge and the glass. The housing stands about 300 mm tall from the dash at the rear.
Instrument cluster hood	Its top must be below the housing's rear step: the CAD steps up at the rear to ride 5 mm over the simulator's hood. Measure the real hood and expect to regenerate the CAD.
Defroster vents and dash sensors	Note their positions. The housing must not cover the sun-load or light sensor the car uses for automatic headlamps and climate, and must leave enough defroster outlet open to clear the glass in front of the driver; if it cannot, the car fails this row.
Rear-view mirror stalk	Reachable for a clamp at the top of the glass on the centreline, roughly 450 mm from the driver's eye, with room for a camera 175 mm wide.
Accessory sockets	At least one switched 12 V socket with a known fuse rating (often 15 or 20 A); two sockets make the power budget comfortable (B7.4).
Steering-column shroud	A flat spot for the driver camera with a clear view of the driver's face through the wheel.
Permission	The owner agrees in writing to a non-permanent installation: nothing drilled, bonded or wired into the car beyond the accessory socket.

B1.3 Fix the driver's eye point

1. Park the car on level ground. Put the angle gauge on the floor beside the car (or on the door sill) and record the reading as the *ground reference*; every later angle is measured relative to it.
2. Seat the person who will drive the demos in their normal driving position: seat, backrest, head restraint and steering wheel as they will be on the day. Record all four settings (seat-track notch, backrest angle, wheel reach and height) in the log; the calibration is only valid for them.
3. Have the driver look straight ahead at a distant point. Hang the plumb bob from a point midway between their eyes (a second person holds it at the bridge of the nose, about 10 mm in front) and mark where it touches the floor mat with tape. That is the eye point's plan position.
4. Measure the height of the eye midpoint above the ground with the tape (or the rangefinder from the floor plus the floor's height). The plan assumes 1.22 m. Record it.
5. Measure the eye point's position relative to three fixed datums that do not move with the seat: the centre of the steering-wheel hub, the base of the mirror stalk, and the driver's B-pillar seat-belt anchor. These let you find the eye point again without the driver.
6. Ask the driver to sway left, right, up and down as they would in normal driving; note the range. The design eye box is 130 mm wide and 70 mm tall around the nominal point.

B1.4 Find and measure the HUD patch on the windshield

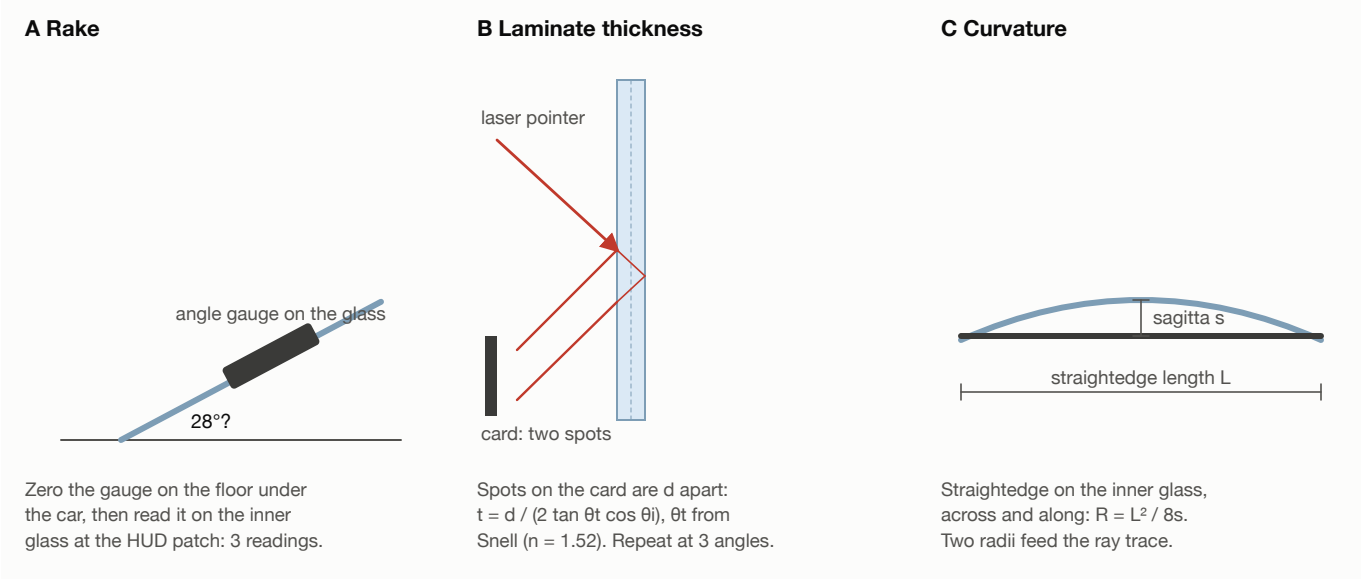


Figure B1.1. The three windshield measurements. All are made on the inner surface at the HUD patch.

1 **Find the patch.** Put the laser pointer on the tripod head at the nominal eye point, aimed straight ahead (parallel to the car's centreline) and 5° below horizontal, set with the angle gauge against the ground reference. Where the spot meets the glass is the centre of the HUD patch. Mark it on the outside of the glass with a small piece of painter's tape (never write on the inner surface). Measure the distance from the eye point to the spot along the beam with the rangefinder or tape; the plan assumes 750 mm.

2 **Rake.** Hold the angle gauge flat on the inner glass at the patch, its long edge in the car's vertical centre plane through the patch (pointing up the glass). Read the angle; subtract the ground reference. Repeat three times, lifting and re-seating the gauge each time, and average. Repeat 100 mm left and right of the patch to see how fast the rake changes.

- ☐ Rake recorded at three points; spread under 0.5°

3 **Laminate thickness.** Look for the glass label in a lower corner (maker, model number, DOT code); a maker's catalogue may list the thickness. Then measure it: in a darkened car, shine the laser at the patch from a known angle θ_i (measure with the angle gauge, 50–70° from the glass normal) and hold the white card where the reflection lands, about 300 mm away. You will see two spots, one from each outer surface (a faint third from the interlayer may appear). Measure their separation d perpendicular to the beam. The thickness is $t = d / (2 \tan \theta_t \cos \theta_i)$, with θ_t from Snell's law, $\sin \theta_t = \sin \theta_i / 1.52$. At $\theta_i = 65°$ and the assumed 4.96 mm, d = 3.1 mm. Average three angles.

- ☐ Thickness recorded, with the label's part number

4 **Curvature.** Lay the straightedge on the inner glass across the patch, horizontal (across the car). The glass curves away from it toward the middle or the ends. Slide feeler gauges under the straightedge at its centre to find the gap s (the sagitta) while both ends touch. The radius is $R = L^2 / (8\ s)$: with L = 300 mm, a 4 m radius gives s = 2.8 mm and an 8 m radius 1.4 mm. Repeat with the straightedge running up the glass along the rake. These are R_h (lateral) and R_v (vertical). Repeat both 100 mm above and below the patch.

- ☐ R_h and R_v recorded at three heights

5 **Cowl and roof.** Measure the height of the windshield's bottom edge (the cowl) and top edge above the ground, and their fore-aft distance from the eye point. The simulator uses a cowl at 0.80 m and the roof edge at 1.385 m.

WHY THE CURVATURE MATTERS SO MUCH

The Cycles cross-check traced the plan's toroid mirror against a windshield curved with 4 m lateral and 8 m vertical radii: the virtual image turned into a real image about 12 m ahead and the window grew to about 10° × 7.5°. A freeform mirror refitted for that glass recovered a flat image at 7.4 to 7.6 m. A real windshield is curved; a mirror designed for flat glass will not work in it. The two radii you measure here are inputs to the mirror's design, not details.

B1.5 Map the dash

1 Mark the housing's footprint with painter's tape: centred on the driver's centreline (the simulator puts the car's centreline 370 mm to the driver's right), its front edge where a vertical from the HUD patch's lower region meets the dash, about 460 mm long and 370 mm wide.

2 Take contour-gauge profiles of the dash top along three fore-aft lines inside the footprint (left edge, centre, right edge), and one lateral profile at the front and one at the rear. Trace each onto paper with the datum marked, or photograph it on a grid.

3 Measure the instrument cluster hood: its top height, its front edge's position, and its width, relative to the eye point.

4 Measure the vertical gap from the dash to the glass at the front of the footprint and at 100 mm intervals backward. The CAD keeps the housing 8 mm under the glass.

5 Photograph the dash from above and from the driver's seat with a rule in view; photograph the defroster vents, sensors and sockets.

6 Optional: scan the dash, cluster hood and the inner glass. `import_scan.py` (in `docs/arhud-blender`) joins, cleans and fits a scan to the simulator's frame from paired landmark points; it reached about 1.4 mm on the simulated test scan.

B1.6 Run the ray trace

The trace takes the measurements and returns the mirror's shape, the positions of the panel, fold and mirror, the image distance across the window, the window's size, the eye box, the ghost, and the sun-entry patch. The project already has the tools; this procedure feeds them the real car. Allow one to three days; record every input and output in the log.

1 **Enter the measurements.** In `docs/arhud-plan-tft-src/calc.py`, replace the assumed values with the measured ones: `RAKE`, `LOOKDOWN` (keep 5° unless the patch had to move), `L_EYE_GLASS`, `T_GLASS`, and `H_CAM` (the eye height). Run `python3 sim_params.py` to regenerate the simulator's parameters and `python3 spec.py` for the specification.

2 **Enter the glass curvature** in `docs/arhud-sim/js/geom.js`: `GLASS = { Rh: …, Rv: … }` in metres. Check that `GLASS_TO_MIRROR` (0.31 m) still clears the glass: the simulator writes the mirror's clearance into its notes (Vehicle mode).

3 **Fit the mirror.** In `docs/arhud-blender`: `node dump_layout.mjs`, then `python3 freeform_mirror.py glassRh=<Rh> glassRv=<Rv> name=freeform_car`. It fits a nine-term polynomial mirror with the panel's tilt and shift free, against a 7.5 m target. Read the fitted image-distance range across the window and the window size from its output.

4 **Cross-check in Cycles** (Blender, headless): `Blender -b -P optics_check.py` with the same glass radii and mirror. It measures the image distance dot by dot by stereo parallax, the ghost, the eye box, and the field independently of the fit. Accept the design only if the two agree.

5 **Sun.** `python3 sun_focus.py` reports the solar spot on the panel (size, irradiance) and the patch of sky whose sun reaches the crop and the whole cell. Record the cell patch: it is the sun-watch rule's input.

6 **Export for the quote.** `Blender -b -P mirror_export.py -- name=freeform_car` writes the sag table (CSV) and an STL of the mirror surface into `export/`.

7 **Regenerate the housing.** In `docs/arhud-cad`: `node dump_layout.mjs`, then `~/.venvs/arhud-cad/bin/python make.py`. The beam check must report 0 blocked rays; the envelope check must clear the measured glass, hood and dash. Edit `hudcad/geo.py` for the real dash profile and hood.

TRACE OUTPUT TO RECORD	USED IN
Mirror figure (sag table), clear aperture, vertex position and orientation	B10 request for quote
Panel position, tilt and shift; fold position and angle; fold size	B10 (fold glass size), B11 (CAD)
Image distance range across the window; window size in degrees	B13 acceptance; renderer set-up
Eye-box extent without vignetting	B13
Ghost separation at the eye	B13 ghost photographs
Sun-entry patch for the crop and for the whole cell; solar spot size and irradiance	B4 sun-watch rule; B12 sun test
Lookup table from eye position to panel point (exported from the simulator's `lut.js` via `dev/export-lut.mjs`)	B13 renderer projection

B1.7 Request mirror quotes

Send the request the day the trace is accepted. A quote does not commit the order; ask for lead time explicitly, because it sets the daylight demo date. Shops that list custom concave optics to 500 mm include Avantier and Shanghai Optics (plan, section 8.2). Include:

- Clear aperture and outline (the plan's 300 × 200 mm, or what the trace gives), substrate thickness (the CAD's strongback assumes about 12 mm), edge finish.
- The surface: the sag table and STL from `mirror_export.py`, with its coordinate frame. If the shop can make only toroids, give R_t and R_s from the trace (for the flat-glass layout: 685 and 636 mm). State the speed: about f/1.1, much faster than catalogue parts; ask whether they will figure it.
- Figure tolerance in the form the image needs. A local slope error α deflects the reflected ray by 2α, which the eye sees directly as an angular error. Keeping that under 0.05° (a sixth of the head-motion budget) asks for local slope errors under about 1.5 arcminutes; ask what the shop can hold and how they measure it.

- Coating: first-surface protected aluminium (or enhanced aluminium), reflectance ≥ 90% across 450–650 nm (the plan's assumed 0.90), and its reflectance in the near infrared if offered.
- Ask the shop to report the solar spot at the panel from their own model of the surface (the plan's request): a figure change can spread the spot several-fold at no cost to the image.
- Quantity 1, plus a price for a second as a spare; lead time; packing for a first-surface mirror.

B1.8 Place the first orders, book the venue, write the checklist

1 Order everything in Chapter 6 marked *week 0*: it does not depend on the car.

2 Agree the venue and dates for recordings (week 1–2), live drives (week 3 onward) and rehearsals. Confirm permission for a lead car and a volunteer pedestrian if the hazard demo needs them.

3 Confirm who signs off the installation (the plan names the university's risk office for anything beyond a closed course) and give them Appendix F as the draft checklist. Agree the safety driver.

B1.9 Gate G0

- ☐ The car is named and its measurements are in the log: eye point, rake, thickness, two radii, cowl and roof, dash profiles, hood.
- ☐ The ray trace and the Cycles check agree, and their outputs are recorded.
- ☐ Mirror quotes are requested with the surface, tolerance and lead-time questions above.
- ☐ Week-0 orders are placed.
- ☐ The venue is agreed, and the installation checklist exists and has been seen by whoever signs it off.

B1.10 If it goes wrong

SYMPTOM	RESPONSE
Rake outside 25–35°	The design can still work, but the mirror moves and the sun patch changes. Run the trace and check the mirror clears the glass and the housing fits the dash before committing.
Double-spot measurement shows three or more spots, or spots not parallel	The glass may have a wedge interlayer (a factory HUD windshield). Measure the angle between the two reflected beams over a longer throw (2 m) and give the wedge angle to the trace.
The housing does not fit under the glass	Shorten the path: a shorter-focal-length mirror with a smaller crop (plan, section 6.6 table: a 40 mm crop needs f ≈ 217 mm), at the cost of fewer pixels per degree. Re-run the trace.
Fit and Cycles disagree by more than about 0.5 m in image distance	Do not order. Check the layout file is current (re-run `dump_layout.mjs`), that both used the same glass radii, and that the mirror mode is the same in both.
No shop will figure an f/1.1 freeform	Ask for a toroid; accept the curved image surface and rely on the lookup-table projection (B13), or fall back to the Stage 1 demo (B8) while a better mirror is sourced.

B2 Prepare the Jetson

SOLVES	WEEKS	CLOSES
The one computer every software stage runs on (Figure 2.1, problems 1–4)	1	No gate; its GPU measurement feeds G1

NEEDS	PRODUCES	PEOPLE
The Jetson (owned), an Ubuntu 22.04 x86 host, an NVMe drive	JetPack 6.2 on NVMe, the ZED SDK and DepthAI installed, TensorRT engines, a Python environment, remote access	Software builder

What this stage solves, intuitively. The Jetson was last seen running JetPack 5.1.3, whose TensorRT 8.5 cannot run the networks the plan chose and for which the current ZED SDK is not built. This stage moves it to JetPack 6.2 (Ubuntu 22.04, Python 3.10, TensorRT 10), installs the camera kits, and measures what the GPU can actually do, because the whole perception design rests on that number.

B2.1 Materials

ITEM	SPECIFICATION
Seeed reComputer J4012	Owned: Orin NX 16 GB on the J401 carrier; one HDMI 2.1, four USB 3.2, one gigabit Ethernet port, an M.2 slot for an NVMe drive, 9–19 V input
NVMe SSD, M.2 2280	1 TB or more (recordings of the course at 1080p, logs); if the unit's drive is smaller, replace it now
Host PC	A physical x86 machine running Ubuntu 20.04 or 22.04 (Seeed: no virtual machine, no ARM host) with 50 GB free; mini PC 1 works
USB-C data cable	For recovery-mode flashing (a charge-only cable will not enumerate)
Bench supply or the unit's adaptor	9–19 V; the kit's adaptor is listed as 12 V 5 A with a 5.5 × 2.5 mm plug
HDMI monitor, keyboard, mouse, Ethernet	For first boot (monitor 1 works)

B2.2 Check the board before touching it

1. Try `ssh jetson` (or `jetson-desktop.local`). If it answers, record `cat /etc/nv_tegra_release`, `df -h` and `sudo nvpmodel -q`, and copy anything worth keeping in the home directory to the host. The plan notes the board was unreachable on 2026-10-21.
2. Open the case (screws on the underside), check the module is seated, the fan's connector is on its header, and note the NVMe drive's size. Fit the new drive if needed: M.2 key M, one screw.

B2.3 Flash JetPack 6.2

1. This unit is the J4012 *Classic* ("Super Mode NOT Supported" on Seeed's product page). Instructions for the "reComputer Super J4012" (a reset switch, `mfi_recomputer-super-…` images) do not apply.
2. On the host: `sudo apt install qemu-user-static sshpass abootimg nfs-kernel-server libxml2-utils binutils -y`.

3 Download Seeed's JetPack 6.2 (L4T 36.4.3) image for the Orin NX 16 GB from the Seeed wiki page "reComputer J4012 Flash JetPack" and check it with `sha256sum` against the wiki's value. Extract it with `sudo tar xpf mfi_….tar.gz` where there are 50 GB free.

4 **Force recovery.** With the power off, fit a jumper between FC REC (pin 3) and GND (pin 2) on the J401's 12-pin button header (pins: 1 PWR BTN, 2 GND, 3 FC REC, 4 GND, 5 SYS RET, 6 GND, 7 AUTO ON, 8 DIS, 9 UART TXD, 10 UART RXD, 11 LED+, 12 LED–). Connect the USB-C port to the host with a data cable, then apply DC power.

5 On the host, `lsusb` must show `0955:7323 NVidia Corp` (the Orin NX 16 GB in recovery). If it does not, the board is not in recovery or the cable is charge-only.

6 Flash from inside the extracted folder: `sudo ./tools/kernel_flash/l4t_initrd_flash.sh --flash-only --massflash 1 --network usb0 --showlogs`. It writes the NVMe drive in 2–10 minutes; do not interrupt it.

7 Power off, remove the recovery jumper, power on, and complete the first-boot set-up on the monitor: user name, host name (for example `hud-jetson`), time zone, network.

B2.4 Configure

```
cat /etc/nv_tegra_release          # record the L4T release; the ZED SDK installer must match it
sudo apt update && sudo apt full-upgrade
sudo apt install nvidia-jetpack        # CUDA, cuDNN, TensorRT, if the image does not already have them
sudo nvpmodel -q --verbose             # list the modes; on the standard configuration 0 = MAXN, 1 = 10 W, 2 = 15 W, 3 = 25 W
sudo nvpmodel -m 0                  # MAXN (confirm the id from the list first)
sudo pip3 install -U jetson-stats      # jtop; reboot afterwards
sudo apt install python3-venv git build-essential cmake
```

Run at MAXN, never MAXN SUPER: Seeed warns that the J401 carrier's cooling is insufficient for it and that enabling it may permanently damage the unit. MAXN SUPER only exists in NVIDIA's "super" flash configurations, so if `nvpmodel -q --verbose` lists it, the wrong image was flashed. Use `jetson_clocks` only for benchmarks.

1 **Remote access.** Install an SSH key; install Tailscale if the course has internet; give the Ethernet port a static address for the audience link (for example 192.168.50.2/24; mini PC 2 will be .3).

2 **Python.** Create one virtual environment with access to the system's TensorRT bindings: `python3 -m venv --system-site-packages ~/hud-venv`. Install `pyzmq msgpack numpy pyubx2 pvlib pyyaml`. Install PyTorch only from NVIDIA's wheels for JetPack 6 if a tool needs it (Ultralytics' export does).

3 **ZED SDK.** Download the installer whose L4T release matches `/etc/nv_tegra_release` exactly: for Seeed's L4T 36.4.3 image, `ZED_SDK_Tegra_L4T36.4_v5.5.1.zstd.run` (the installer aborts on any mismatch). `chmod +x` it and run it; accept the Python API installation into the venv. Check the camera later with `ZED_Explorer`.

4 **DepthAI.** `pip install depthai` in the venv (version 3.x on 2026-10-25, with aarch64 wheels for Python 3.9 and later), and add the udev rule Luxonis documents so the OAK-D enumerates without root.

5 **Renderer.** Install raylib's Python bindings (`pip install raylib`); if no aarch64 wheel is available, build raylib from source for the desktop GL platform and use the bindings' source build VERIFY.

6 **Repository.** Clone the project; create the `hud/` package skeleton (B4.1).

B2.5 Build the engines and measure

```
yolo export model=yolo11s.pt format=engine half=True imgsz=640        # on the Jetson: engines are device-specific
trtexec --onnx=ufldv2.onnx --fp16 --saveEngine=ufldv2.engine          # lane network, from the repo's ONNX export
trtexec --loadEngine=ufldv2.engine --iterations=500                   # per-inference time
tegrastats --interval 1000 --logfile b2_baseline.log                  # idle baseline
```

Record each engine's per-inference time. The full GPU budget with depth running is measured in B4.8 once the camera is connected.

B2.6 If it goes wrong

SYMPTOM	RESPONSE
Host does not see the board	Not in recovery mode, charge-only cable, or a USB hub in between; connect directly to the host.
Flash fails part way	Run it again from recovery mode; check free space on the host; use the exact image for the 16 GB module.
ZED installer refuses	Its L4T version does not match `/etc/nv_tegra_release`; download the matching one.
Thermal throttling in `jtop`	Check the fan runs; you may be in a super mode; give the unit air in the car (B7).

B3 Sensors and recordings

SOLVES	WEEKS	CLOSES
Gives perception real data from the real course (2.2, 2.3, 2.4) before any optics exist	1–2 (recordings), labelling through week 3	No gate; its labelled frames are G1's test set
NEEDS	**PRODUCES**	**PEOPLE**
B2; the ZED 2i, OAK-D Pro, GNSS receiver and antenna; the car and the course	Sensors mounted where they will stay, a configured GNSS receiver, recordings by day, dusk and night, 300 labelled frames	Builder in the passenger seat, a driver, a labeller

What this stage solves, intuitively. Neural networks are only as good as the roads they were trained on. The fastest way to find out whether they work on the demo course is to record the course early, with the camera exactly where it will live, and to label a few hundred frames by hand. Mounting the sensors in their final positions now also means the recordings, the calibration and the demo all share one geometry.

B3.1 Materials

ITEM	QTY	SPECIFICATION
Stereolabs ZED 2i, 4 mm lenses	1	$499. The USB cable is no longer in the box: buy Stereolabs' locking USB-C cable, 1.5 m ($20) or 3 m ($30)
Mirror-stalk clamp for the ZED	1	A rigid clamp lined with rubber that grips the rear-view mirror's stalk, with a ¼-inch-20 camera screw on a short arm; a dash-cam mirror mount can be adapted, or print one in ASA with a steel ¼-20 screw
Luxonis OAK-D Pro	1	$349–429; USB-C cable; Luxonis's Y-adapter for external power (it draws 4–7.5 W); a mini ball head on its ¼-20 thread; a base clamped or strapped to the column shroud (no adhesive on the car)
u-blox ZED-F9R breakout (SparkFun GPS-22660, SMA connector, $299.95) and an L1/L2 multi-band active antenna with at least 17 dB gain	1 each	SparkFun's "GNSS Multi-Band Magnetic Mount Antenna, 5 m (SMA)" fits; the plan's line is $300–380 for both
GNSS antenna ground plane	1	A 100 mm steel disc or plate under a magnetic antenna, on a non-slip pad
Footwell base plate	1	A 4–6 mm aluminium or plywood plate clamped to the passenger seat's rail bolts or strapped down, carrying the Jetson, the GNSS board and the switch; it must not move relative to the car (the GNSS receiver's dead reckoning needs it rigid)
Cable clips, braided sleeve, cable ties, gaffer tape	1 set	Routing
Power for the recording sessions	1	The B7 power chain, or a temporary inverter-free 12 V supply to the Jetson through the DC-DC converter

B3.2 Mount the road camera on the mirror stalk

1 Fit the clamp to the mirror's stalk, as close to the glass as it goes, on the car's centreline. The plan wants the camera within about 0.5 m of the driver's eye and high: every range error turns into an angle at the eye in proportion to this distance, and a dash-top camera doubles every number (plan, 5.4).

2 Screw the ZED 2i onto the clamp's ¼-20 screw. Level it across the car (roll within 0.5°) with the angle gauge on its top; aim it straight ahead (yaw within about 1° of the centreline, sighting along its top at the far end of the course); pitch it down 1–3° so the road from about 5 m out fills the lower half of the image. Lock every joint.

3 Wiggle test: push the camera gently in each direction; it must spring back exactly. A camera that moves invalidates the calibration.

- ☐ Camera returns to position after a push

4 Route its USB cable along the top edge of the windshield into the headliner's edge, then down the side of the A-pillar trim to the footwell, held by clips. Do not route a cable over or inside an A-pillar trim that covers a curtain airbag, and do not remove any trim. Use Stereolabs' own locking cable in the length the route needs (1.5 or 3 m); an unbranded passive USB 3 extension is a common cause of dropped frames.

B3.3 Mount the driver camera

1 Fix the ball-head base to the steering-column shroud with a strap or clamp so the camera looks at the driver's face through the wheel's upper opening, about 0.5–0.7 m from the eyes. The wheel's spokes must not block the face at straight-ahead.

2 Aim at the eye midpoint with the driver seated; check the face fills about a third of the image width in the DepthAI preview.

3 Route its USB-C cable down the column, clear of the wheel's tilt and telescope range and the pedals, to the footwell. Use the Y-adapter: its data leg to the Jetson, its power leg to the 5 V buck (B7). Luxonis rates the camera at 6 W typical and 7.5 W maximum with the IR emitters, more than a 900 mA USB port supplies.

B3.4 Install and configure the GNSS receiver

1 Screw the F9R breakout to the footwell base plate, flat, its x axis forward (the silkscreen shows the IMU's axes), square to the car within about 3°. u-blox requires the module to be mounted rigidly in a fixed position and orientation relative to the car's body; its dead reckoning learns how the IMU sits and assumes it never moves.

2 Put the antenna on its ground plate on the dash top near the glass with a clear view of the sky, away from the HUD footprint. Run its cable along the same route as the camera's to the board. If the windshield has a metallised (heat-reflecting) coating, GNSS reception through it is poor; check the fix quality inside the car before relying on it.

3 Configure it from a Windows laptop with u-blox's legacy **u-center** (its Generation 9 Advanced Configuration view; u-center 2 supports only generation 10 and 20 receivers, not the F9R), or from the Jetson with `pyubx2` configuration messages. Set `CFG-NAVSPG-DYNMODEL = 4` (automotive; automatic alignment works only in this model), `CFG-SFIMU-AUTO_MNTALG_ENA = 1` (automatic IMU-mount alignment), and the priority navigation rate `CFG-RATE-NAV_PRIO` (1–30 Hz; start at 10 Hz). Enable the position-velocity-time and the fusion-status messages on USB. Save the configuration to the board's flash.

4 **Wheel ticks: none, by design.** The F9R is designed to use an odometer (wheel ticks or vehicle speed); without it, it runs in untethered mode, whose accuracy during GNSS outages u-blox describes as compromised. Wheel ticks would mean a wire into the car, which this design forbids. GNSS Doppler speed with a fix is unaffected; plan the course with an open sky.

5 **Calibration drive** (u-blox's procedure): stand with the engine running and a clear sky for at least 3 minutes until a good 3-D fix; drive at 30 km/h or more with about ten left and right turns of at least 90°; drive at least 100 m straight at 40 km/h or more. The fusion status then reads 1 (fusion). Stops of 30 s or more, mixed curves and some driving above 50 km/h refine the alignment.

6 Check speed against a phone's GNSS log on a straight run; they should agree within about 0.2 m/s.

B3.5 Record the course

1 Plan the sessions: the demo course in daylight, at dusk, and at night, and an ordinary public road from the passenger seat (the driver drives normally; the HUD is not shown to the driver). About 30 minutes each.

2 Record the ZED as SVO files (they replay through the same capture call `roadd` uses), and log GNSS and IMU through `logd` with monotonic timestamps. If `logd` is not ready, record GNSS separately with a timestamp marker (cover the camera with a hand for one second at the start: the dark frames and the log's note line up the clocks).

3 Record the OAK-D Pro's view of the driver for a few minutes in each lighting condition, with the driver's consent, to develop the head tracker.

4 After each session, copy the files to the host and check them: no dropped-frame warnings, sharp images, a GNSS fix throughout.

- ☐ Day, dusk and night recordings checked

B3.6 Label 300 frames

1 Pick 300 frames spread evenly across the three course sessions (more from curves and from frames with a lead car or a pedestrian than from empty straights). Export them as images with their SVO frame index.

2 Label in a polygon tool (CVAT or similar): every visible lane line as a polyline, marked ego left, ego right or other; road edges; every vehicle and person within 50 m as a box with its class.

3 Have a second person check 10% of the frames; fix the labelling rules where the two disagree.

4 Store the labels in the repository with the frame indices; B4's scoring tool reads them.

B3.7 If it goes wrong

SYMPTOM	RESPONSE
ZED drops frames or disconnects	USB cable too long or passive extension; try another USB 3 port on the Jetson; check that the ZED and OAK-D together do not saturate the carrier's USB controller (plan, 8.3) by running both at full rate.
Images vibrate or blur	Clamp not rigid; shorten its arm; reduce exposure time in bright light.
GNSS fusion never initialises	Board not rigid or not square; poor sky view; repeat the calibration drive with more turns.
Face not found at night	Turn on the OAK-D's IR flood; check its power.

B4 Perception software and gate G1

SOLVES	WEEKS	CLOSES
Where the road and objects are (2.2), how the car moves (2.4), where the eye is (2.3), where the sun is (2.7)	1–4, in parallel with B5 and B6	Gate G1: perception scored on 300 labelled frames; GPU time known
NEEDS	**PRODUCES**	**PEOPLE**
B2 (Jetson with SDKs), B3 (recordings and labels)	The `hud/` package's sensing processes, the scoring tool, a perception score	Software builder; a labeller for the 300 frames

What this stage solves, intuitively. The renderer needs four answers every frame: where the lane lines are on the ground ahead, where the vehicles and people are, how fast the car is moving and turning, and where the driver's eye is. This stage writes the processes that produce those answers from the sensors, each stamped with the moment its data was captured. Before the HUD depends on them, it scores them on frames a person has labelled by hand, so a weak lane network is found in week 4, not on the demo day.

B4.1 Repository layout

The plan makes the HUD software a new standalone package, `hud/`, with no build dependency on openpilot (the openpilot tree in the repository stays as reference only). A layout that keeps each process small:

```
hud/
  common/       clock.py (monotonic time, ZED-to-monotonic offset), bus.py (ZeroMQ PUB/SUB, msgpack),
                schema.py (topic names and fields, Appendix D), frames.py (car, eye, camera transforms), config.py
  roadd/        ZED capture, depth, ground plane, lifting, IMU stream, SVO recording
  laned/        lane network engine, pre- and post-processing
  detectd/      YOLO11s engine, tracker, sign classifier
  egod/         GNSS reader, speed and yaw fusion, sun position
  headtrackd/   OAK-D Pro pipeline, eye midpoint
  hudd/         renderer (B6)
  controllerd/  Pico serial link (B5)
  logd/  audienced/  supervisor.py
tools/          replay.py, score.py, label_import.py, clip.py, gpu_budget.sh
config/         calibration.yaml, rules.yaml, dimming.yaml, sun_patch.yaml (all committed)
```

B4.2 The message layer (do this first)

1. Use one clock: `time.monotonic_ns()` on the Jetson. Every message carries three fields before its payload: `t_capture` (the monotonic time the sensor frame was captured), `t_publish`, and `seq`.
2. Translate each sensor's own timestamp to the monotonic clock at the moment of capture. For the ZED, read the image timestamp (wall clock, nanoseconds) and subtract the current difference between `time.time_ns()` and `time.monotonic_ns()`, measured at the same instant; log the jitter of that difference for an hour once, to be sure it is stable.
3. Publish on ZeroMQ PUB sockets bound to `ipc:///tmp/hud/<topic>`; encode payloads as msgpack maps. Set a small send high-water mark (for example 2) on camera-rate topics so a slow subscriber drops old messages instead of queueing them.

4 Always publish, even when there is nothing to report: an empty `objects` list means "no objects", while no message at all means the producer is down. Subscribers judge staleness from `t_capture`, never from arrival time.

5 Write `supervisor.py`: starts each process, restarts a crashed one after 1 s, and publishes its liveness in `health`.

6 Write `logd` and `tools/replay.py` now: every later step is tested by replaying recordings at their original timing, and the renderer is developed on the mini PC from those logs.

B4.3 `roadd`: the road camera, depth and the ground plane

1 Open the ZED 2i with the ZED SDK's Python API (`pyzed.sl`): 1080p or 720p at 30 fps, metres, depth mode `NEURAL_LIGHT` to start (Stereolabs: 23% of the Orin NX GPU at 30 fps; `NEURAL` costs 59%). Do not enable positional tracking: the SDK's plane API needs it and it costs about 44% of the GPU, and the SDK's visual-inertial tracking is unreliable in cars (plan, 5.1). Keep both as cross-checks only.

2 Work in the car frame from the start. With the SDK's `IMAGE` coordinate system (x right, y down, z forward), the plan's car frame (x forward, y right, z down, origin at the left lens) is a permutation: $x_{car} = z$, $y_{car} = x$, $z_{car} = y$. Apply the camera's mounting rotation from `calibration.yaml` (B7) after that.

3 **Ground plane, every frame.** Take the depth points in a road region: ahead between about 5 and 25 m, within about ±2 m of the car's axis, below the horizon (starting values; tune on recordings). Sample about 2,000 valid points. Fit a plane by RANSAC (3-point hypotheses, 100–200 iterations, inlier threshold about 0.05 m plus 0.5% of range), then refit by least squares on the inliers.

4 Accept the fit only if the inlier share is above about 40%, the normal is within about 5° of the last accepted plane, and the camera's height above it is within 0.15 m of the calibrated height. Otherwise fall back to the calibrated static plane rotated by the IMU's pitch change, and flag it. Smooth the accepted plane with a light exponential filter so it does not jitter. Publish the plane's parameters with the lanes. (The road simulation found the fitted plane worth about a third of the lead-car error under 7 m/s² braking.)

5 **Lift lanes.** For each lane point (u, v) from `laned`, form the ray through the pixel from the left camera's intrinsics and intersect it with the plane. Keep confidence and line type.

6 **Smooth lanes** (a correction to the plan: the simulator drew raw per-point noise, $\sigma \approx 0.02 + 0.004 \cdot x$ m, and the lines visibly wobbled). Fit each line in the car frame as $y(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$, weighted by confidence, over its confident range. Carry the previous frame's coefficients forward by the car's motion over the frame interval, then blend (a per-frame weight around 0.3–0.5 for the new fit is a starting value; a small Kalman filter is better). Publish points sampled every metre from the smoothed curve, and the raw points under a debug field.

7 **Lift objects.** Range is the median stereo depth over the lower fifth of the box; beyond 35 m (stereo's useful range with the 4 mm lens), intersect the box's bottom-centre ray with the plane instead. Bearing comes from the box centre. Publish position, velocity (from the track's history), class, track id and age.

8 Republish the ZED's IMU samples on `imu` with monotonic timestamps. Add an SVO recording switch (SVO files replay through the same capture call, which is how B3's recordings feed this process on the bench).

B4.4 `laned`: the lane network

1. Start with Ultra Fast Lane Detection v2 (MIT licence; CULane F1 75–76; its repository ships ONNX and TensorRT scripts). Keep CLRNet or CLRerNet (Apache; F1 79.6–81.4) as the second candidate, and YOLOP/YOLOPv2 if one network must do lanes and objects together.
2. Export to ONNX with the repository's script, then build the engine on the Jetson itself (engines are specific to the GPU and TensorRT version): `trtexec --onnx=ufldv2.onnx --fp16 --saveEngine=ufldv2.engine`. Use the input size from the model's own configuration file.
3. Run on the left image. Post-process to up to four lines as image points with a per-line confidence; label the two nearest the image centre as ego left and ego right.
4. Choose between candidates by score on the course recordings (B4.9), not by published benchmark numbers; there are no published Orin figures for any of them.

B4.5 `detectd`: objects and tracks

1. Export YOLO11s on the Jetson: `yolo export model=yolo11s.pt format=engine half=True imgsz=640`. FP16, not INT8: INT8 accuracy collapses on this board (Ultralytics). Ultralytics measures 7.9 ms per 640-pixel image on an Orin NX 16 GB. The licence is AGPL-3.0, acceptable while the repository is public; RF-DETR Nano (Apache) is the alternative.
2. Keep the classes the HUD uses: person, bicycle, motorcycle, car, bus, truck, stop sign, traffic light.
3. Track with ByteTrack or OC-SORT (both MIT, CPU). Require three consecutive detections above 0.5 confidence before a track is published as a hazard; keep it 400 ms after its last detection.
4. Stretch: a small classifier on sign crops for US speed-limit values (LISA or Mapillary data), accepting a value after two consistent reads of one tracked sign. Drop it from the demo if it is not reliable (B14).

B4.6 `egod`: speed, turn rate and the sun

1. Read the ZED-F9R over USB with a UBX parser (for example `pyubx2`): position, velocity, speed accuracy, heading and fix type from the navigation solution at the configured priority rate (B3), and the fusion status. Without wheel ticks the receiver runs untethered: trust its dead-reckoned speed only for short outages, and flag the source.
2. Fuse at 100 Hz. Yaw rate comes from the ZED IMU's gyroscope (low latency), with its bias estimated whenever the car is stationary (speed under 0.2 m/s for two seconds). Speed is GNSS Doppler speed, propagated between GNSS updates by the accelerometer and corrected toward each new GNSS value; pitch and pitch rate come from the IMU.
3. Treat the GNSS latency as a parameter (u-blox does not specify it; one user measured about 12 ms on a similar module) and measure it in B6 with the LED test.
4. Rules from the plan: below 1 m/s, report speed 0 and no prediction; with no fix, publish a dash and a glyph flag and no prediction.

5 **Sun-watch, once a second.** From GNSS time, latitude and longitude compute the sun's azimuth and elevation with a standard solar-position algorithm (NREL's SPA, for example through `pvlib`). Rotate into the car frame using the vehicle heading. Compare with the cell patch from the B1 trace (stored in `config/sun_patch.yaml`), with a 2° margin and hysteresis so the flag does not chatter at the edge. Publish `sunWatch`: azimuth, elevation, inside flag.

B4.7 `headtrackd`: the eye point

1 Build a DepthAI pipeline on the OAK-D Pro that runs face detection and facial landmarks on the camera's own processor and computes stereo depth at the two eye landmarks (MediaPipe is not supported on Jetson; on-camera landmarks are the plan's primary route, OpenSeeFace on the CPU the alternative).

2 Output the eye midpoint in the driver camera's frame with a confidence; transform it to the car frame with the driver camera's measured pose (B7). Publish `eyePose` at 30 Hz.

3 Turn on the IR flood illuminator at night (the ambient level from `health` decides): in DepthAI 3, `setIrFloodLightIntensity(x)` and `setIrLaserDotProjectorIntensity(x)` on the device, with x from 0 to 1 (both are off by default and strobe with the left mono camera). If the tracker's confidence drops or its data is older than 500 ms, the renderer falls back to the nominal eye point and shows a glyph.

B4.8 Measure the GPU budget

With `roadd`, `laned` and `detectd` running together from an SVO replay, run `tegrastats` (or `jtop`) for five minutes and record GPU utilisation, per-process frame times and the SoC temperatures. Compare with the plan's estimate:

JOB ON THE GPU	PLAN ESTIMATE PER FRAME	MEASURED
ZED depth, NEURAL_LIGHT to NEURAL	8–20 ms	
Lane network, FP16 (assumed)	4–8 ms	
Object detector, YOLO11s FP16, 640	6–10 ms	
Face landmarks, only if on the Jetson	4–8 ms	
hudd render pass, 4 draws + warp	2–4 ms	
Serial total (plan estimate)	**24–50 ms**	

If the serial total exceeds 33 ms, run camera-rate topics at 20 Hz, and run depth at half the network rate. Record the result in the log; it closes the second half of gate G1.

B4.9 Score perception: gate G1

1 Import the 300 labelled frames (B3.6) with `tools/label_import.py`: lane polylines and boxes per frame, linked to the SVO frame index.

2 Replay the same SVO segments through `roadd`, `laned` and `detectd`; record the outputs.

3 Run `tools/score.py`. Lift the labelled lane lines onto the same fitted plane so the comparison is in metres, and compute the plan's criteria.

CRITERION (GATE G1)	TARGET	RESULT
Ego lane found	≥ 95% of frames	
Lane lateral error at 10 m / 20 m / 30 m	≤ 0.15 / 0.3 / 0.5 m	
Ego-lane vehicles within 50 m recalled	≥ 90%	
Pedestrians within 30 m recalled	≥ 85%	
Range within 10% of the stereo median	yes	
GPU time per frame measured (B4.8)	known	

B4.10 If it goes wrong

SYMPTOM	RESPONSE
Lane network below the score on the course	Try the second candidate; fine-tune on labelled course frames (label more than 300); as the plan's fallback, use a comma four in passive mode as the perception source (plan, section 12), at the cost of its car constraints.
Plane fit fails in traffic, rain or at night	The static-plane fallback with IMU pitch costs about 0.27° per degree of pitch error at 20 m; tighten the road region; reject frames with too few valid depth points.
GPU over budget	20 Hz, half-rate depth, `NEURAL_LIGHT`, a smaller lane input, or move face landmarks entirely onto the OAK-D (they should already be there).
Overlay drifts with time	Timestamps are not on one clock: check the ZED-to-monotonic offset and that every topic carries `t_capture`.
Lines jitter frame to frame	Lane smoothing is off or too light; check the motion carry-forward uses the right sign.

B5 HUD controller and LED drivers

SOLVES	WEEKS	CLOSES
Brightness control (2.5, 2.11), heat protection (2.6), the shutter and its fail-safes (2.7)	1–3, in parallel with B4 and B6	No gate; its bench tests (B5.6) must pass before the backlight is built (B9)
NEEDS	**PRODUCES**	**PEOPLE**
The parts in B5.1; a bench supply, multimeter, oscilloscope	A tested controller board, two LED-driver channels, firmware with its fail-safe rules, `controllerd`	Builder

What this stage solves, intuitively. The backlight, the shutter and the temperature limits must keep working when the Jetson crashes, reboots or is unplugged, because those are exactly the moments when nobody is watching the panel. So they get their own small computer, a Raspberry Pi Pico, with its own rules: it dims the LEDs on request, but it turns them off and closes the shutter by itself if the cell gets too hot or the Jetson goes quiet. It is wired so that even an unpowered or resetting Pico leaves the LEDs off and the shutter closed.

B5.1 Materials

ITEM	QTY	SPECIFICATION
Raspberry Pi Pico (RP2040)	1	With headers (Pico H) or headers to solder
TSL2591 breakout (Adafruit or equivalent)	1	I²C address 0x29; range to about 88,000 lux (Adafruit's figure). VEML7700 (0x10, to about 140,000 lux) is the alternative
MAX31855K breakout and type-K thermocouples	1 + 3	Read-only SPI, 0.25 °C steps; glass-braid bead thermocouples
NTC thermistor	2	10 kΩ at 25 °C, B ≈ 3950, glass bead
RECOM RCD-24-0.50 LED driver, pinned version (or the /W/X3 wired version with both dimming leads)	2	One per string. Input 4.5–36 V, output 2–35 V, 0.5 A. Pins: 1 +Vin, 2 analogue dimming, 3 PWM/ON/OFF, 4 GND, 5 –Vout, 6 +Vout (datasheet REV 1/2016). The 0.70 A part also works with the level capped in firmware
Dual rail-to-rail op-amp, 5 V supply (MCP6002 class)	1	Buffers and scales the two dimming levels to 0–4.4 V
NPN transistors (2N3904 or BC547)	2	Enable switches for the drivers' PWM/ON/OFF pins
Resistors (1%): 10 kΩ ×6, 3.3 kΩ ×2, 4.7 kΩ ×4, 100 kΩ ×2, 1 kΩ ×2; 10 kΩ 0.1% ×1	1 set	Filters, gains, pull-ups, NTC divider
Capacitors: 1 µF film or X7R ×4, 100 nF ×4, 470 µF 10 V electrolytic ×1, 10 µF ×2	1 set	Filters, decoupling, servo rail
Schottky diode (1N5817 class)	1	VSYS feed from the 5 V buck
5 V, 5 A buck converter, input range covering 16 V	1	Feeds the Pico, servo, fan, panel board and the OAK-D's Y-adapter
SG90 micro servo	2	4.8 V nominal; 50 Hz frame, about 1–2 ms pulses
Perfboard 70 × 90 mm, female headers for the Pico, JST-XH connectors, two-way screw terminals	1 set	Controller carrier

ITEM	QTY	SPECIFICATION
Dummy loads: 22 Ω, 10 W resistors	2	Bench testing the drivers (they must never run without a load)
Shielded 4-core cable, 0.5 mm² twin, heat-shrink	2 m each	Sensor and LED leads

B5.2 The LED electrical design, and why it differs from the plan

Strings. Eight LEDs in two strings of four (Figure B9.2), each string on its own driver channel, so a failed LED darkens half the board, not all of it. A Cree XP-G4 has a typical forward voltage of 2.8 V at 700 mA and 85 °C (3.3 V maximum); a Lumileds HL2X 2.77 V. Four in series need about 11.2 V typical and 13.2 V at worst. The RCD-24 is a step-down driver whose dropout at 0.5–0.7 A is under about 0.5 V (datasheet graph), so a 16 V supply leaves more than 2 V of margin at the worst-case forward voltage. That is why the guide sets the DC-DC converter to 16 V (B7.3), inside the Jetson's 9–19 V input range.

Current. The plan's backlight needs about 6.3 W of LED with crossed prism films for 10,000 cd/m², and the guide aims at about 13,000 (B10.1), about 8 W. Two strings of four at 0.5 A deliver about 11 W at full level, leaving margin for film gain below the assumed 2.0. The firmware caps the level if the thermal test (B9.7) demands it.

DIMMING: ANALOGUE, NOT PWM

The plan asks for PWM dimming above 2 kHz to avoid camera beats and names the RCD-24 class. The RCD-24's datasheet limits its PWM input to 1 kHz, and to 200 Hz for a linear response, so it cannot meet that requirement. The guide dims by analogue current instead (pin 2), which produces no flicker at any level, and uses the PWM/ON/OFF pin (pin 3) only as a static on/off switch. Both of the driver's inputs are inverted: 0 V or an open pin means full on. The analogue input runs from 0.13 V (full) to about 4.2 V (off), more than a 3.3 V Pico can produce, hence the op-amp scaling to 0–4.4 V. Analogue dimming covers roughly 20:1 reliably; for the night range the controller can turn one string off (2:1) and the renderer draws white as a dimmer grey (another 20:1), together about 800:1, enough for 3 cd/m² at the eye.

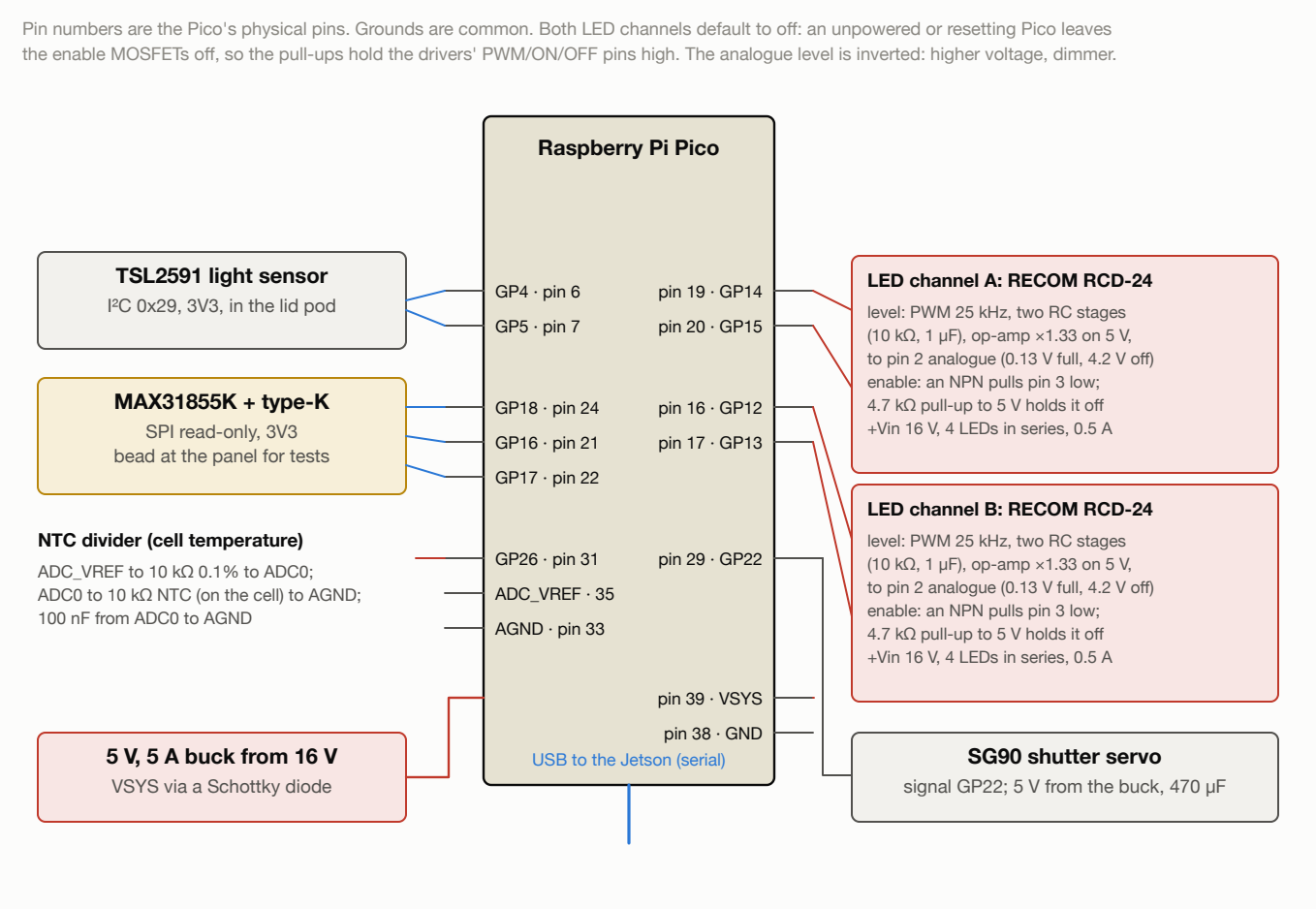


Figure B5.1. Controller wiring. The fail-safe defaults are in the hardware: drivers off and shutter closed whenever the Pico is not actively driving them.

B5.3 Build the controller board

1. Plan the perfboard: Pico on female headers in the middle; the op-amp, transistors and filters on the right toward a 6-way connector to the drivers; the MAX31855 breakout and the sensor connectors on the left; the 5 V input, Schottky diode, 470 µF capacitor and servo header at the bottom. Keep the analogue (NTC, filters, op-amp) away from the servo's supply leads.
2. Solder the female headers and power first: 5 V screw terminal, Schottky diode (anode at 5 V) to VSYS (pin 39), ground to pins 38 and 3. Plug the Pico in, power from the bench supply at 5 V with a 0.5 A limit, and check 3.3 V on pin 36.
3. **NTC divider.** ADC_VREF (pin 35) to the 10 kΩ 0.1% resistor to GP26 (pin 31); GP26 to the NTC connector; the NTC's other lead to AGND (pin 33); 100 nF from GP26 to AGND.
4. **Light sensor.** A 4-way connector: 3V3 (pin 36), GND, SDA to GP4 (pin 6), SCL to GP5 (pin 7). The breakout carries its own pull-ups; use the shielded 4-core to the lid pod, shield grounded at the board end only.
5. **Thermocouple amplifier.** Mount the MAX31855 breakout: VIN to 3V3, GND, CLK to GP18 (pin 24), DO to GP16 (pin 21), CS to GP17 (pin 22). Fit a miniature type-K socket to its thermocouple terminals.

6 **Dimming levels, two channels.** For channel A: GP14 (pin 19) to 10 kΩ, to a node with 1 µF to ground, to another 10 kΩ, to a second 1 µF to ground, to the op-amp's first non-inverting input. Op-amp output to its inverting input through 3.3 kΩ; inverting input to ground through 10 kΩ (gain 1.33). Op-amp output through 1 kΩ to the connector's "A level" pin. Channel B identically from GP12 (pin 16) on the op-amp's second half. Op-amp supply from 5 V with 100 nF at its supply pin.

7 **Enables, two channels.** GP15 (pin 20) through 4.7 kΩ to the base of NPN A; 100 kΩ from base to ground; emitter to ground; collector to the connector's "A enable" pin, which also has a 4.7 kΩ pull-up to 5 V. Channel B the same from GP13 (pin 17).

8 **Servo.** A 3-way header: 5 V, ground, signal from GP22 (pin 29); 470 µF across 5 V and ground right at the header.

9 Wash off flux, inspect every joint under a lens, and check for shorts between 5 V, 3.3 V and ground with the meter before powering again.

B5.4 Wire the LED drivers

1 Mount both RCD-24 modules on a small board or the bay's side, with screw terminals for 16 V in and the LED strings out.

2 For each: pin 1 +Vin to the 16 V rail through a 2 A fuse; pin 4 GND to the rail's ground (never to –Vout); pin 6 +Vout to the string's anode; pin 5 –Vout to the string's cathode; pin 2 to the controller's level pin for that channel; pin 3 to the controller's enable pin.

3 Never connect or disconnect the LED string while the driver is powered, and never run it without a load: switch the 16 V input, not the output. On the bench, use the 22 Ω, 10 W dummy load in place of the string.

B5.5 Firmware

MicroPython is enough (the C SDK is fine too). The firmware talks to `controllerd` on the Jetson over USB serial with one line per message. Appendix B lists the full pin map and protocol.

```
every 10 ms:
   read NTC (average 10 samples), convert with the B equation; MAX31855 every 200 ms; TSL2591 every 500 ms
   if no heartbeat from the Jetson for 5 s:       enables off, shutter closed, flag USB_SILENT
   if NTC > T2 (cut-off):                         enables off, shutter closed, latch flag OVER_TEMP until NTC < T1 − 5 °C and a reset
command
   elif NTC > T1 (derate):                        cap both levels at 50%, flag DERATE
   if a sun-close request is active:              enables off, shutter closed
   else apply the requested levels and shutter state from the last command
   set level PWM: duty = (4.2 V − 4.07 V × level) / (3.3 V × 1.33), PWM at 25 kHz
   shutter open: 50 Hz pulses at the open end-point; closed: stop pulses (the spring closes it)
every 100 ms: send the status line
hardware watchdog: 500 ms (a hang resets the Pico; its outputs default to off and closed)
```

Levels are linearised with the table measured in B5.6 step 4, because a driver's analogue response is rarely a straight line near its ends. The derate and cut-off temperatures T1 and T2 are set from the thermal test (B9.7); until then use 55 °C and 62 °C at the NTC.

On the Jetson, `controllerd` opens the serial port, sends a heartbeat at 5 Hz, computes the level from the ambient light through the dimming curve (`config/dimming.yaml`, tuned in B13.5), sends shutter requests from the power state and `sunWatch`, and republishes every status line as `health`.

B5.6 Bench tests (all must pass)

1. **Safe default.** Power the drivers from the bench supply at 16 V with the dummy loads, the controller unpowered. Measure each driver's pin 3: it must be above 0.6 V (off), and no current flows in the loads.
   - ☐ Drivers off with the Pico unpowered
2. Power the controller with no firmware running (hold BOOTSEL): still off. Flash the firmware: still off until enabled by a command.
3. Enable channel A at level 0.1: the load current rises to about 50 mA. Watch the load current on the oscilloscope across a 0.1 Ω sense resistor: it must be steady, with no PWM ripple.
   - ☐ No flicker on the load current
4. Sweep the level from 0 to 1 in 0.05 steps on each channel, recording the load current at each step. Store the table in the firmware for linearisation. Full level must give 0.5 A ± 5%.
5. **NTC.** Compare the reported temperature with the thermocouple at room temperature and with both taped to a warm aluminium block (60 °C); adjust the B value until they agree within 1 °C.
6. **Thermocouple.** Ice water (0 °C) and boiling water (about 100 °C at sea level): within 2 °C.
7. **Light sensor.** Indoors, compare with a phone light-meter app; outdoors in sun, confirm it does not saturate behind a piece of the dust-cover polariser (it reads about 40% of the open value).
8. **Servo.** Find the open and closed end-points in microseconds with the shutter mechanism (B11) or a mock-up; store them. Stop the pulses with the shutter open: the spring must close it.
9. **Fail-safes.** Stop the heartbeat: both channels off and the shutter closed within 5 s. Warm the NTC past T1: levels halve. Past T2: off and closed, and they stay off when it cools until reset. Send the debug "hang" command: the watchdog resets the Pico within 0.5 s and the outputs go off.
   - ☐ All four fail-safe tests pass

B5.7 If it goes wrong

SYMPTOM	RESPONSE
LEDs on at power-up	An enable transistor is shorted or its pull-up missing; the driver's pin 3 must read above 0.6 V when the Pico is off. Do not proceed until this passes.
Dimming does not reach full brightness	The level voltage at pin 2 must reach about 0.13 V or below; check the op-amp output's low swing and the 1 kΩ series resistor.
Dimming does not reach off	Pin 2 must reach about 4.2 V; check the op-amp supply is 5 V and the gain resistors; turn the channel off with its enable instead.
ADC readings noisy	Average more samples; route the NTC lead away from the servo leads; add 1 µF in parallel with the 100 nF.
Pico resets when the servo moves	Servo supply sagging: the 470 µF must be at the header and the buck must be rated for the peaks.

B6 Stage 0 bench rig, the renderer, and latency: gate G2

SOLVES	WEEKS	CLOSES
Drawing for the eye (problem 4 of Figure 2.1), age compensation (2.4), the first look through glass	1–3	Gate G2: 60 fps from a log on the rig; whole chain ≤ 90 ms by photodiode; prediction sign confirmed
NEEDS	**PRODUCES**	**PEOPLE**
B2 (Jetson), B4's message layer and a first recording; for the latency test, B3's ZED	The Stage 0 rig, `hudd`, a synthetic log generator, the latency probe and its measurement	Software builder

What this stage solves, intuitively. Optical Stage 0 is a teleprompter turned on its side: a panel lies face-up under a sheet of glass at 45°, and you look at the glass and see the panel's picture floating behind it, overlaid on whatever is behind the glass. It has no magnification and no windshield, so it says nothing about the final optics; it exists so the renderer, its colours, its timing and its drawing rules can be built on a desk in week 1, long before any custom part arrives.

B6.1 Materials

ITEM	QTY	SPECIFICATION AND SOURCE
Panel	1	YODOIT 16-inch monitor (unit 1, owned) or the stock 5.5-inch kit once it arrives
Teleprompter beamsplitter glass, 70/30	1	The cheap one of the plan's pair (Neewer 17-inch class, about $53 VERIFY). Confirm the vendor's convention: the first number is usually the transmitted share, so 30% reflects toward the viewer.
Enclosure	1	Black foam board or 6 mm plywood: a base about 450 × 350 mm, two triangular cheeks with a 45° slot for the glass, a backdrop 450 × 350 mm; matte black paint or flock paper; hot glue or wood glue
Camera on a tripod	1	Manual focus; at the viewing position
Latency probe parts	1 set	A spare Raspberry Pi Pico (or the controller's); a bright 5 mm white LED and a 100 Ω resistor; a BPW34 photodiode, a 100 kΩ resistor and an LM393 comparator or the Pico's ADC; a short black tube (a film canister) for the LED; black tape; jumper wires

B6.2 Build the Stage 0 rig

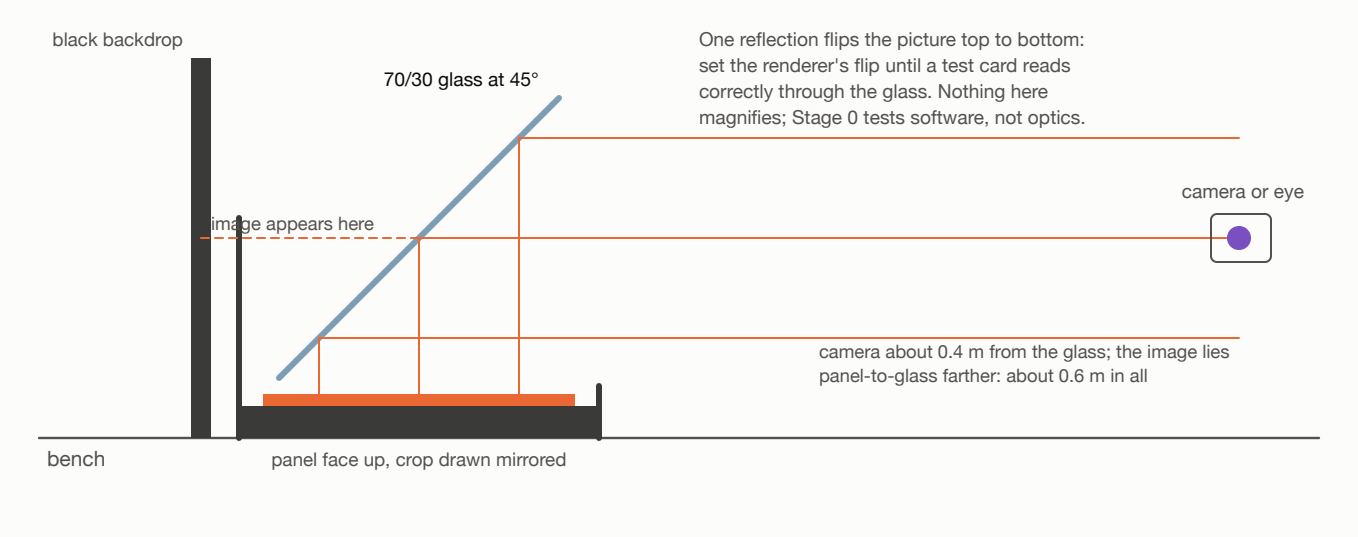


Figure B6.1. The Stage 0 rig in side view. The image appears behind the glass at the same distance from the glass as the panel.

1. Cut the base, the two cheeks and the backdrop. Paint every inside face matte black or cover it with flock paper; reflections from a shiny inside surface show as haze in the image.
2. Glue the cheeks upright on the base, parallel, the panel's width apart plus 10 mm. Their 45° slots face the viewer's side and rise away from the backdrop.
3. Lay the panel face-up in the well between the cheeks; for the 16-inch monitor, block it up so its top surface is level. Check level with the angle gauge.
4. Slide the glass into the slots. Check 45° ± 0.5° with the angle gauge. Handle the glass by its edges; it scratches easily.
5. Stand the backdrop behind the glass on the far side from the viewer.
6. Put the camera on its tripod facing the glass horizontally at the height of the glass's centre, about 0.4 m away. The virtual image is as far behind the glass as the panel is below it, so the image distance is about 0.6 m.
7. Show a test card on the panel: the letter F, an arrow pointing up, and a grid. One reflection flips the picture top to bottom as seen by the camera; set the renderer's output flip (B6.4) until the card reads correctly.
   - ☐ Test card reads correctly through the glass
8. Try the luminance meter on the rig's virtual image (plan: week 1–2). The LS-150 focuses from 1,012 mm to infinity, so stand it back until the image is more than about 1.1 m away. If the spot meter cannot focus on the virtual image, order or borrow the lab meter now; every luminance row in B9 and B13 depends on it.
   - ☐ Meter decision recorded

B6.3 Synthetic and recorded logs

Before `hudd` sees a real recording, give it perfect data. Write `tools/synth.py`, which publishes every topic at its real rate with exact geometry: a straight lane 3.7 m wide, then a curve of 300 m radius, a lead car 30 m ahead that brakes at 5 m/s², a pedestrian crossing at 20 m, constant speed 25 m/s with the matching yaw rate in the curve, and an eye point that sways ±75 mm. Because the truth is known, every error on screen is the renderer's. Then switch to B3's recordings played through `tools/replay.py`.

B6.4 Write `hudd`

1. **Window.** raylib (OpenGL 3.3 core) full screen on the HDMI output at 1920 × 1080, 60 Hz, vsync on. Clear to black every frame: black is "no light" on an LCD with the backlight behind only the crop, so everything outside the crop must stay black.
2. **Two passes.** Draw the content into an off-screen texture in *virtual-image coordinates* (a rectangle representing the window as the driver sees it, at twice the crop's resolution, 1892 × 946, for anti-aliasing). Then draw that texture onto the panel through a *warp mesh*: a grid of quads (start with 64 × 32) whose vertices carry each virtual-image point's position on the panel. For Stage 0 the mesh is the identity plus the flip; in B8 and B13 it comes from calibration.
3. **Crop placement.** Put the crop's origin, size and flips in `config/calibration.yaml`. On Stage 0 the crop can be anywhere; in Stage 2 it must sit exactly over the backlight cavity (found in B9).
4. **Projection.** For each car-frame point, transform it to the eye frame (nominal eye plus the tracker's live offset), then to virtual-image coordinates. On Stage 0 use a simple pinhole whose focal length in pixels is half the crop height divided by the tangent of half the vertical field (the plan's K_{eye}). From B13 on, the projection is the lookup table from the ray trace.
5. **Age compensation, before projection.** For each point captured at t_c, age = $t_{now} - t_c$, where t_{now} is the predicted time the frame will be on the glass (now plus the measured render and scan-out time). Integrate speed v and yaw rate ω from `ego` over the age to get the distance travelled s and the angle turned ψ; then $p_{now} = R_z(-\psi) \cdot (p_{capture} - [s, 0, 0])$. Tracked objects are also advanced by their own velocity over the age.

6 **Drawing rules** (plan section 7.4, to be confirmed on the course in B14):

ELEMENT	RULE
Stale data	`lanes` and `objects` fade over 300 ms once older than 250 ms and are gone at 600 ms. `ego` older than 200 ms: prediction off; older than 1 s: speed shows a dash. `eyePose` older than 500 ms: nominal eye point and a glyph.
Lane lines	Appear above confidence 0.6, full at 0.8; fade below 0.5, gone at 0.3; three consecutive frames before first appearance. At least 6 px wide on the panel (hides the ghost).
Ribbon	Between the two ego lines where both are drawn; one line missing: a standard 3.7 m width from the other, lighter. Far end at the nearer of the two lines' last confident points. Fill at least about 39% of white at noon (the road simulation's figure for a contrast of 1.15); a "blank" mode that skips the fill is useful for checking registration.
Lead chevron	Nearest tracked car, bus, truck or motorcycle whose base lies inside the ribbon, within 80 m; at the bearing of the box's bottom centre and its range, on the ground; gap in metres, and in seconds above 3 m/s. Not drawn when the ribbon is not drawn.
Hazards	Pedestrians, cyclists, signs after three detections above 0.5; held 400 ms after the last, fading in the last 200 ms; inside the window only.
Speed block	Fixed (not conformal): GNSS speed with a fix glyph; last speed-limit value with a sign glyph, timed out after 10 min.
Draw region	Nothing below the window's bottom edge or above the horizon plus 2°.
Clutter mode (demo only)	Everything regardless of confidence, every track outlined, for 10 s on a key press.

7 **Statistics.** Publish `hudStats` at 10 Hz: the age of each topic as drawn, render time, dropped frames, and the last registration residual. Publish a downscaled `hudFrame` for the audience view.

B6.5 Confirm the sign of the prediction

A wrong sign doubles the error instead of removing it, and it looks plausible on a straight road. Check it on a recording before any live drive.

1 Pick a recorded segment through a curve at speed, with clear lane paint.

2 Run `hudd` in "camera overlay" mode: it draws the lines onto the road camera's own image (projected with the camera's intrinsics instead of the eye), delayed by a fixed extra 200 ms to exaggerate the effect.

3 With compensation off, the drawn lines lag the paint (they sit where the paint was 200 ms ago). With compensation on, they should sit on the paint. If they move further away, a sign is wrong: check the direction of ψ and of s.

- ☐ Compensation moves the lines onto the paint in a left curve and a right curve

B6.6 Build the latency probe and measure the chain

1 Wire the probe Pico: GPIO 2 through 100 Ω to the LED's anode, cathode to ground; the BPW34 reverse-biased from 3.3 V through 100 kΩ to ground, the junction to the LM393's input against a threshold set by a trimmer (or to ADC 0 directly). Comparator output to GPIO 3.

2 Put the LED in its black tube, pointed into the ZED's left lens from 20–50 mm, taped so it cannot move. In `roadd`, add a probe mode that watches the mean brightness of that small image region and publishes a flag with the frame's capture time when it crosses a threshold.

3. In `hudd`, draw a 60 × 60 px white square at a fixed panel position whenever the newest flag is on. Tape the photodiode over that square, shielded from room light.
4. Firmware: flash the LED at random intervals of 0.5–1.5 s, timestamp the flash and the photodiode's rising edge in microseconds, and print the difference over USB serial. Collect 200 samples.
5. For the renderer and panel alone, let the Pico send a byte over serial that `hudd` answers with the square: the difference is render plus HDMI board plus panel response. This also gives the HDMI board's input lag (plan, Phase 5).
6. Report the median and the 95th percentile, and fill in the plan's budget:

STAGE OF THE CHAIN	PLAN ESTIMATE	MEASURED
Camera exposure, readout, transfer	10–30 ms	
Perception inference (lanes, objects, depth)	10–25 ms	
Publish and pick-up by hudd	1–3 ms	
hudd render (60 Hz, vsync)	8–16 ms	
Panel scan-out via the HDMI board	8–25 ms	
Total, uncompensated	**37–99 ms**	

Estimate the GNSS latency by recording a few hard stops and cross-correlating GNSS speed with the speed integrated from the accelerometer; the lag at the correlation peak is the GNSS delay. Enter it into `egod`'s configuration.

B6.7 Gate G2

- ☐ Ribbon, lines, chevron and speed block render at a steady 60 fps from a recorded log on the Stage 0 rig (no dropped frames in `hudStats` over 10 minutes).
- ☐ The whole chain, LED at the camera to light from the panel, measures ≤ 90 ms (95th percentile recorded alongside).
- ☐ The prediction's sign is confirmed on a recording (B6.5).
- ☐ `hudd` runs full screen on the Jetson itself, not only on the mini PC.

B6.8 If it goes wrong

SYMPTOM	RESPONSE
Chain above 90 ms	Find the stage with the probe's split measurement. A panel board that buffers a frame adds 16 ms: try another board. Perception slower than the budget: lower its rate or input size. Rendering: check vsync is not stacking frames (triple buffering adds a frame).
Frame drops on the Jetson	Check the power mode (MAXN), that nothing else draws to the display, and that the warp mesh is not rebuilt each frame.
Haze or a second image on the rig	Shiny inside surfaces or the glass's second surface; blacken the enclosure, tilt the backdrop slightly.
The probe triggers on room light	Shield the photodiode, raise the comparator threshold, run the test with the room lights off.

B7 Install in the car and drive with the overlay on a monitor (gate G3)

SOLVES	WEEKS	CLOSES
Safe power and shutdown (2.12), the audience view (2.13), the first live test of perception, compensation and calibration (2.2–2.4)	3–5	Gate G3: overlay within 0.5° of the paint and the lead car for 20 min at up to 25 m/s with prediction on; GNSS speed within 1 m/s of a phone
NEEDS	**PRODUCES**	**PEOPLE**
B3 (sensors mounted), B4 (perception), B5 (controller), B6 (renderer, G2 passed)	The permanent power chain and footwell installation, the audience view, the camera and eye calibration, a passing G3 drive	Builder, driver, safety driver, a passenger watching the monitor

What this stage solves, intuitively. Everything except the optics goes into the car now, wired the way it will stay. The overlay is first drawn onto the road camera's own picture on a monitor in the passenger footwell: if the lines sit on the paint in that picture, perception, timing and compensation all work, and any error later in Stage 1 or Stage 2 belongs to the optics or the eye calibration. Nobody in the driver's seat watches the monitor.

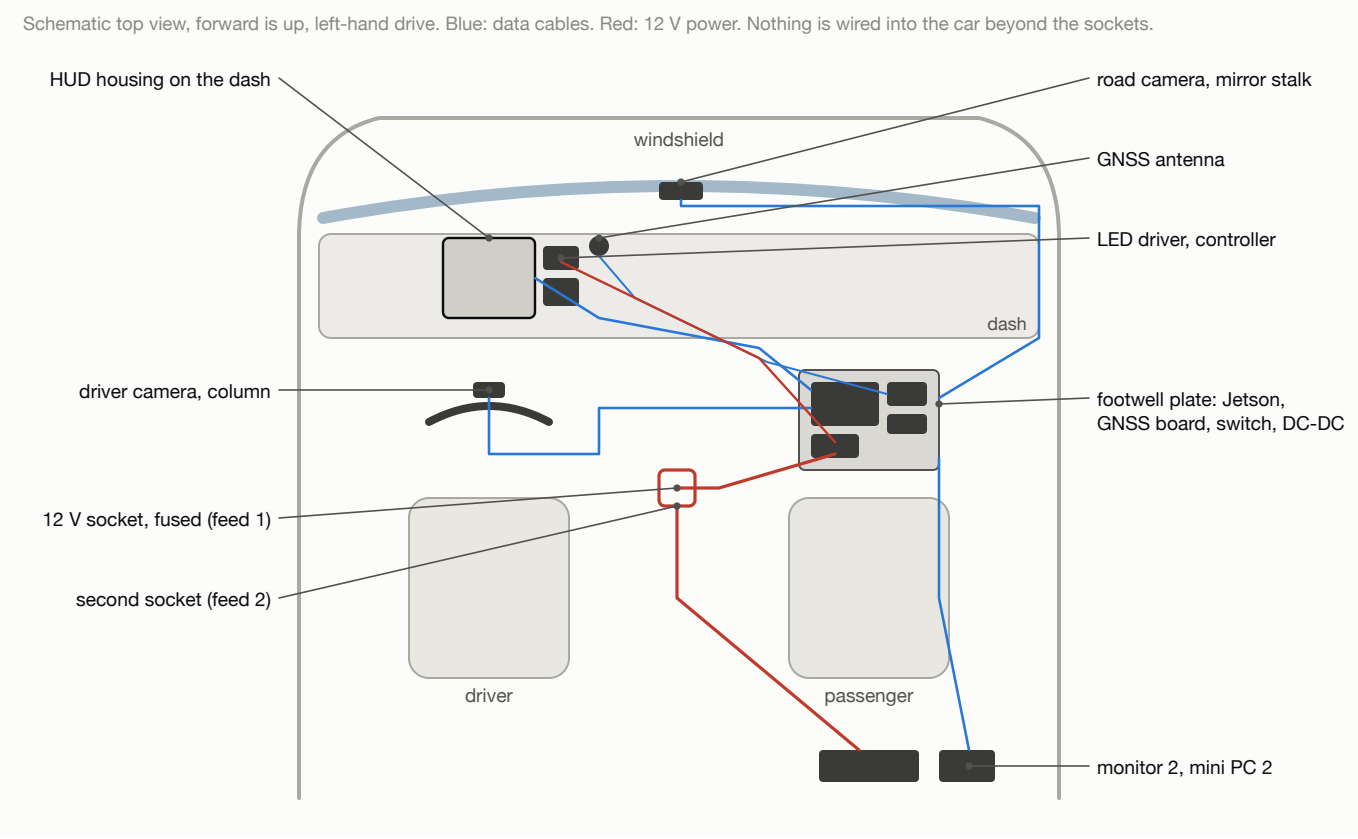


Figure B7.1. Where everything goes, seen from above.

B7.1 Materials

ITEM	QTY	SPECIFICATION
Mini-Box DCDC-USB-200	1	Input 6–34 V, output 5–24 V set by jumpers or USB, 160 W peak; input and output on 4-pin Mini-Fit Jr connectors; deep-discharge cut-off at 11.2 V
12 V accessory plugs with inline blade-fuse holders	3	Feed 1 (always-live), the ignition-sense lead, feed 2 (audience side); 10 A fuses on feeds 1 and 2, 1 A on the ignition lead
Wire	—	1.5 mm² (16 AWG) for 12 V in; 1.0 mm² for 16 V to the Jetson; 0.5 mm² for the LED drivers' 16 V and the 5 V buck; red and black; ferrules
DC plug 5.5 × 2.5 mm with lead	1	For the Jetson (check against the unit's own adaptor: distributors list 12 V 5 A, 5.5/2.5 mm) VERIFY
Two-wire lead with a 2-pin connector	1	DC-DC's J8 pulse output to the Jetson's button header (pin 1 power button, pin 2 ground)
OAK-D Pro Y-adapter	1	Luxonis's USB-C Y-adapter: data to the Jetson, power from the 5 V buck (the camera draws 4–7.5 W, more than a 900 mA port supplies)
Gigabit switch (12 V or USB powered), Ethernet cables ×2	1	Audience link
USB-C PD car charger, 100 W	1	Feed 2: mini PC 2 and monitor 2
Headrest monitor bracket	1	Clamps to the passenger headrest's posts
Monitor 1 and a small stand	1	In the passenger footwell for gate G3 only

B7.2 Find the right sockets

The converter can only shut the Jetson down cleanly after the key is turned off if its input stays live after the key is turned off. Many cars switch their front accessory socket with the ignition; some have a socket that is always live (often in the boot or the centre console), and some keep sockets live for a few minutes after key-off.

1. With a multimeter, test every socket: key out, key in accessory position, engine running, and for ten minutes after key-off. Note each socket's fuse rating from the owner's manual or the fuse box lid.
2. Use an always-live socket for feed 1 (the converter's VIN) and any switched socket for the ignition-sense lead. Use a third socket (or the switched one, if its fuse allows) for feed 2.
3. If no socket stays live after key-off, the converter loses its input with the key: then shut the HUD down from the laptop (or a button that runs `poweroff`) before turning the key, and set the converter's off delay to 0 so it never waits on a missing input. Record which applies.

B7.3 Configure the converter

Set the DCDC-USB-200's jumpers on its configuration header (J7), with the converter unpowered, then power-cycle it. The manual's table and its USB tool agree on these settings; the product page differs on one output step, so confirm the output with the meter before connecting anything.

SETTING	JUMPERS ON J7	WHY
Output 16 V	positions 5 and 3	Inside the Jetson's 9–19 V range, and enough headroom for four LEDs in series on the RCD-24 (B5.2)
Automotive mode	position 7	Uses the ignition input; pulses J8 at ignition-on and after the off delay
Off delay 1 min, hard off 1 min	positions 2 and 1	A short stop does not reboot the HUD; after a real key-off the Jetson gets the shutdown pulse after a minute and power is cut a minute later

1 On the bench, power the converter's VIN from the bench supply at 12.5 V and its ignition input from the same supply through a switch. With no load, measure 16.0 V ± 0.3 V on the output; fine-tune over USB with Mini-Box's tool in 0.25 V steps if needed.

- ☐ Output 16 V measured

2 On the Jetson, make the power button shut the system down: in `/etc/systemd/logind.conf` set `HandlePowerKey=poweroff` and restart `systemd-logind`. Wire J8 to the J401 button header's pin 1 (power button) and pin 2 (ground).

3 Bench test the sequence: ignition on (the Jetson boots); ignition off; after one minute the Jetson shuts down cleanly; a minute after the pulse the output turns off. Then ignition on again. The pulse at ignition-on must not shut a running or booting Jetson down; if it does, use the converter's USB tool to change the pulse behaviour or remove the J8 lead and rely on the hard-off delay plus a `poweroff` from the Jetson's own watchdog script VERIFY.

- ☐ Key-off leads to a clean shutdown, then power off

B7.4 Power budget and fuses

LOAD (16 V RAIL UNLESS STATED)	TYPICAL	PEAK	NOTE
Jetson J4012 at MAXN with ZED 2i and GNSS on its USB	15–20 W	about 35 W	Measure with the meter in B7.8 ASSUMED
LED channels A and B (8 LEDs, 0.5 A, about 95% efficient)	0–11 W	about 12 W	Full only in daylight
5 V buck: panel board (5 V, up to 2 A), OAK-D (up to 7.5 W), Pico, servo, fan	8–12 W	about 22 W	Through a 90% buck
Converter output	**25–45 W**	**about 70 W**	Above 60 W the converter needs forced air: mount it in the Jetson fan's airflow
Feed 1 at 12.5 V	**2–4 A**	**about 6 A**	10 A fuse; 1.5 mm² wire
Feed 2: PD charger, mini PC 2 and monitor 2	40–60 W	about 100 W	Its own socket and 10 A fuse; check the socket's own fuse rating

B7.5 Install the footwell plate

1 Fix the base plate in the passenger footwell so it cannot move (B3.1): the GNSS board's dead reckoning needs it rigid. Leave the area under the passenger's feet free if someone will ride there.

2 Mount on it: the Jetson (vents unobstructed, fan side open), the DC-DC converter in the Jetson fan's airflow, the GNSS board (already square to the car from B3.4), the switch, and the 5 V buck.

3 Wire feed 1: accessory plug, 10 A fuse within 150 mm of the plug, 1.5 mm² to the converter's VIN; ignition lead with its 1 A fuse to the converter's ignition pin (follow the board's silkscreen labels, not wire colours: the manual's colours disagree). Common ground.

4 Converter output (16 V) to: the Jetson's DC plug (centre positive: check with the meter before plugging in), the LED drivers' 2 A fused feed (run to the dash beside the HUD housing), and the 5 V buck.

5 5 V buck outputs to: the panel board's USB power input (the Wisecoco kit is powered by 5 V over USB; its connector differs by board revision), the controller's 5 V terminal, the OAK-D's Y-adapter power leg, and the housing fan.

6 Data: ZED 2i and GNSS board to the Jetson's USB 3 ports; OAK-D's data leg to a third; controller's USB to the fourth; HDMI from the Jetson to the panel board (or to monitor 1 for gate G3); Ethernet from the Jetson to the switch, switch to mini PC 2.

7 Tie every cable to the plate or along the trim with clips; nothing loose near the pedals or the seat rails; no cable across the passenger's airbag door.

- ☐ Cables secured; nothing near the pedals

B7.6 The audience view

1 Clamp the headrest bracket to the passenger headrest's posts and mount monitor 2 facing the rear seats; mount mini PC 2 behind it. Power both from the PD charger on feed 2.

2 Give mini PC 2 the static address 192.168.50.3/24. On the Jetson, `audienced` serves the MJPEG stream (the rendered HUD frame, the driver camera with the eye point, and the live numbers); mini PC 2 opens it full screen in a browser in kiosk mode at boot.

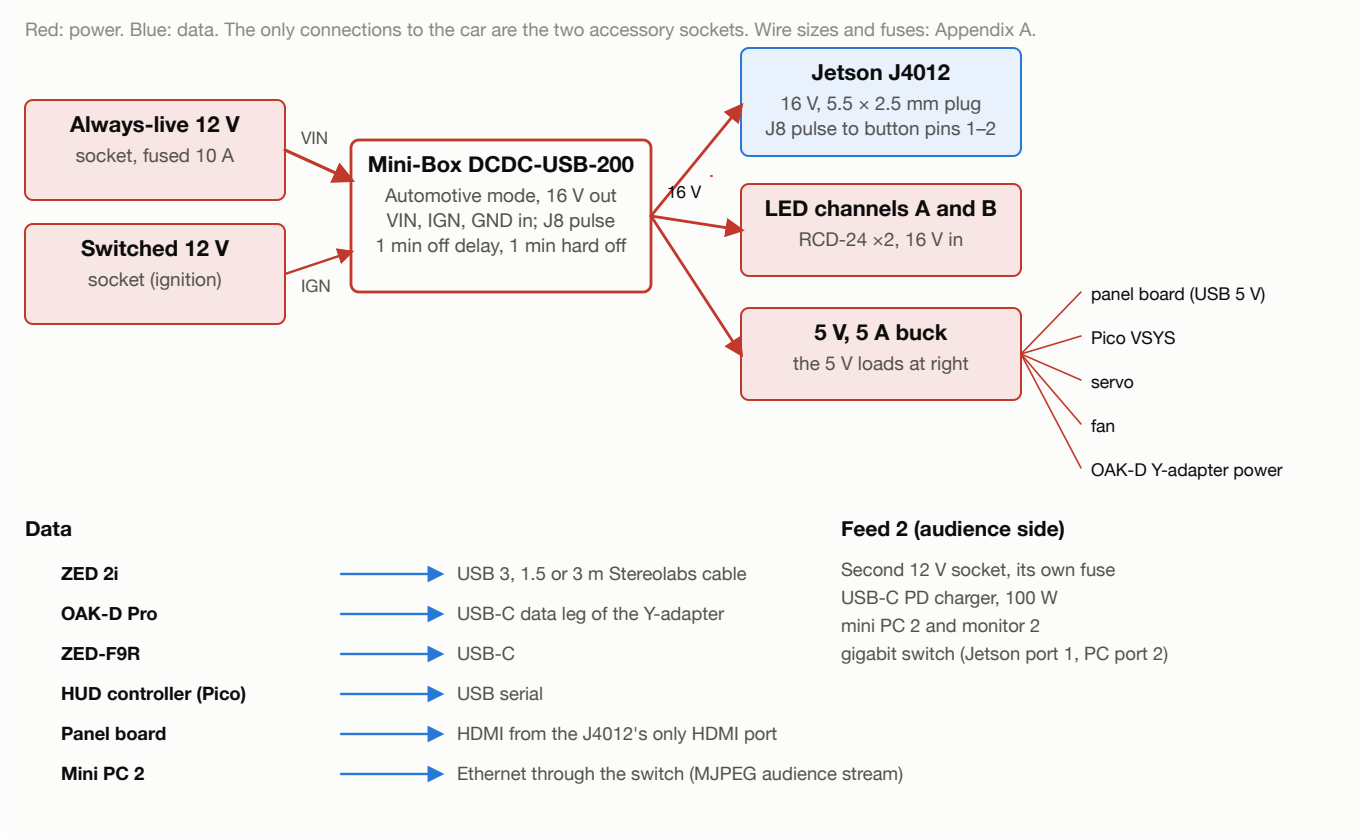


Figure B7.2. Power and data wiring. Appendix A lists every cable with its wire size, fuse and route.

B7.7 Calibrate the camera and the eye point

1. Road camera in the car frame: the 30 m target board procedure (B13.3 A). It is the same procedure as Stage 2's, done now so G3 measures perception with the final camera geometry.
2. Driver camera pose and the nominal eye point (B13.3 B and B1.3).
3. Write `calibration.yaml` and commit it.

B7.8 Drive with the overlay on the monitor

1. Sign the installation checklist (Appendix F).
2. Set monitor 1 on its stand in the passenger footwell, facing the passenger. Run `hudd` in camera-overlay mode on it: the road camera's live image with the ribbon, lines and chevron drawn on it by the full pipeline, with prediction on.
3. Measure the converter's input current with a clamp meter at idle, driving, and with the backlight at full (when it exists) and update B7.4.
4. Drive the course for 20 minutes at up to 25 m/s. The passenger watches the overlay against the paint and the lead car and logs any excursion; `hudStats` logs the residuals and ages. A phone records its own GNSS speed for comparison.

B7.9 Gate G3

- ☐ Over 20 minutes at up to 25 m/s with prediction on, the overlay stays within 0.5° of the lane paint and the lead vehicle in the camera image.
- ☐ GNSS speed agrees with the phone's within 1 m/s while moving.
- ☐ Key-off shutdown works in the car, and the checklist is signed.

B7.10 If it goes wrong

SYMPTOM	RESPONSE
Jetson resets when cranking the engine	Battery voltage dips below the converter's input range; with the always-live feed the converter rides through short dips; otherwise start the engine before booting.
Overlay lags in curves	Compensation sign or yaw-rate axis (B6.5); GNSS or IMU timestamps off.
Overlay offset by a constant angle	Road-camera extrinsics: repeat the board calibration; check the camera did not move.
Fuse blows	Measure the current; check for a short in the LED or 5 V wiring before replacing it with the same rating, never a larger one.

B8 Stage 1 combiner in the car (gate G4)

SOLVES	WEEKS	CLOSES
A real in-car virtual image before the custom mirror exists (2.1), the first registration on the road (2.3, 2.10)	4–8	Gate G4: static error at 30 m ≤ 1.0°; ribbon on the lane through curves at 25 m/s, at night
NEEDS	**PRODUCES**	**PEOPLE**
B7 (everything else in the car), B6 (renderer)	A working night and overcast HUD at about 4.2 m; the demo fallback if the mirror is late	Builder, driver, safety driver

What this stage solves, intuitively. Optical Stage 1 makes the virtual image with a cheap flat Fresnel lens instead of the custom mirror, and shows it on a separate glass (the *combiner*) standing on the dash, instead of on the windshield. The panel sits just inside the lens's focal length, so the lens works as a magnifying glass and the image appears about 4 m ahead. It is dim (night and overcast only) and has ring artefacts, but it puts a registered overlay on a real road weeks before Stage 2 can, and it is the demonstration if the mirror is late.

B8.1 Materials

ITEM	QTY	SPECIFICATION AND SOURCE
Fresnel lens, 300 mm square, f ≈ 300 mm	1 (2 bought)	Knight Optical LFQ300300 (£118.50) or Edmund 11-inch (£121.60); the second is a spare
Panel	1	The YODOIT monitor (unit 1) with an 80 × 40 mm crop drawn in its centre; or the Newhaven 5-inch 1,100-nit HDMI module ($74.75) for night; or the 5.5-inch kit with the custom backlight once B9 is done
Combiner glass, about 30% reflecting	1	The second teleprompter glass; the better 30R/70T glass (twowaymirrors.com, about $514 for 16 × 21 inch) only if Stage 1 will be shown to an audience
Enclosure	1	Matte-black box, 280 mm inside height from the panel's face to the lens, footprint to suit the panel; 6 mm plywood or printed panels; a ledge at the top for the lens
Combiner mount	1	Two side arms with friction hinges (or a printed hinge with a clamping screw) so the glass tilt can be set and locked
Anti-slip mat, two webbing straps	1 set	Holding the enclosure on the dash
Edge protection	1 m	U-channel rubber edging for the combiner glass

B8.2 Build and install

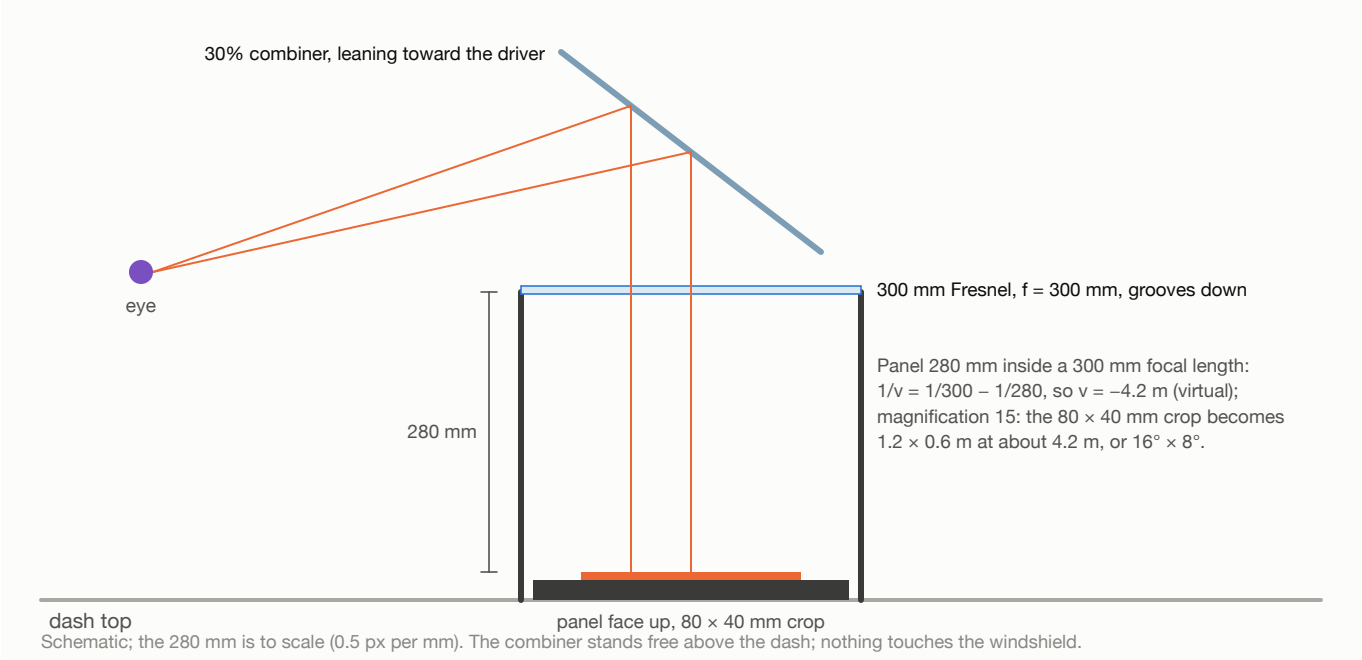


Figure B8.1. Optical Stage 1. The numbers are the plan's thin-lens result for a 300 mm lens and a 280 mm panel distance.

1. Build the enclosure. Blacken every inside face. The panel lies face-up at the bottom; the lens rests on the top ledge, 280 mm above the panel's face (measure to the lens's grooved structure, not to its frame).
2. Orient the Fresnel: lens makers specify which face goes toward the longer conjugate (here the viewer's side). Follow the maker's note; if in doubt, try both ways on the bench and keep the one that gives the sharper grid.
3. Fit the combiner's hinge arms to the enclosure's front. Edge the glass with the rubber U-channel and clamp it in the arms. Set its tilt to about 40–45° from horizontal, top toward the driver: light rising from the lens then reflects back toward the eye.
4. Set the enclosure on the anti-slip mat on the dash in front of the driver, low enough that the combiner's top edge stays below the driver's line of sight to the road beyond 10 m except through the glass. Strap it. Pull-test it as in B13.2.
   - ☐ No movement at 1.5 × weight
5. Configure `hudd` for Stage 1: crop 80 × 40 mm on the monitor (about 872 × 436 px at 277 ppi), one reflection (set the flip with the test card), focal length from the 16° × 8° field.
6. At night, from the eye point, tilt the combiner until the window is centred in the line of sight. Measure the image distance by parallax (Figure B12.1) with the pole at 4.2 m; move the panel up or down a few millimetres to bring the image to about 4 m. Expect about 15× magnification: the 80 × 40 mm crop appears 1.2 × 0.6 m wide.

A LOOSE COMBINER IS A PROJECTILE

Glass standing on the dash can strike the driver in a hard stop. Edge it, lock its hinge, strap the enclosure, pull-test it, and keep it out of the steering wheel's and the passenger airbag's paths. Stage 1 is also the more legally exposed configuration (plan, 11.2): closed course only.

B8.3 Calibrate and drive

1. Set up the 30 m board (B13.3 A); camera at the eye point. Photograph the panel grid through the combiner against the ground grid; fit the warp mesh (a 2-D correction like B13.3 C, but on a single plane: Stage 1 has no custom mirror to trace).
2. Refine the eye point with the board's rings.
3. Measure static registration at 30 m with the camera, then with the camera moved ±75 mm on the rail, with the head tracker off and on.
4. Drive the course at night and in overcast at up to 25 m/s with prediction on. The passenger watches the audience view's residuals; the driver reports where the ribbon leaves the lane.

B8.4 Gate G4

- ☐ Static error at 30 m ≤ 1.0° from the eye point.
- ☐ The ribbon stays on the lane through the course's curves at 25 m/s, at night.
- ☐ Decision recorded: whether Stage 1 will be shown, and whether Stage 2 continues on schedule.

B8.5 If it goes wrong

SYMPTOM	RESPONSE
Image too near or too far	Panel distance: 1 mm closer to the lens moves the image about 0.2 m nearer around this point. Shim the panel.
Rings or colour fringes	Expected from a Fresnel; check its orientation; keep lines thick.
Double image	Both faces of the combiner reflect; tilt slightly, or use glass with an anti-reflection coating on the far face.
Error grows toward the window's edges	Refit the warp with more grid points.

B9 The picture generator: measure the panel, build the backlight (gate G5)

SOLVES	WEEKS	CLOSES
Daylight brightness (2.5), heat (2.6), polarisation (2.8)	4–8; must be finished before the mirror arrives	Gate G5: ≥ 7,716 cd/m² on axis (guide: aim for 13,000), ±20% over the crop, cell under its limit after 30 min at 40 °C, polarisation oriented, input lag measured
NEEDS	**PRODUCES**	**PEOPLE**
B5 (LED driver and controller), the luminance meter, the reference polariser	The finished PGU: cell on a custom backlight, with thermistor, retarder if needed, and its HDMI board	Builder

What this stage solves, intuitively. The liquid-crystal panel is a stencil, not a lamp: it passes only a few per cent of the light behind it. Its own backlight is a thin light guide lit from one edge, made for indoor viewing. This stage measures exactly how much the stencil passes, throws the stock light away, and builds a small, very bright lamp behind just the 60 × 30 mm patch the optics use: eight LEDs in a white box that blends them into one even glow, a diffuser, two prism films that aim the light toward the eye box instead of wasting it sideways, and a heat path to the outside.

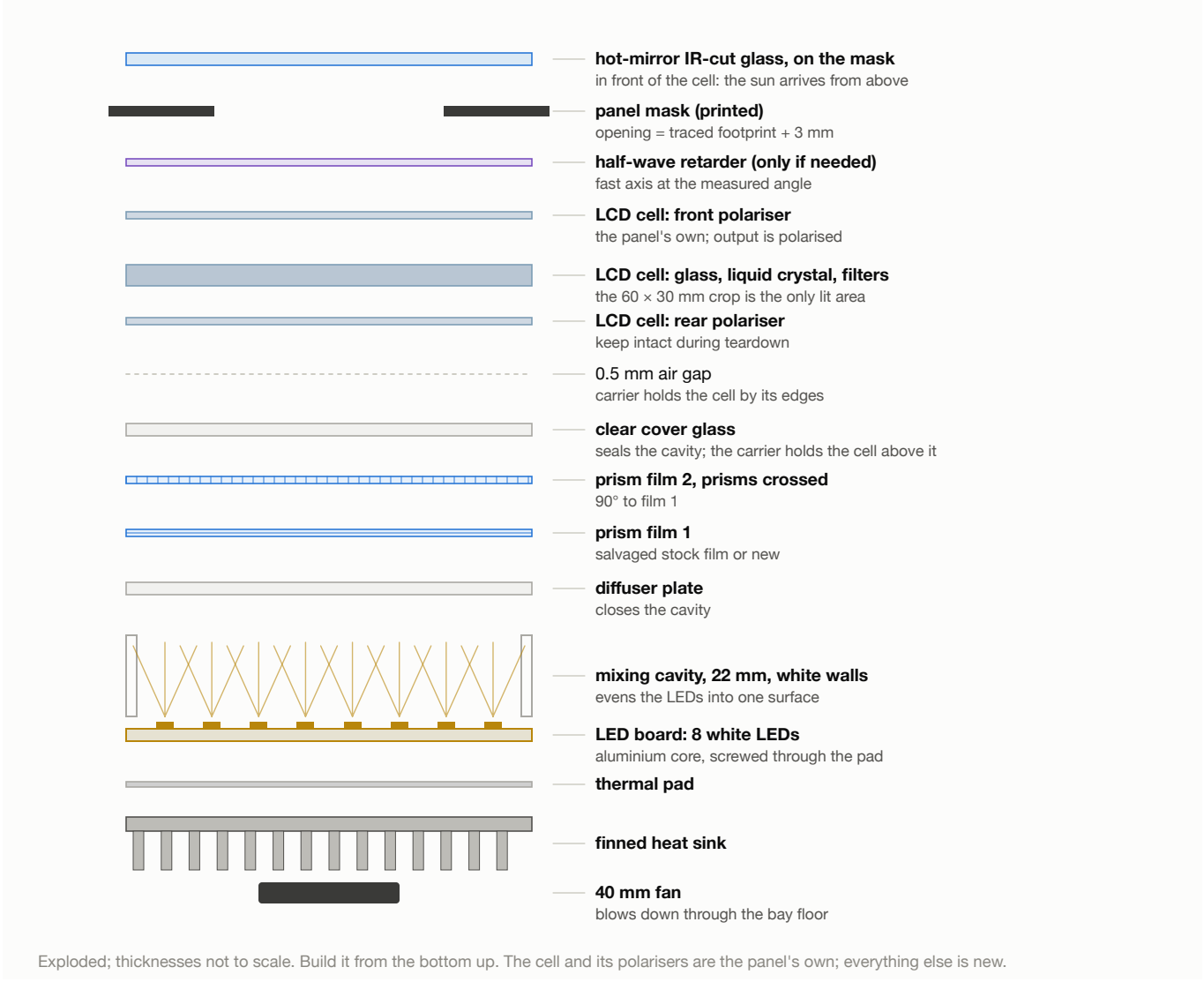


Figure B9.1. The PGU stack in assembly order. The hot-mirror plate sits in front of the cell, where the sun arrives (design basis); a plain cover glass seals the cavity.

B9.1 Materials

ITEM	QTY	SPECIFICATION AND NOTES
Wisecoco 5.5-inch 1920 × 1080 IPS HDMI kit	2	One stays stock (Stage 0, spare, reference); one is taken apart. $95 each (plan)
High-power white LEDs	8 (+4 spare)	Cree XP-G4 or Lumileds LUXEON HL2X class, 5,000–5,700 K, on 20 mm star boards or one custom aluminium-core board (Figure B9.2); see B5 for the drive current and string layout
Heat sink	1	Finned aluminium, about 80 × 50 mm base, 25–30 mm fins; the CAD bay holds it on four posts between its outer fins
Fan	1	40 mm, 12 V (or 5 V), low noise; blows down through the bay floor
Thermal interface	1	Thermal pad 0.5 mm or thin thermal paste; M3 screws with nylon washers for the LED board

ITEM	QTY	SPECIFICATION AND NOTES
Mixing cavity walls	1 set	A printed frame 22 mm tall with a 60 × 30 mm inner opening, lined with high-reflectance white diffuse film (the stock backlight's white reflector sheet can be reused)
Diffuser plate	1	2 mm opal (light-diffusing) acrylic, high transmission, cut to the cavity's outer size; or the stock diffuser sheet over clear acrylic
Prism films	2	The two prism sheets salvaged from the stock backlight, or 3M BEF-class film; cut to the cavity; handle by the edges
Cover glass	1	1–2 mm clear glass, cavity size
Half-wave retarder film	1 piece	Only if the axis measurement calls for it; cut 70 × 40 mm from the sheet (plan: Edmund WP280, American Polarizers, Knight Optical)
NTC thermistor	1	10 kΩ at 25 °C, B ≈ 3950, glass bead with leads; Kapton tape
Insulated test box, heater, thermostat	1	For the 40 °C thermal run (Chapter 5)

B9.2 Measure the stock panel (before taking anything apart)

1. Power the stock kit from the Jetson or mini PC at 1920 × 1080. Show full white. Put the luminance meter 1 m away on axis, focused on the panel. Read nine points (3 × 3 grid) over the area the crop will use. The vendor says 450 cd/m².
   - ☐ Stock luminance and uniformity recorded
2. **Polarisation axis.** Hold the reference polariser in front of the meter's lens and rotate it until the reading is highest; read the polariser's axis angle against the panel's long edge with the angle gauge. Note it with the panel's orientation (which edge will face forward in the car).
   - ☐ Axis angle recorded
3. **Contrast.** Read full black at the same points, in a dark room. Contrast = white / black. The plan assumes 1,000:1 for IPS.
4. **Input lag.** With the latency probe (B6.6), measure the HDMI board's lag from command to light.

B9.3 Take the second panel apart

> **ESD AND THE FLEX**
> Wrist strap on. The cell's driver chips and flexible circuit are the fragile parts. Never lift the cell by the flex, never fold the flex near the glass, never press on the front polariser.

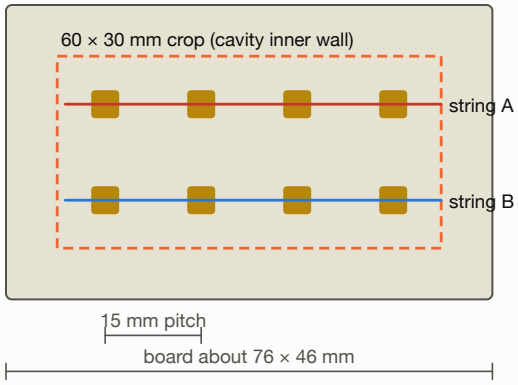
1. Unplug the panel's flex from the HDMI board's connector (lift the connector's latch first).
2. The cell and backlight are held in a metal frame by clips or tape. Release the clips around the frame with a plastic pry tool; work around the edge slowly. If the cell is taped to the backlight frame, warm the tape gently with a hair dryer and lift the cell from one corner with a plastic card, never a blade.
3. Lift the cell out with both polarisers attached, flex supported, and set it face down on a clean lint-free cloth.

4 Lift out the stock optical sheets (diffuser, two prism sheets) by their edges and keep them flat between paper: they can be reused. Keep the white reflector sheet. The light guide and the LED strip are not needed.

5 **Transmittance.** Reassemble the stock backlight on its own (light guide, LED strip, sheets) and connect it to the board, with the cell set aside but still connected so the board runs. Read the bare backlight's luminance at the same points. The cell's transmittance is the stock panel's white reading divided by this. It replaces the plan's assumed 6%: a 450 cd/m² panel over a 7,500 cd/m² backlight would confirm it. Recompute the backlight target with the measured value.

☐ Transmittance measured; backlight target recomputed

B9.4 Build the backlight, bottom up



Eight LEDs on a 15 mm grid fill the crop evenly once the 22 mm cavity mixes them: cavity depth at least the LED pitch is the rule of thumb. Two strings of four, each on its own constant-current channel, so one failed LED darkens half the board, not all of it, and each string's voltage stays below what a 19 V supply can drive.

Keep the outermost LED centres 7.5 mm from the cavity wall; the white wall acts as a mirror and fills the edge. Measure uniformity over the crop before closing the stack; move to a 3 × 5 grid if corners read low.

Figure B9.2. The LED board behind the crop: eight LEDs on a 15 mm grid in two strings.

1 **Heat sink and fan.** Screw the 40 mm fan to the heat sink's fin side so it blows away from the fins (down, out of the housing's floor in the car). Run the fan once to check its direction.

2 **LED board.** Clean the heat sink's face and the LED board's back with alcohol. Lay the thermal pad (or a thin, even layer of paste). Screw the board down with M3 screws and nylon washers, tightening alternately; the board must sit flat.

3 **Leads.** Solder the two strings' leads (B5 gives the wiring: string A and string B to separate driver channels), strain-relieve them, and check each string's polarity with the multimeter's diode test: each LED should glow faintly.

4 **Cavity.** Line the printed 22 mm frame's inner walls with the white reflector film, reflective side in. Place the frame over the LEDs, centred so the 60 × 30 mm opening is over the LED grid with 7.5 mm from the outer LED centres to the walls.

5 **Diffuser, films, cover.** Lay the diffuser plate on the frame. Lay prism film 1 on it, prisms up, ridges running across the short side; lay prism film 2 on top with its ridges at 90° to film 1. Lay the cover glass on top. Tape the stack's edges together with black Kapton or aluminium tape, leaving the opening clear.

6 **First light, low current.** Connect the strings to a current-limited bench supply (or the driver at its lowest duty) and bring each string up slowly; never look into the bare cavity at power (welding glass). Check every LED lights and the cavity glows evenly.

7 **Uniformity before the cell.** Read the cover glass's luminance on a 5 × 3 grid at moderate current. Corners more than 20% below the centre: move the outer LEDs outward, raise the cavity, or use a stronger diffuser. Repeat until even.

- ☐ Bare uniformity within ±20%

B9.5 Put the cell back, and fix the polarisation

1 Tape the NTC bead to the back of the cell, just outside the crop on its warmer side (above the LED grid's edge), with Kapton tape; lead its wires away flat.

2 Lay the cell on the cover glass (in the housing, the carrier holds it 0.5 mm above), the crop over the cavity. Reconnect the flex.

3 **Find the crop in pixels.** In `hudd`, draw a one-pixel border at the intended crop coordinates. Move it in the configuration until it sits exactly on the cavity's lit edge; record the crop's pixel origin and size (946 × 473 px nominal) in `calibration.yaml`. Mark the crop's corners on the cell's frame with a fine pen for B12.

4 **Decide on the retarder.** The light must leave the panel polarised across the car (90° from the car's fore-aft axis) once the panel is in its car orientation. If B9.2 measured that, no retarder. Otherwise, a half-wave film turns an input angle a into 2f – a: for a panel at 45°, set the film's fast axis at 67.5° (Figure 2.8). Cut the film 70 × 40 mm with its fast axis marked, lay it over the crop, and confirm with the reference polariser that the output is now across.

- ☐ Output polarisation across the car (after the retarder if fitted)

B9.6 Measure the finished PGU

1 Full white, controller at full duty. Read the on-axis luminance at the crop's centre and on a 5 × 3 grid. Target: at least 7,716 cd/m², and about 13,000 to cover the cover-glass, polariser, IR-cut and retarder losses (B10.1). Uniformity within ±20%.

2 Read the luminance at ±15° horizontally and ±10° vertically: at least 50% of the on-axis value at ±15° horizontally (the eye box needs about ±11.6° × ±6.3°).

3 Record the LED current, voltage and power at full duty, and the duty-to-luminance curve at ten steps; the controller's dimming table uses it.

B9.7 Thermal test: 30 minutes at 40 °C

1 Put the PGU in the insulated box with the heater and thermostat set to 40 °C, the fan running and its air path open to the box. Thermocouple on the front of the cell in the crop (black-taped bead), the NTC read by the controller, a second thermocouple for the box air.

2 Run full white at full duty for 30 minutes, logging all three at 1 Hz.

3 The cell must stay under its rating. The kit's rating is not published: use 70 °C (the industrial modules' operating limit) unless a datasheet says otherwise. Set the controller's derate threshold (half duty) and cut-off threshold (backlight off, shutter closed) from the curve, for example 55 °C and 62 °C at the NTC if the cell front runs 5 °C hotter than the NTC reads; record the offset you measured.

- ☐ Cell under its limit after 30 min; thresholds set

B9.8 Gate G5

- ☐ On-axis luminance ≥ 7,716 cd/m² (record against the guide's 13,000 aim), uniform within ±20% over the crop.
- ☐ Cell under its limit after 30 minutes at full power in the 40 °C box.
- ☐ Polarisation oriented across the car.
- ☐ HDMI board input lag measured.

B9.9 If it goes wrong

SYMPTOM	RESPONSE
Short of the luminance target	Check the films' orientation (crossed, prisms up); raise the drive current within the LEDs' rating; add a reflective polariser film (DBEF class) above the prism films, its pass axis matching the cell's rear polariser (3M quotes about 2.8 for such a stack); move toward a collimated design (plan, 6.6); or the HUD-grade module experiment.
Hot spots over each LED	Cavity too shallow for the pitch: raise it, add a second diffuser sheet, or use more, weaker LEDs.
Cell over temperature	Improve the fan's air path (the bay's grilles must be open), add a small fan across the cell's back, lower full duty and accept a lower daylight margin, or move to a collimated backlight (less light wasted means less heat for the same luminance).
Colour shift with temperature or a dark patch that grows	The liquid crystal is near its clearing point: stop and lower the thresholds.
Visible pixel grid of the films (moiré)	Rotate one film a few degrees; add the diffuser sheet above the films.

B10 Mirror, fold and filters: specify, order, receive, inspect

SOLVES	WEEKS	CLOSES
The magnifier (2.1), the sun defences in the optical path (2.7), the polarisation at the cover (2.8)	Quote in week 0–1, order after the trace, arrival after the unquoted lead time	No gate; every part passes its incoming inspection before B12
NEEDS	**PRODUCES**	**PEOPLE**
B1's trace outputs; the luminance meter and reference polariser for the checks	An inspected concave mirror, cold-mirror fold with absorber, hot-mirror IR-cut plate, dust cover with laminated polariser, shutter leaves	Builder; the optics shop

What this stage solves, intuitively. These are the only custom or specialised optical parts. The concave mirror is the magnifier and sets the image distance and shape; the fold, the IR-cut plate and the dust-cover polariser are the optical half of the sun defence. Each one is checked on arrival against what was ordered, because a wrong figure or a reversed coating is cheap to catch on the bench and expensive to discover in the car.

B10.1 What to order

PART	SPECIFICATION	SOURCE (PLAN)
Concave mirror	The B1 trace's figure (freeform sag table, or toroid R_t / R_s), clear aperture about 300 × 200 mm, substrate about 12 mm, first-surface protected aluminium ≥ 90% at 450–650 nm, local slope error under about 1.5′ (B1.7). One, plus a quote for a spare.	Avantier, Shanghai Optics (custom to 500 mm, by quote): $1,000–4,000
Cold-mirror fold	Dielectric cold mirror on 3 mm glass, reflecting ≥ 90% from about 425 to 675 nm and transmitting ≥ 80% from about 750 to 1,200 nm, at the fold's 35° incidence (most catalogue cold mirrors are specified at 45°; ask for the curve at 35°). Size: the CAD's fold glass, about 149 × 88 mm; cut from a larger plate if needed.	Optics catalogues or a dichroic-glass supplier: with the IR-cut plate, $150–400
Absorber	2 mm aluminium, black anodised, the fold glass's size, mounted 3 mm behind it on the bracket's tabs	Any sheet-metal service
Hot-mirror IR-cut plate	Dielectric hot mirror (reflects about 750–1,200 nm, transmits ≥ 85% of visible), anti-reflection coated, for normal incidence, 1–2 mm glass, large enough to cover the panel mask's opening (measure the printed mask; about 70 × 40 mm)	As above
Dust-cover glass	3 mm float glass, 270 × 270 mm, edges arrised; anti-reflection coated if affordable (see the budget note)	Local glazier
Polariser for the dust cover	Linear polariser film or sheet, at least 300 × 300 mm, with the highest available transmission for light along its pass axis	Edmund XP42-18 300 × 300 mm (£162); polarization.com film ($20 per foot)

PART	SPECIFICATION	SOURCE (PLAN)
Optically clear adhesive sheet (if the polariser has no adhesive)	300 × 300 mm OCA	Phone-repair suppliers
Shutter leaves	0.5 mm 5052 aluminium, laser cut from `docs/arhud-cad/out/shutter-leaves.dxf`, black anodised	Laser-cutting service

THE LIGHT BUDGET DOES NOT YET INCLUDE THESE PARTS

The plan's path from panel to eye, 19.4%, counts the two mirrors and the windshield only. Light also passes the dust-cover glass (about 92% uncoated), its polariser (perhaps 85–90% for light along its axis ASSUMED), the hot-mirror plate (about 90% ASSUMED) and, if fitted, the retarder (about 92%). Together that is roughly 0.65–0.75 of the plan's figure, and the 10,000 cd/m² panel would deliver about 1,250–1,450 cd/m² at the eye, under the 1,500 target. Mitigations, in order: buy the anti-reflection-coated cover glass and the highest-transmission polariser; measure each part's transmission on arrival (B10.4) and recompute; and build the backlight to about 13,000 cd/m² rather than 10,000 (B9), which the crossed prism films make possible at about 8 W of LED if their assumed gain holds.

B10.2 Handling first-surface mirrors

- The reflecting coating is on the front, exposed. It scratches under a fingernail and a fingerprint etches it within days. Wear gloves; hold the mirror by its edges or its back.
- Store face down on a clean lint-free cloth, or in its shipping case. Never face-up toward a window.
- Dust: blow it off. Fingerprints: a single, light drag of lens tissue wet with isopropyl alcohol, no pressure, never a dry wipe and never a circular rub.

B10.3 Inspect the concave mirror

1. Unpack it gloved on the optics bench. Check the outline and thickness with calipers against the order; check the edges for chips.
2. Look at the coating under raking light (a torch nearly parallel to the surface, in a dark room): note scratches, digs and coating voids. Photograph them; anything inside the traced footprint larger than about 0.5 mm goes back to the shop.
3. **Check the curvature.** Make a point source: a white LED behind a 0.5 mm hole in black card. Stand the mirror upright on the bench; put the source near its axis at about its radius of curvature (about 0.64–0.69 m for this design) and look at the light returned onto a white card held just beside the source. Slide source and card together along the axis. A sphere returns a point at one distance, its radius. A toroid returns a horizontal line at one distance and a vertical line at another: these are its two radii. Measure each from the mirror's vertex.
4. Compare with the order: within about 1% of each radius for a toroid. For a freeform, compare the shop's measurement report with the sag table instead; the functional test is B12's image.

   ☐ Radii or report match the order

5 Check the reflectance: light a white card evenly, read its luminance with the meter directly, then via the mirror at a small angle from the same distance. The ratio is the reflectance (the plan assumes 0.90).

- ☐ Reflectance recorded

B10.4 Inspect the fold, the IR-cut plate and the cover parts

1 Cold mirror: identify the coated face (pen-tip test, B12.3). Look at a white lamp in it at about 35°: the reflection should look neutral white; a strong colour cast means a coating designed for a different angle. Measure its reflectance for s light: reference polariser in front of the meter, axis across the plane of incidence.

2 Hot-mirror plate: look at an incandescent lamp or the sun's reflection off a wall through it; it should look almost clear. Measure its visible transmission with the meter against a lit white card.

3 Polariser: find and mark its pass axis. Look through it at the glare off a glossy horizontal surface (a table, a wet road) at a low angle: that glare is mostly horizontally polarised (s). It is brightest when the pass axis is horizontal. Mark the axis on the film's edge with an arrow.

4 Measure the polariser's transmission for polarised light: the panel's own output (stock kit) through the film with the axes aligned, against the panel alone. Record it; it enters the budget above.

B10.5 Laminate the dust cover

1 Clean the glass on both faces with isopropyl alcohol and lint-free wipes; work in the least dusty room available, with the air still.

2 Lay the polariser over the glass's underside with its pass axis parallel to the glass edge that will lie across the car. Mark that edge "across".

3 Peel back 30 mm of the liner (or of the OCA's liner), tack that edge, and squeegee slowly toward the far edge while peeling, so no air is trapped.

4 Trim the film 1 mm inside the glass's edge with a fresh blade. Small bubbles at the dust cover are out of focus and only scatter a little light; large ones or creases mean starting again.

5 Check: through the cover, a polarised-sunglasses lens rotated to vertical blocks the bench rig's image if the cover passes s (horizontal) light.

- ☐ Cover's pass axis across the car

B10.6 If it goes wrong

SYMPTOM	RESPONSE
Radii off by more than about 1%	Reject, or re-run the trace with the measured radii: the panel's distance can absorb a small focal error, at the cost of the image distance and size.
Mirror late	Run the demo on Stage 1 (B8) and keep building everything else to be ready the day it arrives.
Cold mirror tinted at 35°	Usable if the colour shift is small; the dimming curve and white balance in `hudd` can compensate a little; otherwise order a coating designed for 35°.

B11 Print and assemble the housing

SOLVES	WEEKS	CLOSES
Holds the optics in alignment, keeps stray light and dust out, carries the shutter, lifts off the dash (2.1, 2.7, 2.12)	5–9 (about 248 printer hours; start as soon as the CAD is regenerated for the car)	No gate; B12 checks the assembled housing with the optics
NEEDS	**PRODUCES**	**PEOPLE**
B1's trace and dash measurements in the CAD; the fold-to-panel decision below	An assembled, empty housing with its shutter, dust cover and light-sensor pod	Builder; access to an enclosed printer

What this stage solves, intuitively. The optics only work if four parts hold their positions to a fraction of a millimetre and a fraction of a degree: the panel, the fold, the mirror and the opening under the glass. The housing is the frame that holds them, a dark box that stops sunlight and cabin light from reaching the eye by any path except the intended one, and the support for the shutter that protects the panel. It is printed in 39 pieces because the envelope (459 × 300 × 370 mm) is larger than any desktop printer.

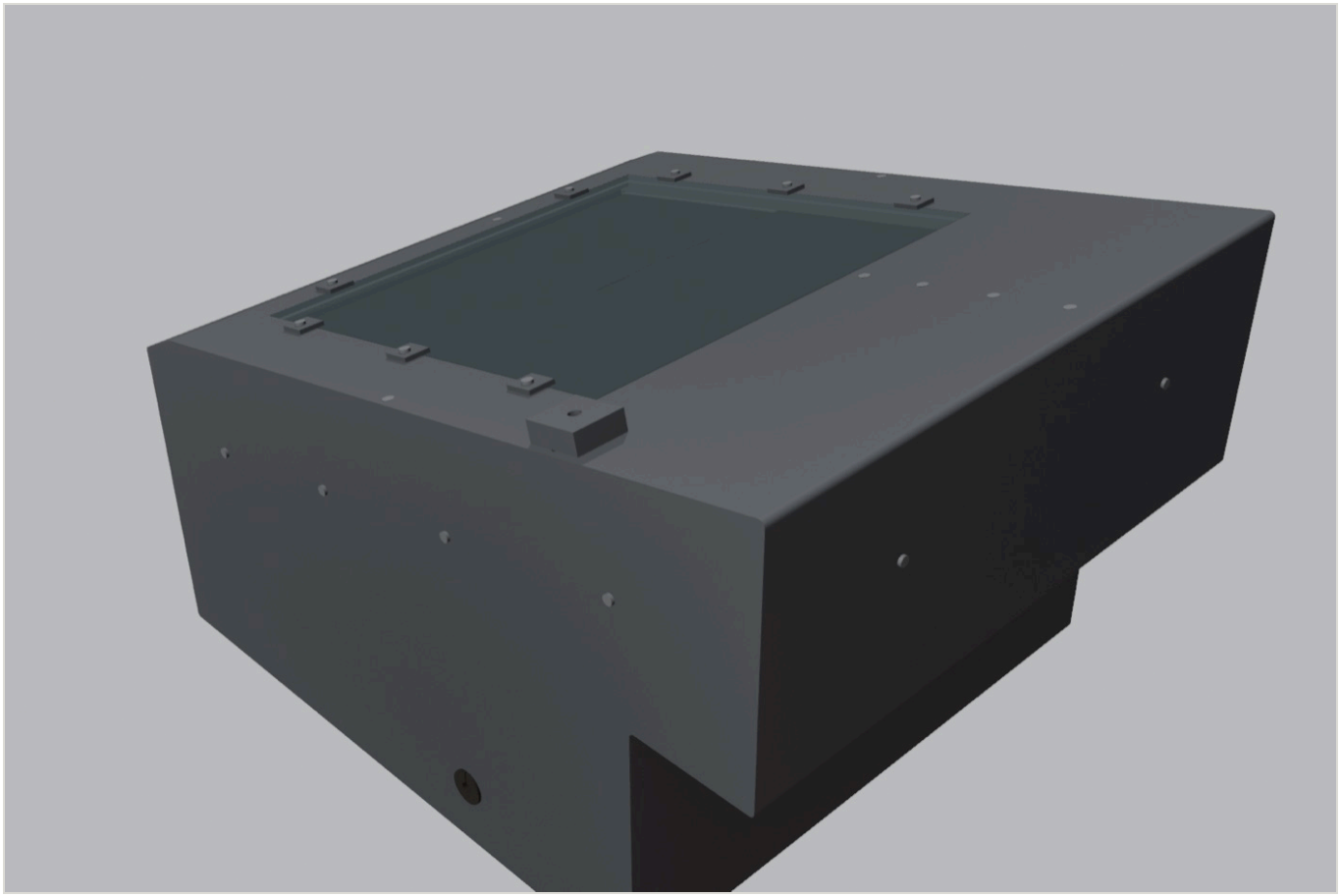


Figure B11.1. The assembled housing (CAD render). The lid's large opening carries the dust-cover glass; the light-sensor pod sits behind it.

B11.1 Decide the fold height, then regenerate

DECISION **FOLD-TO-PANEL SPACING: 50 MM (AS MODELLED) OR 65–70 MM**

The CAD's beam check finds 42 of its 690 traced rays passing through the front edge of the panel's own cell glass. They are rays from the top of the eye box to the top of the window: with the fold only 50 mm over the panel, the mirror-to-fold leg skims the panel's front edge. At 65–70 mm the leg clears with margin; the panel moves about 14–19 mm forward and 20–27 mm down. Recommended: 65 mm. Change `FOLD_TO_PANEL` in `docs/arhud-sim/js/geom.js` (with B1's car values already in), then in `docs/arhud-cad` run `node dump_layout.mjs` and `make.py`, and confirm the report shows 0 blocked rays and 0 panel-glass rays. Keeping 50 mm means the top of the window vignettes from the top of the eye box.

1. Confirm `make.py` reports: beam 0 blocked; panel-glass rays 0 (after the decision); windshield clearance ≥ 8 mm; hood and dash clearances positive for the measured car; no overlapping parts; every printed part fits the printer.
   - ☐ `make.py` report saved in the log
2. Open `out/BOM.md`; it is the list to print and to buy against. The STL files in `out/stl/` are already oriented on their print faces and yawed to fit the bed.

B11.2 Printing

PART	QTY	BED FOOTPRINT, MM	SUPPORTS	MATERIAL	G	STL
Body, rear left quarter	1	229 × 247 × 185	none	ASA	542	bodyRearLeft.stl
Body, rear right quarter	1	229 × 247 × 185	none	ASA	537	bodyRearRight.stl
Body, front left quarter	1	230 × 196 × 185	none	ASA	417	bodyFrontLeft.stl
Body, front right quarter	1	196 × 230 × 185	none	ASA	413	bodyFrontRight.stl
Lid, rear left	1	247 × 227 × 70	none	ASA	311	capRearLeft.stl
Lid, rear right	1	247 × 155 × 70	none	ASA	236	capRearRight.stl
Lid, front left	1	246 × 227 × 70	none	ASA	191	capFrontLeft.stl
Lid, front right	1	246 × 155 × 70	none	ASA	164	capFrontRight.stl
PGU bay	1	86 × 208 × 135	none	ASA	331	bay.stl
Panel carrier	1	86 × 190 × 7	none	ASA	70	carrier.stl
Panel mask	1	85 × 140 × 2	none	ASA	15	mask.stl
Fold bracket	1	82 × 190 × 84	tree supports under the bottom rail	ASA	82	foldBracket.stl
Mirror stand	1	134 × 249 × 133	tree supports under the triangle	ASA	61	mirrorStand.stl
Mirror strongback	1	150 × 240 × 14	none	ASA	268	strongback.stl
Mirror edge clip, left	1	44 × 38 × 16	none	ASA	5	mirrorClipL.stl
Mirror edge clip, right	1	44 × 38 × 16	none	ASA	5	mirrorClipR.stl

PART	QTY	BED FOOTPRINT, MM	SUPPORTS	MATERIAL	G	STL
Light-trap vanes, left	1	206 × 108 × 6	none	ASA	95	vaneL.stl
Light-trap vanes, right	1	206 × 108 × 6	none	ASA	95	vaneR.stl
Grommet, left wall	1	20 × 6 × 20	none	TPU 95A	1	grommetWall.stl
Grommet, HDMI exit	1	26 × 23 × 4	none	TPU 95A	2	grommetHdmi.stl
Grommet, LED and fan leads	1	12 × 12 × 4	none	TPU 95A	0	grommetLed.stl
Adjuster access plug 1	1	13 × 13 × 5	none	ASA	0	accessPlug1.stl
Adjuster access plug 2	1	13 × 13 × 5	none	ASA	0	accessPlug2.stl
Adjuster access plug 3	1	11 × 13 × 8	none	ASA	0	accessPlug3.stl
Shutter rail, left rear	1	236 × 18 × 8	none	ASA	32	railLeftRear.stl
Shutter rail, left front	1	177 × 18 × 8	none	ASA	24	railLeftFront.stl
Shutter rail, right rear	1	236 × 22 × 8	none	ASA	41	railRightRear.stl
Shutter rail, right front	1	177 × 22 × 8	none	ASA	31	railRightFront.stl
Servo bracket	1	48 × 33 × 17	none	ASA	5	servoBracket.stl
Spring anchor post	1	8 × 8 × 6	none	ASA	0	springAnchor.stl
Glass clamps (8)	8	16 × 15 × 2	none	ASA	1	clamps.stl
Light-sensor pod	1	30 × 28 × 11	none	ASA	4	alsPod.stl

SETTING	VALUE AND REASON
Material	ASA, matte black, about 5 kg (the solid mass is 4.0 kg before infill savings; buy margin for failed prints). ASA holds to about 95 °C; dash tops pass 80 °C in summer sun and PETG softens near 75 °C. Grommets in TPU 95A.
Printer	Enclosed, 250 mm cube or larger, hardened or brass 0.4 mm nozzle, with extraction or in a ventilated room
Temperatures	Follow the filament maker's sheet; ASA typically prints at 240–260 °C nozzle, 90–110 °C bed, enclosure warm, part-cooling fan low
Layers and walls	0.2 mm layers; at least 4 perimeters everywhere an insert goes (the inserts need solid plastic around them); 4 top and bottom layers; 25–40% gyroid infill
Adhesion and warping	Brim of 5–8 mm on the body quarters and lid pieces; draught-free enclosure; let parts cool in the closed printer
Supports	Only where the table says so (fold bracket and mirror stand: tree supports under one feature)
Check each part	Measure two critical dimensions per part against the STL with calipers; reprint any part warped by more than 0.5 mm over its length

B11.3 Bought and sheet parts, and fasteners

PART	QTY	SPECIFICATION
Levelling feet	4	M5 threaded rubber levelling foot, 20 mm pad
Shutter leaves, rear and front	2	0.5 mm aluminium 5052, laser cut from `out/shutter-leaves.dxf`, black anodised (send to a laser-cutting service; deburr the edges)

PART	QTY	SPECIFICATION
Dust-cover glass	1	3 mm float glass, 270 × 270 mm (the BOM rounds to 270 × 270), arrised edges, s-pass polariser laminated underneath (B10)
Foam gasket tape	2 m	3 × 2 mm closed-cell, black, for the glass rebate
VHB tape	1 roll	For the light-trap vanes
Black flock paper (optional)	1 sheet	Inside the mirror bay's side walls if the vanes are not enough

FASTENER OR FIXING (FROM THE CAD)	QTY
M3 x 10 button	39
M3 x 16 button	4
M3 x 20 button	2
M3 x 3 heat-set insert (short)	30
M3 x 5.7 heat-set insert	53
M3 x 6 button	16
M3 x 6 csk	7
M3 x 8 button	29
M4 x 30 socket	3
M4 x 8.1 heat-set insert	3
M5 x 9.5 heat-set insert	4
RTV silicone bonding pad, 25 mm x 1 mm	3
compression spring 9 x 15 mm, music wire	3

Two rows of the CAD's hardware list carry M3 sizes for screws its own names describe differently: the HDMI board's four screws ("M2.5 self-tapping") and the servo's two tab screws ("M2 self-tapping"). Buy M2.5 × 6 and M2 × 8 self-tapping screws for those (an SG90 usually comes with its own).

B11.4 Assembly, step by step

The eighteen steps below are the CAD's own build order. Each lists the parts and fasteners it uses, drawn directly from the CAD data. Work on a clean, soft surface; the body quarters are large and their seam strips are the light seals, so keep them free of debris.

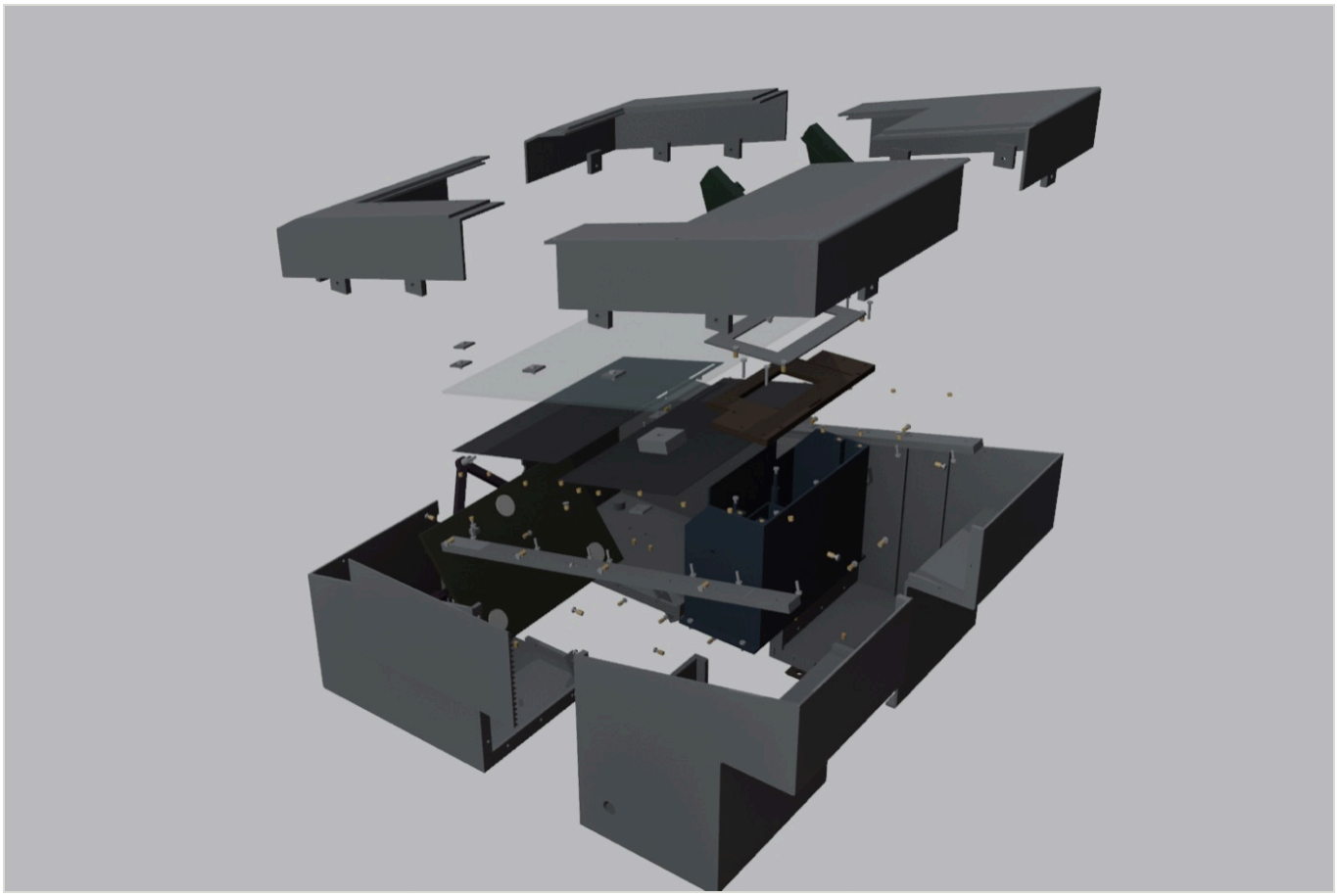


Figure B11.2. Exploded view (CAD render): body quarters at the bottom, the PGU bay and fold bracket in the middle, the mirror stand and strongback at the front, the lid pieces, rails, leaves and glass above.

Step 1 · Heat-set inserts

Print the four body quarters (side wall down, no supports). Press brass inserts into every boss with the insert tip at 245 °C: seams, cap tabs, feet, bay and stand mounts. Hold the iron square to the hole, let the insert sink under its own weight plus light pressure until it is flush, lift the iron straight out, and press the insert flat with a cool flat tool for five seconds. An insert that tilts more than about 2° is reheated and straightened before it cools.

ITEM	KIND	QTY	SPECIFICATION
Body, rear left quarter	printed	1	Floor, walls and seam strips; prints on its side wall.
Body, rear right quarter	printed	1	Floor, walls and seam strips; prints on its side wall.
Body, front left quarter	printed	1	Floor, walls and seam strips; prints on its side wall.
Body, front right quarter	printed	1	Floor, walls and seam strips; prints on its side wall.
Inserts: floor mounts	hardware	8	8 x M3 x 5.7 heat-set insert
Inserts: body z seam	hardware	7	7 x M3 x 5.7 heat-set insert
Inserts: body x seam	hardware	14	14 x M3 x 5.7 heat-set insert

Step 2 · Join the body

Bolt the quarters together along the internal seam strips with M3 button screws into the inserts. Snug the screws in a cross pattern, then tighten; do not strip the inserts. The rear quarters' front strip is the stray-light baffle, 6 mm under the lowest mirror-to-fold ray; the continuous strips keep light out of the seams. Hold a torch inside in a dark room and look for light at the seams.

☐ No light leaks at the seams

ITEM	KIND	QTY	SPECIFICATION
Fixings: body z seam	hardware	7	7 x M3 x 10 button
Fixings: body x seam	hardware	14	14 x M3 x 10 button

Step 3 · Levelling feet

Screw the four M5 rubber feet into the floor inserts. Set them on the car's dash later (B13) so the housing sits level and still, with about 14 mm under the floor for the fan's air.

ITEM	KIND	QTY	SPECIFICATION
Levelling feet (4)	bought	4	M5 threaded rubber levelling foot, 20 mm pad
Inserts: foot	hardware	4	4 x M5 x 9.5 heat-set insert

Step 4 · PGU bay

The sealed bay for the backlight's heat sink, fan and HDMI board bolts to the floor over the floor opening. The fan blows out through the grille in the duct; fresh air comes in through the slots fore and aft. The HDMI board stands on edge against the bay's right wall, its HDMI socket toward the floor. (The PGU itself goes in during B12.)

ITEM	KIND	QTY	SPECIFICATION
PGU bay	printed	1	Sealed tub for heat sink, fan and HDMI board; duct and grilles to the floor.
Inserts: bay top	hardware	4	4 x M3 x 5.7 heat-set insert
Fixings: bay to floor	hardware	4	4 x M3 x 8 button
Fixings: heat sink to posts (self-tapping)	hardware	4	4 x M3 x 10 button
Fixings: HDMI board (M2.5 self-tapping)	hardware	4	4 x M3 x 6 button

Step 5 · Cable grommets and access plugs

Push the TPU grommets into the HDMI and LED-lead exits in the bay floor and the sensor-lead port in the left wall. The three printed plugs close the front-wall holes the ball-end key uses to reach the mirror adjusters; leave them out until B13's alignment is done.

ITEM	KIND	QTY	SPECIFICATION
Grommet, left wall	printed	1	TPU 95A, slit for the cable.
Grommet, HDMI exit	printed	1	TPU 95A, slit for the cable.
Grommet, LED and fan leads	printed	1	TPU 95A, slit for the cable.
Adjuster access plug 1	printed	1	Closes a front-wall hole the ball-end key uses.

ITEM	KIND	QTY	SPECIFICATION
Adjuster access plug 2	printed	1	Closes a front-wall hole the ball-end key uses.
Adjuster access plug 3	printed	1	Closes a front-wall hole the ball-end key uses.

Step 6 · Panel carrier

The carrier closes the bay. Its pocket holds the cell by its edges at the traced height, 0.5 mm over the backlight's cover glass. Test-fit it now; it is installed with the PGU in B12.

ITEM	KIND	QTY	SPECIFICATION
Panel carrier	printed	1	Holds the cell by its edges over the backlight.

Step 7 · Panel mask

The mask clamps the cell's edges; its opening is the traced footprint on the panel plus 3 mm, so it never clips a ray from the eye box. In this guide the hot-mirror IR-cut plate sits on the mask over the opening (B10), held by the mask's screws through two thin printed strips or by black foam tape at its edges. Test-fit only.

ITEM	KIND	QTY	SPECIFICATION
Panel mask	printed	1	Clamps the cell; opening = traced footprint + 3 mm.
Inserts: mask	hardware	4	4 x M3 x 5.7 heat-set insert
Fixings: mask	hardware	4	4 x M3 x 6 button

Step 8 · Fold bracket

The bracket holds the cold mirror over the panel at 35° incidence. Its cheeks stand on the carrier and share the bay's corner screws, so panel and fold are one rigid module. The absorber plate mounts on the tabs behind the glass with M3 screws and nuts. Test-fit; the glass goes in during B12.

ITEM	KIND	QTY	SPECIFICATION
Fold bracket	printed	1	Cheeks, side rails with lips, absorber tabs; 35° incidence, 50 mm over the panel.
Fixings: fold bracket and carrier to bay	hardware	4	4 x M3 x 16 button
Fixings: absorber to bracket (with nut)	hardware	4	4 x M3 x 8 button

Step 9 · Mirror stand

The stand screws to the front floor across the centre seam. Its legs sit outboard of the adjusters so the key can reach every screw head.

ITEM	KIND	QTY	SPECIFICATION
Mirror stand	printed	1	Triangle of adjuster bosses on two outboard legs.
Fixings: mirror stand to floor	hardware	4	4 x M3 x 8 button

Step 10 · Strongback and adjusters

Press the three M4 inserts into the strongback. Fit the three M4 × 30 socket screws with their springs through the stand into the strongback (bed-levelling style: the springs push the strongback away from the stand, the screws pull it back). Set all three to the same height for now. The mirror is bonded to the strongback in B12 on

three RTV pads.

ITEM	KIND	QTY	SPECIFICATION
Mirror strongback	printed	1	Ribbed plate bonded to the mirror's back; M4 inserts for the adjusters.
Fixings: mirror adjuster	hardware	6	3 x M4 x 30 socket; 3 x compression spring 9 x 15 mm, music wire
Inserts: mirror adjuster	hardware	3	3 x M4 x 8.1 heat-set insert
Fixings: mirror bond	hardware	3	3 x RTV silicone bonding pad, 25 mm x 1 mm

Step 11 · Mirror edge clips

Two clips catch the mirror's bottom edge outside the traced footprint, in case the bond lets go. Fit them after the mirror (B12).

ITEM	KIND	QTY	SPECIFICATION
Mirror edge clip, left	printed	1	Catches the mirror's bottom edge outside the traced footprint.
Mirror edge clip, right	printed	1	Catches the mirror's bottom edge outside the traced footprint.
Inserts: mirror clip	hardware	2	2 x M3 x 5.7 heat-set insert
Fixings: mirror clip	hardware	2	2 x M3 x 8 button

Step 12 · Light-trap vanes

Stick the saw-tooth vanes to the side walls of the mirror bay with VHB tape, teeth facing the mirror. They trap sunlight that would otherwise light the walls the driver sees around the mirror.

ITEM	KIND	QTY	SPECIFICATION
Light-trap vanes, left	printed	1	Saw-tooth light trap, held by VHB tape.
Light-trap vanes, right	printed	1	Saw-tooth light trap, held by VHB tape.

Step 13 · Lid pieces

Print the four lid pieces face down. Press the short inserts for the shutter rails, glass clamps and lid laps, and the cap-to-body inserts. Join the pieces with their half laps and countersunk screws, so the underside the shutter runs on stays flat; check it with the straightedge.

- ☐ Lid underside flat within 0.3 mm under the straightedge

ITEM	KIND	QTY	SPECIFICATION
Lid, rear left	printed	1	Lid with the glass rebate, skirt and spigot tabs; prints face down.
Lid, rear right	printed	1	Lid with the glass rebate, skirt and spigot tabs; prints face down.
Lid, front left	printed	1	Lid with the glass rebate, skirt and spigot tabs; prints face down.
Lid, front right	printed	1	Lid with the glass rebate, skirt and spigot tabs; prints face down.

ITEM	KIND	QTY	SPECIFICATION
Inserts: cap to body	hardware	12	12 x M3 x 5.7 heat-set insert
Inserts: shutter rail	hardware	14	14 x M3 x 3 heat-set insert (short)
Inserts: glass clamp	hardware	8	8 x M3 x 3 heat-set insert (short)
Fixings: lid lap	hardware	7	7 x M3 x 6 csk
Inserts: lid lap	hardware	7	7 x M3 x 3 heat-set insert (short)
Inserts: spring anchor	hardware	1	1 x M3 x 3 heat-set insert (short)

Step 14 · Shutter rails and brackets

Screw two rails per side under the lid, each with a slot for each leaf. Hang the servo bracket under the right rail and the spring's anchor post under the front rim. Mount the SG90 in its bracket; the servo body sits behind the pivot, out of the beam, and its lever runs under the leaves.

ITEM	KIND	QTY	SPECIFICATION
Shutter rail, left rear	printed	1	Two leaf slots; screws to the lid from below.
Shutter rail, left front	printed	1	Two leaf slots; screws to the lid from below.
Shutter rail, right rear	printed	1	Two leaf slots; screws to the lid from below.
Shutter rail, right front	printed	1	Two leaf slots; screws to the lid from below.
Servo bracket	printed	1	Holds the SG90 under the right rail.
Spring anchor post	printed	1	Holds the constant-force spring's end.
Fixings: shutter rail	hardware	14	14 x M3 x 10 button
Inserts: servo bracket	hardware	2	2 x M3 x 5.7 heat-set insert
Fixings: servo bracket	hardware	2	2 x M3 x 20 button
Fixings: servo tabs (M2 self-tapping)	hardware	2	2 x M3 x 8 button
Fixings: spring anchor	hardware	1	1 x M3 x 8 button

Step 15 · Shutter leaves

Slide the two black-anodised leaves into the rails; the front leaf carries the slot the servo's lever pin drives. Hook the return spring between the front leaf and the anchor post: with no power, the spring pulls the shutter closed. Work the shutter by hand through its full travel twenty times; it must close on its own from any position.

- ☐ Shutter closes under the spring alone from fully open, every time

ITEM	KIND	QTY	SPECIFICATION
Shutter leaf, rear	sheet metal	1	0.5 mm aluminium 5052, black anodised, laser cut
Shutter leaf, front	sheet metal	1	0.5 mm aluminium 5052, black anodised, laser cut

Step 16 · Lid onto the body

Lower the lid so its spigot tabs drop inside the body walls; drive the M3 button screws through the walls into the tabs. No tools are needed to lift the whole housing off the dash, only to open it.

ITEM	KIND	QTY	SPECIFICATION
Fixings: cap to body	hardware	12	12 x M3 x 8 button

Step 17 · Dust-cover glass

Lay foam gasket tape in the lid's rebate. Place the glass, polariser side down and its pass axis across the car (B10), flush with the top. Fit the eight clamps over its edges and snug their screws evenly; do not stress the glass.

ITEM	KIND	QTY	SPECIFICATION
Dust-cover glass	bought	1	3 mm float glass, 270 x 270 mm, arrised edges, s-pass polariser laminated underneath
Glass clamps (8)	printed	8	Hold the glass edges on its gasket.
Fixings: glass clamp	hardware	8	8 x M3 x 6 button

Step 18 · Light-sensor pod

The TSL2591 board sits in its pod on the lid behind the glass, looking up through its 6 mm window; its lead runs down the left wall's channel to the grommet.

ITEM	KIND	QTY	SPECIFICATION
Light-sensor pod	printed	1	TSL2591 board under a 6 mm window.

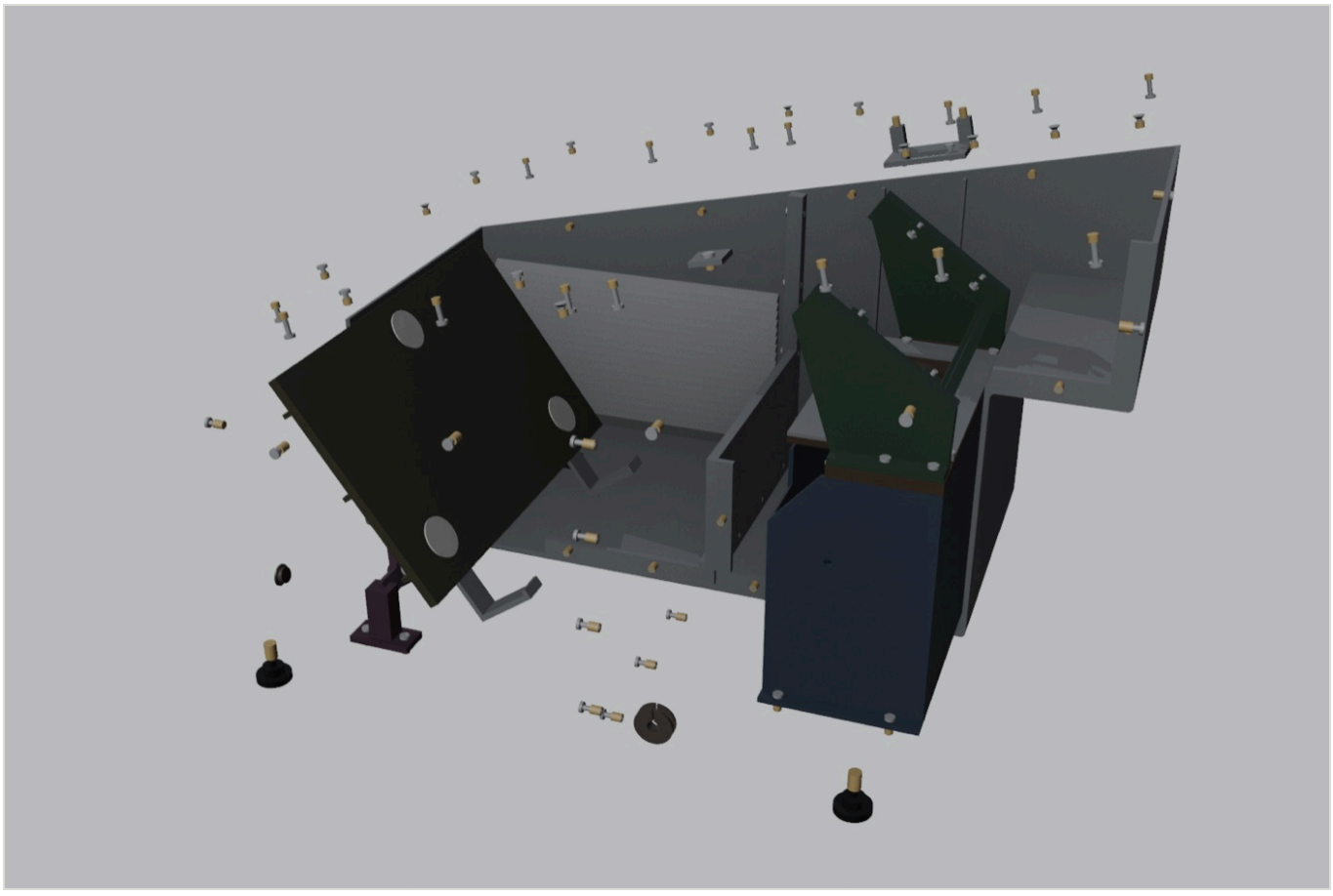


Figure B11.3. Cutaway (CAD render): the PGU bay at the rear under the fold, the mirror on its stand at the front, the shutter under the lid opening.

B11.5 If it goes wrong

SYMPTOM	RESPONSE
Body quarters warp or the seams gap	Lower the part-cooling fan, add brim, keep the enclosure closed until cool; sand seam strips flat; a bead of black silicone on the outside of a seam as a last resort.
Insert pulls out or spins	The boss needs more perimeters; reprint, or fill with epoxy and redrill for a self-tapping screw.
Shutter sticks	Rails not parallel or the lid not flat; shim a rail, deburr the leaves, a dry PTFE lubricant on the rails.
Housing does not fit the dash	Re-measure the profile, edit `hudcad/geo.py`, regenerate; the feet take up a few millimetres.

B12 Optics into the housing, first light, and the sun test

SOLVES	WEEKS	CLOSES
The virtual image (2.1) for the first time with the real mirror; proves the sun defences (2.7) before the panel sees daylight	8–10, the week the mirror arrives	No gate; the empty-housing sun test is a hard prerequisite for B13
NEEDS	**PRODUCES**	**PEOPLE**
B9 (finished PGU), B10 (mirror, fold, filters, inspected), B11 (housing), B5 (controller and shutter)	An aligned HUD unit, a bench image at about the design distance, a logged sun test	Builder and a helper (mirror handling, sun test)

What this stage solves, intuitively. Up to now the optics have existed as a trace and a pile of parts. Here they become one instrument: the panel, fold and mirror go into the housing in the positions the trace chose, and a sheet of glass at the windshield's angle stands in for the car so the image can be seen and aligned on the bench. Before the panel is ever exposed to the sun, a thermocouple sits where the panel will be and the empty housing is pointed at the sun from the dangerous direction, so the protection is proven by measurement rather than assumed.

B12.1 Materials

ITEM	SPECIFICATION
Neutral-cure RTV silicone	Alkoxy or oxime cure, black or translucent. Not acetoxy (vinegar-smelling) silicone: it releases acetic acid that corrodes aluminium mirror coatings.
1 mm shims	Three pieces of 1 mm sheet or feeler stock, to set the bond line
Bench "windshield"	A sheet of 4–6 mm glass, at least 400 × 300 mm, in a frame that holds it at the measured rake over the housing at the trace's glass position (two printed or plywood side frames)
Dummy panel target for the sun test	A 2 mm aluminium plate the size of the cell, painted matte black, with the thermocouple bead bonded at the crop's centre with thermal epoxy
Thermocouple and MAX31855 on the controller (B5), a second thermocouple for ambient	Type K
Test target for the image distance	A pole or a printed cross on a stand, placed at 7.5 m from the eye point (a corridor or the car park)

B12.2 Bond the mirror to its strongback (day 1, cures overnight)

1. Wear gloves. Lay the mirror face down on a clean, soft, lint-free cloth over a flat board. Never touch the coated face; never wipe it dry.
2. Clean the mirror's back and the strongback's three pad areas with isopropyl alcohol; let them dry.

3. Set the three 1 mm shims beside the pad positions. Apply a 25 mm round of RTV at each pad position on the strongback, slightly proud of 1 mm.
4. Lower the strongback onto the mirror's back, centred on the mirror's outline (mark the centre on the mirror's edge with tape first), until it rests on the shims. The RTV must not reach the edge of the mirror. Do not press further: a thin bond line transmits stress and can distort the figure.
5. Leave it untouched for 24 hours at room temperature. Then remove the shims.

B12.3 Install the fold and the mirror

1. Open the housing (lid screws out, lid off). Keep the shutter closed.
2. **Find the cold mirror's coated face.** Touch a pen tip to the glass: on the coated face, the reflection of the tip touches the tip; on the uncoated face a gap the thickness of the glass appears. The coated face goes toward the panel and the concave mirror.
3. Drop the cold-mirror glass into the fold bracket, coating as above, its long edge across the car, so its plane of incidence (the vertical fore-aft plane) matches the windshield's. Fit the four clips without stressing the glass. Screw the black absorber plate to the tabs behind it.
4. Set the strongback with its bonded mirror on the stand's three adjusters and turn the three M4 screws in evenly until the springs are about half compressed. Fit the two edge clips under the mirror's bottom edge, clear of its face.
5. With the dummy target in the panel position (sun test first, B12.4), or the PGU (B12.5), check the clearances: nothing within 2 mm of the traced beam (the CAD's nearest part is 1.8 mm from it).

B12.4 Sun test of the empty housing (before the panel ever sees daylight)

> **THIS TEST FOCUSES SUNLIGHT ON PURPOSE**
> Nothing flammable near the housing; no hands, cables or labels near the focus with the shutter open; a shade cloth ready to throw over the dust cover; an observer who watches the temperature and calls the stop. Stop at the panel's rated temperature or at 10 minutes, whichever comes first.

1. Fit the dummy target in the panel's position (carrier and mask in place, the hot-mirror IR-cut plate on the mask as it will be). Route its thermocouple out through a grommet to the controller's MAX31855. Tape the second thermocouple in the shade inside the cabin for ambient.
2. Install the housing on the car's dash (B13.2 describes the mounting) with the dust cover, polariser and shutter in place.
3. Choose the time and heading: run `egod`'s sun-watch computation (or the trace's sun patch with any solar-position calculator) for the venue's coordinates and pick a time when the sun stands at the patch's centre elevation behind the car (about 51° for a 28° windshield; the B1 trace gives the real value), then park the car facing directly away from the sun's azimuth.
4. Log both temperatures at 1 Hz. Open the shutter with the controller. Watch the target's temperature rise; stop at 10 minutes or at the panel's rated temperature.

5 Close the shutter and log the cool-down. Then turn the car slowly through the patch (or wait for the sun to move) with sun-watch running and confirm it closes the shutter when the sun enters the patch.

ACCEPTANCE (PLAN, SECTION 1)	RESULT
Target temperature stays under the cell's rating for 10 min with the shutter open at the worst sun position	
Sun-watch closes the shutter inside the entry patch	
Peak temperature rise and time to peak recorded	

If the target exceeds the rating, the panel must never be exposed with the shutter open in that sun position: widen the sun-watch patch, add an absorbing IR filter, or ask the mirror shop for a figure that spreads the solar spot (B1.7). Record the decision.

B12.5 Install the PGU

1 Remove the dummy target. Lower the finished PGU (B9) into the bay: heat sink onto its four posts (self-tapping screws), fan over the floor grille blowing down, HDMI board on edge against the right wall with its socket toward the floor. Lead the HDMI cable and the LED and fan leads out through their grommets.

2 Fit the carrier; lay the cell in its pocket with the 60 × 30 mm crop centred over the backlight cavity (use the crop marks you made in B9). Route the thermistor lead.

3 Lay the retarder over the crop if B9 called for it, fast axis at the measured angle. Fit the mask; lay the hot-mirror IR-cut plate on the mask over its opening; fit the fold bracket's cheeks over the carrier and drive the shared corner screws.

B12.6 First light and coarse alignment on the bench

1 Set the bench windshield over the housing at the measured rake, at the trace's glass position relative to the housing (measure from the housing's front top edge).

2 Mount the camera (or your eye) at the nominal eye point relative to the glass. Connect the panel to the Jetson; `hudd` shows the alignment pattern: a border around the crop, a centre cross, and a 5 × 3 grid of crosses. Bring the backlight up to about 10% through the controller. Open the shutter.

3 Look for the pattern in the glass. If it is not visible, hold a white card in the beam to follow it: on the fold, on the mirror, out through the opening, on the glass. Adjust the three M4 screws through the front-wall access holes with the long ball-end key: turning the two lower screws together tilts the mirror up and down; turning one side tilts it left and right.

4 Centre the window in the line of sight 5° below horizontal. Check that the whole border is visible from the centre of the eye box, then from 65 mm left and right and 35 mm up and down.

- ☐ Whole border visible across the eye box

5 Measure the image distance by parallax (Figure B12.1): put the pole at 7.5 m on the line of sight and move the camera sideways by 50–75 mm. If the centre cross stays on the pole, the image is at 7.5 m. The trace predicts the range across the window; at this stage, centre within about ±1 m is good. A gross error (an image at 2 m or at infinity) means the panel is at the wrong distance from the mirror or the mirror's figure is not what was ordered: stop and check.

- ☐ Centre image distance recorded

6 Look for a second, faint window above the main one from the upper eye-box positions (the Cycles check found one 3.5° above the window, from light reflected off the glass back into the mirror) and for haze around the mirror; add flock paper where the haze comes from.

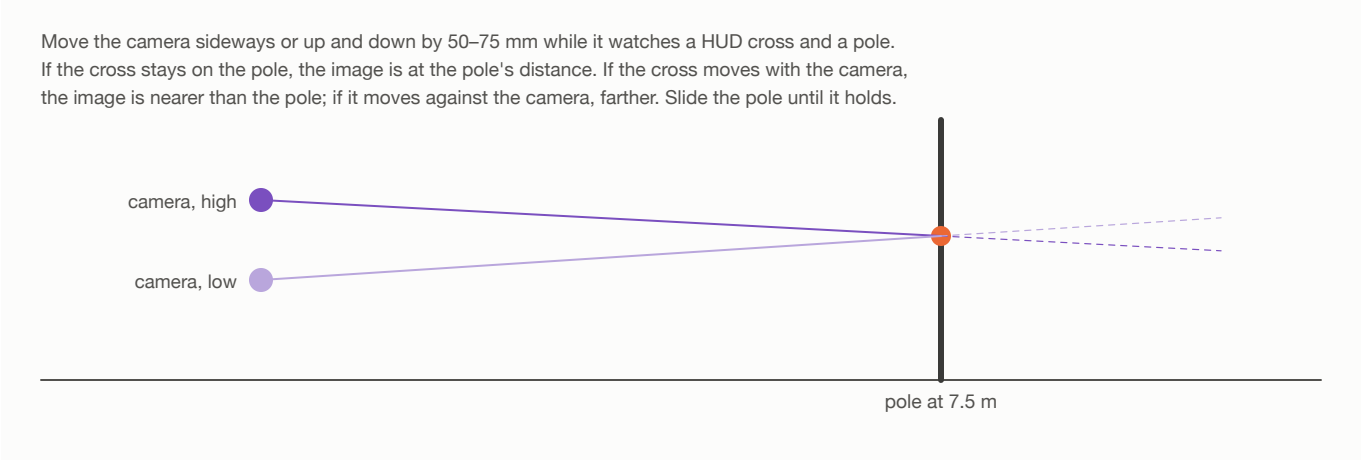


Figure B12.1. Measuring the virtual image distance by parallax. When a mark has no parallax against an object, they are at the same distance.

B12.7 Close up and test the shutter rules

1 Refit the lid and the dust cover; leave the access plugs out until B13's fine alignment is done.

2 With the controller: command open and close; then unplug its USB lead with the shutter open and time how long it takes to close (the rule: USB silent 5 s, backlight off, shutter closed); then cut the controller's power and confirm the spring closes the shutter.

- ☐ Shutter closes within 5 s of USB loss and immediately on power loss

3 Heat the cell's thermistor with a hair dryer (or simulate it with a resistor) past threshold 1: the backlight halves; past threshold 2: backlight off and shutter closed.

- ☐ Thermal rules act at the set thresholds

B12.8 If it goes wrong

SYMPTOM	RESPONSE
Image doubled or smeared in one direction	Astigmatism: the mirror is rotated about its axis (a toroid's axes must lie across and along the car), or tilted far from the traced angle. Re-seat it.
Image keystoned or bowed	Expected; B13's pre-distortion removes it. Large asymmetry left to right means the mirror is twisted.
Window clipped at one edge	The mirror's tilt is off, or a part of the housing blocks the beam (check with the card); for the top edge from high eye positions, the fold-height decision (B11.1).
Image much dimmer than expected	Polarisation: the panel's light is p at the glass. Check with the reference polariser (B9.4) and the retarder's orientation.

B13 Stage 2 in the car: calibrate it, then prove it at noon (gate G6)

SOLVES	WEEKS	CLOSES
Registration (2.3, 2.10), brightness against daylight (2.5), night comfort (2.11), the ghost decision (2.9)	8–12	Gate G6: every acceptance row passes at noon
NEEDS	**PRODUCES**	**PEOPLE**
B7 (car installation), B12 (aligned unit, sun test passed), B4 and B6 (software), the luminance meter	A calibrated HUD, `calibration.yaml`, the dimming curve, a completed acceptance sheet	Builder, a helper for the target board, the regular driver, the safety driver for drives

What this stage solves, intuitively. Calibration teaches the software three facts it cannot know from the design: exactly where the road camera points relative to the car, exactly where the driver's eye sits, and exactly where each panel pixel appears to that eye through this particular mirror and this particular windshield. With those, a mark computed for a lane point lands on the lane. Then every target from Chapter 1 is measured with an instrument, at noon against the sky and the road, and at night.

B13.1 Materials and instruments

ITEM	USE
30 m target board, 2.0 × 1.4 m plywood, matte white, with a printed checkerboard and five ring targets; two stands	Camera-to-car and eye calibration; static registration
Calibration checkerboard, A3, on a rigid board	Camera intrinsics; driver-camera pose
Camera at the eye point on a tripod, with a 300 mm rail for lateral moves	Grid photographs, registration and eye-box measurements
Laser rangefinder, 30 m tape, chalk line, string	Centreline, board survey
Spot luminance meter on a tripod	Luminance and contrast rows
Ground grid: white tape lines at 10, 15, 20 and 30 m and at ±1.85 m from the centreline	Visual check of registration on the ground
Anti-slip pad shaped to the dash; two webbing straps with cam buckles; a luggage scale	Restraint test of the housing

B13.2 Put the unit on the dash

1 Lay the anti-slip pad on the dash footprint marked in B1. Set the housing on it, front edge toward the glass. Adjust the four levelling feet until the housing is level across the car and at the traced pitch fore-aft (angle gauge on the lid, against the ground reference), stable with no rocking, and at least 8 mm below the glass everywhere.

- ☐ Glass clearance ≥ 8 mm at the front edge

2 Restrain it so it cannot move in a hard stop. Pull the housing horizontally with the luggage scale, forward, backward and sideways: it must not slide at 1.5 times its own weight (a 5 kg unit: about 75 N). Add the straps to anchor points agreed with the car's owner if it does; never to an airbag cover, never across the passenger airbag's door or the steering wheel's path, and never by screws into the dash.

- ☐ No movement at 1.5 × weight in four directions

3 Mount the LED driver and the controller beside or behind the housing (B7.6); connect the HDMI, LED, fan, thermistor, servo, light-sensor and USB leads per Appendix A. Tie every cable so nothing can reach the pedals or the steering wheel.

4 Power up; open the shutter; show the alignment pattern. From the nominal eye point, fine-tune the three adjusters until the window is centred on the line of sight 5° down. Push the three access plugs into the front wall.

B13.3 Calibrate: the three transforms

The plan's projection chain is `D @ K_eye @ view_from_eye @ eye_from_car`; this guide replaces `D @ K_eye` by a lookup table from the ray trace corrected by photographs (the design-basis box explains why). All results go into `config/calibration.yaml` (Appendix C), and calibration is redone for every car, every windshield change, and every time the road camera is remounted.

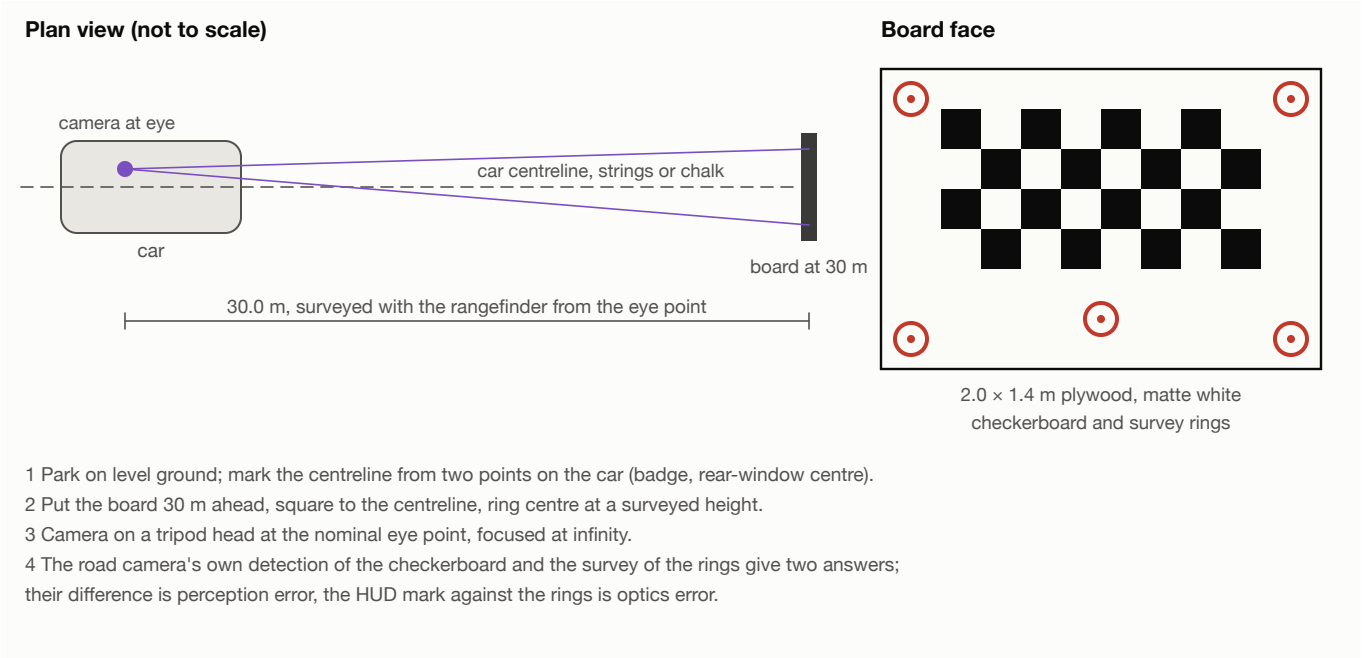


Figure B13.1. The 30 m set-up and the board's face.

A. The road camera in the car frame

1 Park on level ground with 35 m clear ahead. Snap a chalk line for the car's centreline from two points on the car's axis (front badge and rear-window centre, both found with a plumb bob), extended 32 m ahead.

2 Stand the board 30.0 m from the road camera's left lens along the centreline, square to it (check that both lower rings are the same distance from the lens with the rangefinder). Survey each ring's centre: distance, height above the ground, and lateral offset from the chalk line.

3 Run `tools/calib_camera.py`: it detects the checkerboard in the ZED's left image, solves its pose, and combines it with the survey to give the camera's rotation (pitch, yaw, roll) and position in the car frame. The camera's height comes from the ground-plane fit at rest. Save it.

4 Check: the ring centres projected through the new extrinsics land within 2 pixels of their detected positions in the camera image.

- ☐ Ring reprojection ≤ 2 px

B. The driver camera in the car frame

1 Hold the A3 checkerboard at the nominal eye point (taped to a stand on the driver's seat, its centre at the eye point, square to the car's axis). Measure its position from the datums of B1.3.

2 Run `tools/calib_drivercam.py`: the OAK-D Pro sees the board, solves its pose; the measured board position gives the camera's pose in the car frame. Save it.

3 Check: with the driver seated, `eyePose` reports the eye midpoint within 20 mm of the taped nominal point.

- ☐ Tracked eye within 20 mm of the nominal point

C. The eye and the panel: lookup table and photographs

1 Calibrate the eye-point camera's own lens first (checkerboard, 20 views) so each of its pixels is a known direction.

2 Put the camera at the nominal eye point (lens entrance pupil there, within 5 mm), level, looking straight ahead. Show the panel grid: crosses at known panel pixels across the crop. Photograph it with the board and the road in the background.

3 Run `tools/calib_hud.py`: it finds each cross, converts its image position into a direction from the eye, compares it with the direction the trace's lookup table predicts for that panel pixel, and fits a smooth correction (a low-order 2-D polynomial or a thin-plate spline). Residual after the fit should be under about 2 arcminutes.

4 Repeat from five eye positions on the rail and tripod: centre, ±50 mm left and right, ±30 mm up and down. The correction is interpolated between them, so the projection stays right as the head tracker moves the eye.

5 Refine the nominal eye point: show the five ring targets as HUD marks computed from their survey; adjust the eye-point offset in the file until the marks sit on the rings from the camera at the eye point.

6 Commit `calibration.yaml`.

B13.4 Measure everything: the acceptance sheet

Measure each row with the instrument named; record the value, the conditions (time, sky, sun elevation, temperature), and pass or fail. The plan's gate G6 is every row passing at noon. Rows marked † are the guide's additions.

QUANTITY	REQUIREMENT	HOW TO MEASURE	RESULT
Virtual image distance	≥ 6 m, target 7.5 m	Camera-focus method (focus the eye-point camera on the HUD cross, then find the real target at the same focus) or the parallax method of Figure B12.1, at the centre and four corners	
Field of view	10.8° × 5.4° (10° × 4° to 13° × 5°)	Window border photographed from the eye point; angles from the calibrated camera. Expect the traced value (about 8.7–9.1° wide); record it, and note if it falls outside the band	
Eye box	≥ 120 × 60 mm untracked	Camera on the rail: the whole border visible at every position of a 5 × 3 grid over 130 × 70 mm	
Luminance, sky behind, noon	≥ 1,500 cd/m²; ratio (overlay + background) / background ≥ 1.15 against 10,000	Spot meter at the eye point on a full-white patch against the sky; then the sky alone in the same direction with the patch off	
Luminance, sunlit road behind	≥ 750 cd/m² (ratio 1.15 against 5,000)	Same, lower part of the window against the road; reported separately	
Night	White ≤ 50 cd/m² at the night setting (guide: start at 3); unlit window ≤ 0.1	Same meter at the night setting; then a black frame with the backlight at the night setting	
Panel luminance and uniformity	≥ 7,716 cd/m²; ±20% over the crop	From B9 (G5); re-check one point	
Static registration at 30 m	≤ 0.5°	Camera at the eye point; HUD marks for the surveyed rings against the rings; marks also drawn from the road camera's own detection, to separate perception from optics error	
Registration with head motion	≤ 0.3° over ±75 mm	Same, camera moved on the rail with the head tracker following a face target fixed above the camera (or the camera's position fed in), tracker on	
Motion to photon	≤ 90 ms; compensated error ≤ 0.3°	Latency probe (B6) re-run in the car; `hudStats` ages cross-checked	
Ghost image	Separation within 2× the predicted 1.4′; acceptable on the demo graphics at three eye heights	Photograph a thin cross from the eye point at the nominal height and ±35 mm; measure the copies' separation	
Sun focus	Under the cell rating for 10 min, shutter open, worst sun; sun-watch closes inside the patch	B12.4 result; then a logged drive with sun-watch events	
Cell temperature	Under the rating after 30 min at full backlight in a 40 °C box	B9 (G5) result	

QUANTITY	REQUIREMENT	HOW TO MEASURE	RESULT
Speed shown	Within 1 m/s of a reference GNSS trace while moving; dash without a fix	Logged drive against a phone's GNSS log	
Installation	Checklist signed	Appendix F	
Lane-line stability †	No visible wobble on a straight road	Video from the eye point at 25 m/s; lines' frame-to-frame lateral jitter under 1 arcminute	
Ribbon visible at noon †	Contrast ≥ 1.15	Meter on the ribbon against the road beside it	

B13.5 Tune the dimming curve

1. At dusk, drive the course (passenger operating the laptop) with the controller logging lux and duty. At each light level, the passenger adjusts the duty until the driver calls the white "clearly visible, not glaring". Record pairs of lux and duty from full daylight to full dark.
2. Fit a smooth curve through them (piecewise linear in log lux is enough), from the night level to full duty at 10,000 lux or more; store it in `config/dimming.yaml` and in the controller's firmware table (B5). Update once a second with a one-second ramp; give the driver a one-step offset either way.
3. At night on a dark stretch, check the grey rectangle of the black level. If it is visible, lower the night duty; if it is still visible at a white level the driver finds too dim, the VA-cell panel is the alternative (plan, 6.6).

B13.6 Decide on the windshield

Look at the three ghost photographs with the demo graphics (lines at least 6 px, filled ribbon, heavy text). If the ghost is objectionable to the driver and to two other viewers, order the wedge-PVB windshield for this car ($600–1,500 installed) and repeat B1's measurements and B13 for the new glass. Otherwise record "stock windshield accepted".

B13.7 Gate G6

Every row of the B13.4 table passes at noon, with the results and conditions in the log. If a row fails, the demo format falls back to dusk (luminance rows) or the gate returns to the stage that owns the failing part.

B13.8 If it goes wrong

SYMPTOM	RESPONSE
Static error larger at the edges of the window than the centre	The lookup-table correction is too stiff or was fitted from too few crosses; use more crosses and a higher-order fit, and check the eye-point camera's own lens calibration.
Error grows with head motion	Driver-camera pose wrong (B13.3 B) or the correction not interpolated across eye positions (B13.3 C, step 4).
Error differs between the camera-detection marks and the survey marks	That difference is perception error: road-camera extrinsics or the plane fit (B13.3 A).

SYMPTOM	RESPONSE
Sky contrast below 1.15	Check polarisation (the image should vanish through a polariser with its axis vertical), the backlight at full duty, the cell's temperature derating; then films or collimation (plan, 6.6), or a dusk demo.
Window glows at night	Night duty too high; black-level leak; VA cell.

B14 Hazards, polish and rehearsal (gate G7)

SOLVES	WEEKS	CLOSES
Turns a working instrument into a demonstration that runs without its builder at the keyboard	8–12	Gate G7: the script runs twice in a row without an operator touching the Jetson
NEEDS	**PRODUCES**	**PEOPLE**
B13 (or B8 for a Stage 1 demo); the venue, a lead car, a volunteer pedestrian	Tuned hazard rules, the final audience layout, a rehearsed script, the demo-day checklist	Driver, safety driver, presenter, a volunteer pedestrian, a lead-car driver

B14.1 Tune the hazard rules on the course

1. With the volunteer walking slowly along the roadside at 20–30 m and the car at walking pace, then 10 m/s, record how often the pedestrian outline appears late, flickers, or persists after the person is gone. Adjust the three-detection rule and the 400 ms hold until the outline appears within about half a second and never flickers. The volunteer never steps into the car's path; a cone line marks the boundary.
2. Repeat with a cyclist if the venue allows, and with parked cars that must not be outlined as hazards.
3. Run the lead car ahead at 10–25 m/s with gentle braking. Watch the chevron and the gap number from the eye point; the chevron must stay under the car's rear and the gap must count down smoothly.
4. Train and test the speed-limit classifier on the course's signs (if they exist). If it misreads even once in ten passes, remove it from the speed block for the demo.

B14.2 Remove clutter

Drive the course with the presenter and one person who has never seen the HUD in the passenger seat watching the audience view. Remove or dim anything that is neither conformal nor safety-relevant. Keep the clutter mode as a ten-second contrast on a key press only.

B14.3 The audience view

Finalise the layout on monitor 2: the rendered HUD frame, the driver camera with the tracked eye point, and the live numbers beside their targets: registration residual (target 0.5°), data age (target under the 90 ms budget), contrast at noon (target 1.15), latency, GNSS fix, sun-watch state, cell temperature. A number outside its target shows its target next to it, not a colour alone.

B14.4 Rehearse

1. Write the script: start-up, the straight, the curve, the lead car, the pedestrian, a night or dusk segment if shown, the clutter contrast, shut-down. Each step has the presenter's words and what the audience view shows.
2. Make start-up automatic: the supervisor starts on boot; the HUD comes up with the shutter closed, opens it once the controller reports healthy and sun-watch is clear; no command line on the day.
3. Rehearse the script three times on the demo course. Log every intervention.

B14.5 Demo-day checklist

- ☐ Installation checklist (Appendix F) signed for today.
- ☐ Calibration file dated after the last camera or housing move; a 30 m board check from the eye point this morning.
- ☐ Weather and sun: sun-watch patch times for the venue checked; the planned headings avoid the patch.
- ☐ Nobody in the car wears polarised sunglasses.
- ☐ Safety driver briefed; lead-car driver and volunteer briefed; cones out.
- ☐ Laptop charged with SSH access; spare fuse; spare HDMI cable; the Pico firmware (UF2 file) on the laptop.
- ☐ Fallback ready: Stage 1 rig in the boot for a night or overcast demo.

B14.6 Gate G7

The script runs twice in a row without anyone touching the Jetson. Record the date; it sets the demo date.

APPENDICES

Reference

Wiring, pin maps, file formats, message fields, test sheets, the installation checklist, the formulas behind the numbers, a glossary, and where every number comes from.

A Wiring and connector schedule

Every cable in the installation. Label both ends of each cable with its ID before installing it.

ID	FROM	TO	CABLE	FUSE	ROUTE AND NOTES
Power					
P1	Always-live 12 V socket	DC-DC VIN (J-input pin VIN)	1.5 mm² red/black twin, plug at one end, Mini-Fit Jr at the other	10 A blade, ≤ 150 mm from the plug	Under the console trim edge to the footwell plate
P2	Switched 12 V socket	DC-DC IGN	0.5 mm²	1 A blade	With P1
P3	DC-DC output (16 V)	Jetson DC jack	1.0 mm² twin, 5.5 × 2.5 mm plug, centre positive	converter's own	On the plate; check polarity before first connection
P4	DC-DC output (16 V)	LED drivers A and B, +Vin and GND	0.5 mm² twin	2 A inline at the converter	Footwell to dash beside the housing, under the dash's lower edge
P5	DC-DC output (16 V)	5 V, 5 A buck input	0.5 mm² twin	3 A inline	On the plate
P6	5 V buck	Panel board power (USB-C or micro-USB, by revision)	USB cable rated 3 A	—	Into the housing through the HDMI grommet
P7	5 V buck	Controller 5 V terminal; fan	0.5 mm² twin	—	To the dash
P8	5 V buck	OAK-D Y-adapter power leg	USB cable rated 3 A	—	Along the column's underside
P9	LED driver A (+Vout, –Vout)	LED string A	0.5 mm² twin, JST-VH	—	Through the LED grommet; never plug or unplug live
P10	LED driver B (+Vout, –Vout)	LED string B	0.5 mm² twin, JST-VH	—	As P9
P11	Second socket (feed 2)	USB-C PD charger	The charger's own plug	10 A blade if the plug has none	To the passenger headrest
Control					
C1	DC-DC J8	J401 button header pins 1 (PWR BTN) and 2 (GND)	2-core 0.25 mm², 2-pin header connector	—	On the plate
C2	Controller level and enable A, B	LED drivers pins 2 and 3	Shielded 4-core plus ground	—	Short, beside the housing
C3	Controller servo header	SG90 under the lid	The servo's lead, extended with 3-core	—	Down the left wall's channel
C4	Controller NTC connector	NTC on the cell	Shielded twin	—	Through the LED grommet

ID	FROM	TO	CABLE	FUSE	ROUTE AND NOTES
C5	Controller I²C connector	TSL2591 in the lid pod	Shielded 4-core	—	Down the left wall's channel
C6	MAX31855 socket	Type-K bead (tests only)	Type-K extension	—	Temporary
Data					
D1	ZED 2i	Jetson USB 3	Stereolabs USB-C cable with screw locks, 1.5 or 3 m (not included with the camera)	—	Windshield top edge, headliner edge, beside (never over) the A-pillar airbag trim
D2	OAK-D Pro	Jetson USB 3 (Y-adapter data leg)	USB-C	—	Column underside
D3	ZED-F9R	Jetson USB	USB-C	—	On the plate
D4	GNSS antenna	ZED-F9R antenna connector (SMA on GPS-22660)	The antenna's 5 m lead	—	With D1
D5	Controller (Pico)	Jetson USB	Micro-USB	—	Dash to footwell
D6	Jetson HDMI	Panel board HDMI	High-speed HDMI, 2 m	—	Through the HDMI grommet
D7	Jetson Ethernet	Switch port 1	Cat 6	—	On the plate
D8	Switch port 2	Mini PC 2	Cat 6, 3 m	—	Under the passenger seat's edge to the headrest

B Controller pin map, protocol and firmware notes

Pin map (Raspberry Pi Pico)

PIN	GPIO	FUNCTION	CONNECTION
6	GP4	I²C0 SDA	TSL2591 SDA (address 0x29)
7	GP5	I²C0 SCL	TSL2591 SCL
16	GP12	PWM 25 kHz	Level B: RC filter, op-amp B, driver B pin 2
17	GP13	Output	Enable B: NPN B base (high = LEDs enabled)
19	GP14	PWM 25 kHz	Level A: RC filter, op-amp A, driver A pin 2
20	GP15	Output	Enable A: NPN A base
21	GP16	SPI0 RX	MAX31855 DO
22	GP17	SPI0 CSn	MAX31855 CS
24	GP18	SPI0 SCK	MAX31855 CLK (up to 5 MHz)
29	GP22	PWM 50 Hz	Servo signal (about 1–2 ms pulses; end-points measured)
31	GP26	ADC0	NTC divider midpoint
33	AGND	Analogue ground	NTC low side, filter capacitor
35	ADC_VREF	ADC reference (3.3 V, filtered)	NTC divider top (10 kΩ 0.1%)
36	3V3	3.3 V out (keep under 300 mA)	TSL2591, MAX31855
39	VSYS	Power in (1.8–5.5 V)	5 V buck through the Schottky diode
3, 38	GND	Ground	Common ground

Serial protocol (USB CDC, 115200, one ASCII line per message)

MESSAGE	DIRECTION	MEANING
`HB`	Jetson to Pico	Heartbeat, 5 Hz. Five seconds without one: LEDs off, shutter closed.
`LV a b`	Jetson to Pico	Requested levels of strings A and B, 0.000–1.000 (0 also disables the string)
`SH open` / `SH close`	Jetson to Pico	Shutter request
`SUN 1` / `SUN 0`	Jetson to Pico	Sun-watch inside the patch: forces off and closed while 1
`TH t1 t2`	Jetson to Pico	Derate and cut-off temperatures at the NTC, °C (stored in flash)
`RST`	Jetson to Pico	Clears a latched over-temperature once the NTC is below t1 − 5 °C
`ST t=… lux=… ntc=… tc=… a=… b=… en=.. sh=… fl=…`	Pico to Jetson	Status at 10 Hz: time, ambient lux, NTC °C, thermocouple °C, applied levels, enables, shutter state, flags (USB_SILENT, DERATE, OVER_TEMP, SUN, TC_FAULT)

Conversions

NTC: $R = 10\ k\Omega \times n / (4095 - n)$ for ADC count n (the NTC is the lower leg); $1/T = 1/298.15 + \ln(R / 10\ k\Omega) / B$, T in kelvin, $B \approx 3950$ (calibrate, B5.6)
Driver level (inverted analogue input, 0.13 V full, about 4.2 V off): $V = 4.2 - 4.07 \times level$; $PWM\ duty = V / (3.3 \times 1.33)$
MAX31855 frame (32 bits): D31–D18 thermocouple temperature, 0.25 °C per bit, signed; D16 fault; D15–D4 cold junction, 0.0625 °C per bit; D2, D1, D0: short to VCC, short to GND, open
Servo: 20 ms period; pulse width between the measured closed and open end-points (about 1.0 and 2.0 ms)

C Calibration file

`config/calibration.yaml` on the Jetson, committed to the repository after every calibration. Every transform is in the plan's car frame (x forward, y right, z down, metres, origin at the road camera's left lens). Angles in degrees. Example structure with placeholder values:

```
car: "make model year, VIN last 6"      # which car this file belongs to
date: 2026-11-20
driver_seat: {track_notch: 7, backrest_deg: 22, wheel_reach: 3, wheel_height: 2}
windshield: {rake_deg: 28.0, thickness_mm: 4.96, Rh_m: 4.0, Rv_m: 8.0}
road_camera:                        # B13.3 A
  height_above_ground_m: 1.30
  rotation_deg: {pitch: -1.2, yaw: 0.3, roll: 0.1}
  intrinsics_from: zed_factory      # ZED factory calibration, left camera
driver_camera:                      # B13.3 B, OAK-D Pro pose in the car frame
  position_m: [0.95, -0.37, 0.62]
  rotation_deg: {pitch: 12.0, yaw: 178.0, roll: 0.0}
eye_nominal_m: [-0.36, -0.37, 0.08]  # B1.3, refined in B13.3 C step 5
projection:
  mode: lut                         # lut (Stage 2) | plane (Stages 0 and 1)
  lut_file: config/hudd_lut_car.json  # from the B1 trace
  correction: {type: tps, file: config/hud_correction.json, eye_points: 5, residual_arcmin: 1.6}
panel:
  crop_px: {x: 487, y: 303, w: 946, h: 473}   # B9.5 step 3
  flip: {x: false, y: true}                   # per optical stage, set with the test card
stage1:                             # B8 only
  plane_focal_px: 1700
  warp_file: config/stage1_warp.json
```

Redo: the whole file for a new car or windshield; `road_camera` after any camera remount; `driver_camera` after it is moved; `projection` after any change to the housing's position or the mirror's adjusters; `panel.crop_px` after the cell is reseated.

D Topics and messages

ZeroMQ PUB/SUB over `ipc://`, msgpack maps. Every message carries `t_capture`, `t_publish` (monotonic nanoseconds) and `seq`. Empty messages are still published. The graph is a tree with `hudd` at the root.

TOPIC	RATE	FIELDS
`lanes`	30 Hz (20 if GPU-bound)	Per line: car-frame points on the fitted plane (smoothed), confidence, type (ego left, ego right, other, edge), raw points (debug); plane parameters and fit status; capture time
`lanes2d`	30 Hz	Per line: image points, confidence
`objects`	30 Hz	Per track: id, class, bearing, range, car-frame position, velocity, image box, track age, sign value
`objects2d`	30 Hz	Per detection: box, class, confidence, track id
`imu`	400 Hz (batched)	Gyroscope and accelerometer samples in the car frame with monotonic times
`ego`	100 Hz	Speed, yaw rate, pitch, pitch rate, longitudinal acceleration, fix quality, source flags
`gnss`	10 Hz	Position, course, speed, accuracies, time, fusion status
`eyePose`	30 Hz	Eye midpoint in the car frame, confidence, capture time
`sunWatch`	1 Hz	Sun azimuth and elevation in the car frame, inside-patch flag
`health`	2 Hz	Ambient lux, backlight duty, cell temperature, shutter state, controller flags, process liveness, SoC temperatures
`hudStats`	10 Hz	Age of each topic as drawn, render time, dropped frames, last registration residual
`hudFrame`	15 Hz	Downscaled rendered frame for the audience view

E Test record sheets

Print one per run. Fill every cell; "not measured" is a valid entry, a blank is not.

Gate G2 · Latency (B6.6)

DATE	SET-UP (BENCH / CAR)	SAMPLES	MEDIAN, MS	95TH PCT, MS	RENDER + PANEL ONLY, MS	PASS ≤ 90

Gate G5 · Picture generator (B9.6, B9.7)

POINT (5 × 3)	CD/M²	POINT	CD/M²	POINT	CD/M²
top left		top centre		top right	
upper left		upper centre		upper right	
middle left		centre		middle right	
lower left		lower centre		lower right	
bottom left		bottom centre		bottom right	

MINUTE	BOX °C	NTC °C	CELL FRONT °C	MINUTE	BOX °C	NTC °C	CELL FRONT °C
0				20			
5				25			
10				30			
15				cool-down 5			

Gate G6 · Registration (B13.4)

EYE POSITION	TRACKER	RING 1, °	RING 2	RING 3	RING 4	RING 5	MAX
nominal	on						
−75 mm lateral	on						
+75 mm lateral	on						
−75 mm lateral	off						
+75 mm lateral	off						

Gate G6 · Luminance (B13.4)

DATE, TIME	SKY	SUN ELEV.	OVERLAY + BG	BACKGROUND	RATIO	TARGET	PASS
						sky 1.15	
						road 1.15	
						night ≤ 50	
						unlit ≤ 0.1	

F Installation checklist

Signed before every drive by the driver and the person responsible for the installation. A drive does not start with an unticked box.

	CHECK
☐	No connector to the OBD-II port, the CAN bus, or any vehicle wiring other than the accessory socket(s).
☐	Inline fuse fitted within 150 mm of the plug, correct rating, holder closed.
☐	Nothing bonded, screwed or drilled into the car; nothing attached to the glass except the clamped road camera on the mirror stalk.
☐	HUD housing restrained (pull test passed at installation) and at least 8 mm below the glass; not over an airbag door; not in the steering wheel's path.
☐	All mounts removable without tools; all cables tied clear of the pedals, the steering column's movement and the seat rails.
☐	Driver's view of the road and mirrors unobstructed outside the HUD window; defroster outlet in front of the driver open.
☐	Shutter closes on power loss (tested today); controller reports healthy.
☐	Sun-watch running with the current patch; today's headings checked against it.
☐	No polarised sunglasses in the car.
☐	Venue confirmed; safety driver present and briefed; the presenter is not driving.
☐	Stage 1 only: combiner hinge locked, edges protected, enclosure strapped.

DATE	DRIVER (SIGNATURE)	INSTALLATION RESPONSIBLE (SIGNATURE)	SAFETY DRIVER

G Formulas used in this guide

Virtual image of a lens or mirror (object at u inside f): $1/v = 1/f - 1/u$, v negative (virtual); magnification $M = |v| / u$
Plan's Stage 2: f = 330 mm, u = 316 mm, image at 7.5 m, M = 23.7, 60 × 30 mm crop becomes 1.42 m × 0.71 m
Plan's Stage 1: f = 300 mm, u = 280 mm, v = −4.2 m, M = 15
Off-axis mirror at incidence i (Coddington): tangential radius $Rt = 2f / \cos i$, sagittal $Rs = 2f \cdot \cos i$ (both foci at f)
Field of view for a window w at distance D: $2 \cdot \operatorname{atan}(w / 2D)$
Pinhole focal length in pixels for the virtual plane: (crop height in px / 2) / tan(vertical field / 2)

Luminance at the eye: $L_{eye} = L_{panel} \times R_{fold} \times R_{mirror} \times R_{windshield(s)} \times$ (cover, polariser, IR-cut, retarder transmissions)
Plan path: 0.90 × 0.90 × 23.6% = 19.4%; contrast ratio $C = (L_{overlay} + L_{background}) / L_{background} \geq 1.15$
Backlight: $L_{backlight} = L_{panel} / T_{cell}$; flux $\Phi = \pi \cdot L \cdot A$ (Lambertian); LED power = Φ / (efficiency × lm/W)

Fresnel reflectance (n = 1.52, 65°): s 23.6%, p 1.2%
Ghost separation (plain laminate t, incidence θi, refraction θt): $d = 2t \cdot \tan \theta t \cdot \cos \theta i$; seen at distance D as d / D
Laminate thickness from the laser double spot: $t = d / (2 \cdot \tan \theta t \cdot \cos \theta i)$, $\sin \theta t = \sin \theta i / 1.52$
Curvature from a sagitta s under a straightedge L: $R = L^2 / (8s)$

Age compensation: $age = t_{now} - t_{capture}$; $s = \int v \, dt$, $\psi = \int \omega \, dt$ over the age; $p_{now} = Rz(-\psi) \cdot (p_{capture} - [s, 0, 0])$
Registration error from a range error Δz at range z, camera-to-eye baseline b: $\approx b \cdot \Delta z / z^2$
Parallax of an untracked head shift h for a point at distance d (image at 7.5 m): $\approx h \cdot |1/7.5 - 1/d|$ (radians, h and d in metres)
Half-wave retarder: output angle = 2 · (fast axis) − input angle

The parallax line uses the simulator's correction of the plan's table: the error is zero for a point at the image distance and grows for points nearer or farther, rather than following atan(h / d).

H Glossary

Age compensation
Moving each measured point by the car's own motion since the point's camera frame was captured, so it is drawn where it is now.

Backlight
The light source behind an LCD. Here, a custom LED box behind the 60 × 30 mm crop.

Cold mirror
A dielectric mirror that reflects visible light and transmits infrared. Used as the fold, it dumps solar infrared into an absorber.

Combiner
The partly reflecting glass that superimposes the image on the view: the windshield in Stage 2, a separate glass in Stages 0 and 1.

Conformal
Drawn so that it lies on the real object it describes, from the driver's eye.

Crop
The 60 × 30 mm (946 × 473 px) region of the panel that is lit and drawn; everything else stays black.

Eye box
The region the driver's eye can move in while still seeing the whole window: 130 × 70 mm by design.

Fold
The flat mirror that turns the beam so the optical path fits under the dash.

Gate
A pass/fail test (G0 to G7) that decides whether the next phase starts.

Ghost
The second, fainter copy of the image reflected from the windshield's other surface.

Hot mirror
A dielectric filter that reflects infrared and transmits visible light; in front of the cell it keeps focused solar infrared off it.

Lookup table (LUT)
The precomputed map, from the ray trace, from an eye position and a viewing direction to the panel pixel that appears there.

PGU
Picture generation unit: the panel with its backlight.

RANSAC
A fitting method that tries many minimal fits and keeps the one most points agree with, ignoring outliers.

Registration
How well a drawn mark lines up with its real object, as an angle at the eye.

Retarder (half-wave)

A film that rotates linear polarisation: output angle = 2 × fast axis − input angle.

s and p polarisation

Light polarised perpendicular (s) or parallel (p) to the plane of incidence. At the windshield, s is across the car and reflects far better.

Sagitta

The depth of a curve below a straight chord; gives the radius.

Sun-watch

The rule that closes the shutter when the computed sun direction enters the patch of sky the optics can focus onto the panel.

Toroid, freeform

A mirror surface with different curvatures in two directions (toroid), or a general polynomial surface (freeform), used to correct astigmatism and flatten the image.

Virtual image

An image that light only appears to come from; the eye can focus on it but it cannot be caught on a screen.

Where the numbers come from

SOURCE IN THE REPOSITORY	WHAT THIS GUIDE TAKES FROM IT
`docs/arhud-plan-tft-src/calc.py` and `docs/ARHUD-PLAN-TFT.pdf`	Every number in double braces in this guide's source: optics, luminance, sun, latency, GPU, costs, requirements; the plan's figures reused here (mockup, sun entry, architecture, schedule)
`docs/arhud-sim` (Three.js simulator)	The installed layout (`geom.js`): 1.06 m path, toroid radii, fold angle and height; the sphere's astigmatism; traced window and distance; lane-noise finding; lookup-table projection
`docs/arhud-blender` (Cycles cross-check)	Image-surface curvature; freeform fits for flat and curved glass; eye-box extent; ghost; the faint second window; sun spot and the cell patch; luminance survey
`docs/arhud-cad` (build123d housing)	Every printed part, its print orientation and mass; the 18 assembly steps; fasteners; beam, envelope and printer checks; the fold-height finding; CAD renders
`docs/arhud-road` (MetaDrive road simulation)	Ribbon contrast at noon (39% of white), the value of compensation and of the plane fit, tracker on versus off
Vendor pages and datasheets read on 2026-10-25	Connector types, pin functions, dimming inputs and configuration steps for the bought electronics (marked where unverified)

Nothing in this guide has yet been measured on the hardware. Every ASSUMED value names the measurement that replaces it; when a measurement disagrees with this guide, the measurement wins and this guide's source should be updated.